

Lecture notes on stability theory

with chapters on NIP and simplicity

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To Xavier Caicedo

Introduction

Model theory, in the part of it that these notes describe, is a study of the ways in which a formula can fail to be complicated. A formula in two variables is, after all, nothing but a relation between two sets, and whether it is complicated depends on what one is prepared to count as complication. Shelah proposed to count configurations. A formula is stable if it cannot order arbitrarily long sequences; it is NIP if it cannot shatter arbitrarily large finite sets; and a theory is simple if none of its formulas can grow a certain kind of tree. These are the dividing lines of the subject. On the near side of each, a theory admits a structure theory; on the far side, examples show that none is to be had.

I have tried to present each line twice: once from within a single formula, by the configuration that the formula forbids, and once from without, by the relation of independence that the theory then carries. The two presentations are not of equal difficulty. That a stable formula has definable types is a theorem about one formula; that a stable theory carries exactly one notion of independent extension is a theorem about all formulas at once, and it rests on the first. In the same way, the tree property of a single formula and the independence theorem of Kim and Pillay stand at the two ends of the treatment of simplicity. Between the ends lie arguments that I have tried to give in full.

A second theme runs alongside the first. A formula is a function with values 0 and 1, and nothing in the definition of stability requires the values to be so restricted. When they are allowed to be real, stability becomes the statement that two iterated limits may be exchanged, and the subject meets Grothendieck's double limit criterion, Rosenthal's dichotomy and the reflexive Banach spaces. I have followed this theme wherever it leads to a proof, and no further; several chapters close with their real-valued counterparts, and the analytic background is collected in an appendix. The thesis, if one is required, may be put briefly: the types of the logician and the limits of the analyst are the same objects seen from two sides, and each side sometimes proves with ease what the other proves only with effort.

The notes are addressed to a reader who knows the compactness theorem and is not afraid of ultrafilters; nothing else is presupposed. The historical remarks at the end of each chapter record, as accurately as I have been able, where the ideas came from, and the mathematical line is kept free of them. The Leitfaden that follows the table of contents shows how the chapters depend on one another, for the reader who prefers not to read them in order, which is to say, for most readers.

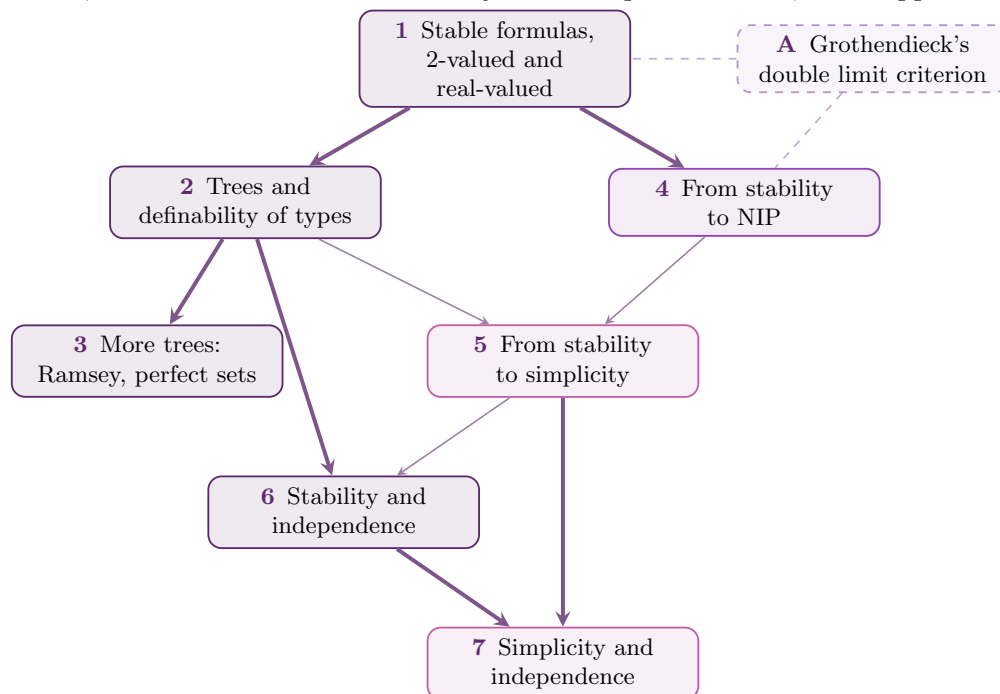
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Leitfaden

The chapters do not form a single line. Chapter 1 is the trunk: it introduces stable formulas, in the 2-valued and in the real-valued form, and states the definability of types, which Chapter 2 proves. From there the notes branch three ways: toward trees and the perfect set dichotomy (Chapter 3), toward NIP (Chapter 4), and toward simplicity and independence (Chapters 5–7). The first five chapters study the dividing lines from inside the formulas: definability and trees for stability, shattering for NIP, dividing and the tree property for simplicity, which meet in the theorem that the stable theories are the simple NIP theories (see Theorem 5.3.4). The last two describe stability and simplicity globally, by the independence relations they carry: uniqueness of independent extensions for stability, amalgamation for simplicity. In the diagram, an arrow from one chapter to another means that the second uses results of the first, a heavy arrow marks heavy use, and arrows that follow from others are omitted. A reader interested in simple theories may read Chapters 1, 2, 5, 6 and 7; a reader interested in NIP may read Chapters 1 and 4, with Appendix A.



The appendix (dashed) collects the analytic background shared by Chapters 1 and 4. Chapter 6 takes from Chapter 5 only the conventions on monster models (see Section 5.1) and the axioms for independence (see Definition 5.4.7); its real source is the definability of types of Chapters 1 and 2.

Chapter 1

Stable formulas, both 2-valued and real-valued

Stability is a property of a single formula $\varphi(x, y)$, and this chapter looks at stability from two sides. On the logical side, a stable formula is one that cannot order arbitrarily long sequences of elements, and the central fact about such formulas is that all their types are definable: Section 1.1 sets up this side and states the definability theorem, Theorem 1.1.6, whose proof occupies Chapter 2. On the analytic side, a formula is simply a $\{0, 1\}$ -valued function of two variables, and nothing prevents us from replacing such functions by bounded real-valued ones. Once we do, stability becomes the statement that two iterated limits can be exchanged, and in Section 1.2 we show how closely this property is tied to reflexive Banach spaces.

1.1. The 2-valued case: φ -types and type definability

Throughout this section and Chapters 2 and 3, we fix a first-order theory T in a language L and an L -formula $\varphi(x, y)$ consistent with T , where x and y are finite tuples of variables. Tuples of elements are written $a, b, \dots$ and are always of the appropriate length. We write $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$.

Definition 1.1.1. Let M be a model of T .

- (a) A φ -formula over M is a formula $\varphi(x, b)^u$ with $b \in M$ and $u \in \{0, 1\}$.
- (b) A complete φ -type over M (in the variables x) is a set p of φ -formulas over M such that
 - (i) the set p is consistent with the elementary diagram of M , and
 - (ii) for each $b \in M$, exactly one of $\varphi(x, b)$ and $\neg\varphi(x, b)$ belongs to p .

The set of complete φ -types over M will be denoted $S_\varphi(M)$.

If $N \succeq M$ and $c \in N$, the set

$$\text{tp}_\varphi(c/M) = \{\varphi(x, b)^u : b \in M, u \in \{0, 1\}, N \models \varphi(c, b)^u\}$$

is a complete φ -type over M . Conversely, every complete φ -type p over M is consistent with the elementary diagram of M , so it is realized by some element c of some $N \succeq M$, and then we have $p = \text{tp}_\varphi(c/M)$.

For $p \in S_\varphi(M)$ we write $D_p = \{b \in M : \varphi(x, b) \in p\}$. By condition (ii), we have $p = \{\varphi(x, b) : b \in D_p\} \cup \{\neg\varphi(x, b) : b \in M \setminus D_p\}$, so the type p is determined by D_p . The set D_p records, for each parameter b , whether $\varphi(x, b)$ holds of the elements that realize p .

A type over M need not be realized in M , but each finite part of it is realized there.

Lemma 1.1.2 (finite satisfiability). *Let $p \in S_\varphi(M)$ be given. For every finite subset $\{\varphi(x, b_l)^{u_l} : l < k\}$ of p , there is $a \in M$ with $M \models \bigwedge_{l < k} \varphi(a, b_l)^{u_l}$.*

Proof. The $L(M)$ -sentence $\exists x \bigwedge_{l < k} \varphi(x, b_l)^{u_l}$ is consistent with the elementary diagram of M . That diagram is a complete $L(M)$ -theory, so the sentence belongs to the diagram, that is, holds in M . ☺

Lemma 1.1.2 gives φ -types a natural topological home. The sets $\varphi(M, b) = \{a \in M^{|x|} : M \models \varphi(a, b)\}$, for $b \in M^{|y|}$, generate a Boolean algebra $\mathcal{B}_\varphi(M)$ of subsets of $M^{|x|}$. For $p \in S_\varphi(M)$, the sets $\varphi(M, b)$ with $\varphi(x, b) \in p$ and the complements $M^{|x|} \setminus \varphi(M, b)$ with $\neg\varphi(x, b) \in p$ have the finite intersection property by Lemma 1.1.2, and they generate an ultrafilter on $\mathcal{B}_\varphi(M)$. Conversely, an ultrafilter on $\mathcal{B}_\varphi(M)$ determines a set of φ -formulas over M that is finitely satisfiable in M , hence consistent with the elementary diagram of M , and so a complete φ -type. A complete φ -type over M is therefore the same thing as an ultrafilter on $\mathcal{B}_\varphi(M)$, and we regard $S_\varphi(M)$ as the Stone space of $\mathcal{B}_\varphi(M)$: a compact, Hausdorff, zero-dimensional space whose basic clopen sets consist of the types containing a given finite set of φ -formulas over M .

Definition 1.1.3. Let $p \in S_\varphi(M)$ be given.

- (a) The type p is *definable* if there is an $L(M)$ -formula $\psi(y)$, a φ -*definition* of p , such that for all $b \in M$, we have

$$\varphi(x, b) \in p \iff M \models \psi(b).$$

Equivalently, the set $D_p = \psi(M)$ is a definable subset of M .

- (b) A *positive Boolean combination of instances of φ over M* is a formula of the form

$$\bigvee_{i < n} \bigwedge_{a \in W_i} \varphi(a, y),$$

where $n < \omega$ and $W_0, \dots, W_{n-1}$ are finite subsets of M . By convention the empty disjunction is $\perp$ and the empty conjunction is $\top$.

In general the set D_p is just a subset of M , with no reason to be describable inside M : when M is infinite there are far more subsets of M than formulas over M , and a definable type is one whose set D_p happens to be captured by a single formula. Realized types are the model case, since for $c \in M$ the type $\text{tp}_\varphi(c/M)$ is defined by the formula $\varphi(c, y)$. A definable type thus behaves like the type of an element of M , even when the type is realized only in an elementary extension.

Definability also has a practical side. A formula $\psi(y)$ makes sense in every elementary extension $N \succeq M$, so a definition of p tells us how to extend p to parameters in N : put $\varphi(x, b)$ in the extension exactly when $N \models \psi(b)$. This is the source of the canonical extensions used to define independence in Chapter 6.

Definability typically fails because of an order. In $(\mathbb{Q}, <)$, let $\varphi(x, y)$ be $x < y$ and let p be the type of $\sqrt{2}$ over $\mathbb{Q}$. The set $D_p = \{b \in \mathbb{Q} : b > \sqrt{2}\}$ is a cut with an irrational edge, whereas the subsets of $\mathbb{Q}$ definable in $(\mathbb{Q}, <)$ with parameters are finite unions of intervals with rational or infinite endpoints; hence the type p is not definable. The type of an element just above a rational q , by contrast, is defined by $y > q$.

In $(\mathbb{Q}, <)$ the cut at $\sqrt{2}$ escapes every formula because the formula $x < y$ orders the rationals. The next definition isolates this behavior in finite pieces.

Definition 1.1.4. Let M be a model of T , and let $n \geq 1$ be given. An *n -ladder for φ in M* consists of tuples $a_1, \dots, a_n$ and $b_1, \dots, b_n$ of elements of M such that, for all $1 \leq i, j \leq n$, we have

$$M \models \varphi(a_i, b_j) \iff i \leq j.$$

We say that φ has the *order property* if for every n some model of T contains an n -ladder for φ , and that φ is *stable* otherwise.

Figure 1.1 explains the name. Put the a_i on one rail and the b_j on the other, both in increasing order, and join a_i to b_j whenever $\varphi(a_i, b_j)$ holds; the “horizontal” pairs (a_i, b_i) are

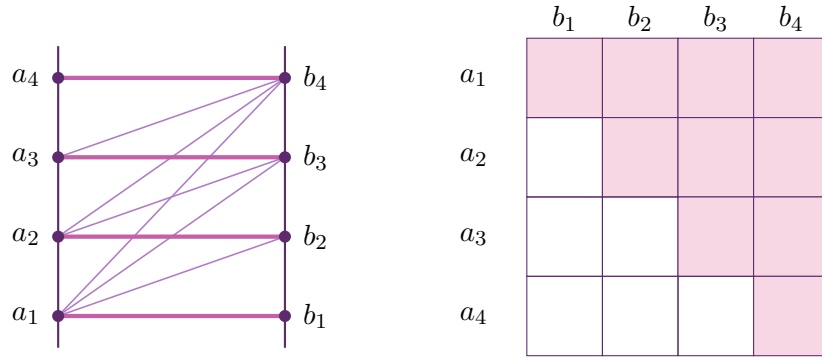


FIGURE 1.1. A 4-ladder. Left: the elements a_i and b_j are joined when $\varphi(a_i, b_j)$ holds, i.e. when $i \leq j$. The pairs (a_i, b_i) are the rungs (thick, pink), and each a_i also reaches every b_j with $j > i$ (purple). Right: the truth table of $\varphi(a_i, b_j)$, with pink cells where φ holds. The table is a staircase, which is why an n -ladder is also called a *half-graph*.

then the rungs of a ladder. What matters is the asymmetry between the two directions: each a_i is related to every b_j with $j \geq i$ and to none with $j < i$, so the formula φ linearly orders the rungs, in the sense that $i \leq j$ if and only if $\varphi(a_i, b_j)$.



Remark 1.1.5. An $(n+1)$ -ladder contains an n -ladder. Hence, by compactness, the formula φ has the order property if and only if some model of T contains an infinite sequence $(a_i, b_i)_{i < \omega}$ with $\varphi(a_i, b_j) \iff i \leq j$. The choice of “ $\leq$ ” is immaterial. Suppose that $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are tuples in a model of T such that $\varphi(a_i, b_j) \iff i > j$ for $1 \leq i, j \leq n$. Reversing the indices by putting $a'_i = a_{n+1-i}$ and $b'_j = b_{n+1-j}$ gives $\varphi(a'_i, b'_j) \iff i < j$. Shifting by putting $b''_j = b'_{j+1}$ then gives $\varphi(a'_i, b''_j) \iff i \leq j$ for $1 \leq i, j \leq n-1$. Thus a “ $>$ ”-pattern of length n yields an $(n-1)$ -ladder.

The main theorem of this chapter confirms that the order property is the only obstruction to definability, and that when the obstruction is absent the definitions can be chosen of the simplest possible kind: positive Boolean combinations of instances of φ (see Definition 1.1.3(b)). These are, up to equivalence, all the formulas built from instances $\varphi(a, y)$ with $a \in M$ using only $\wedge$ and $\vee$, by distributivity; if φ is quantifier-free, so is every such combination.

Theorem 1.1.6 (Definability of types). *The following are equivalent:*

- (1) *The formula φ does not have the order property.*
- (2) *For every model M of T , every $p \in S_\varphi(M)$ has a φ -definition that is a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$.*
- (3) *For every model M of T , every $p \in S_\varphi(M)$ is definable.*

In short: *all φ -types are definable if and only if φ does not have the order property.* The implication (2) $\Rightarrow$ (3) is trivial, and the implication (1) $\Rightarrow$ (2) is proved in Section 2.2. The remaining implication (3) $\Rightarrow$ (1) takes a detour through binary trees: in Sections 2.3–2.4 we show that if φ has the order property, then the formula φ encodes tall trees, and that tall trees produce a φ -type that is not definable.

Chapter Roadmap

The rest of this chapter, Section 1.2, looks at stability from the analytic side: there a formula becomes a bounded real-valued function, stability becomes the exchange of two iterated limits, and spaces of types become compact spaces of the kind studied in functional analysis.

Chapters 2 and 3 do not depend on Section 1.2. A reader mainly interested in the classical theory may go directly to Chapter 2, where Theorem 1.1.6 is proved, and return to Section 1.2 later; the analytic proof of $(1) \Rightarrow (3)$ given there (see Corollary 1.2.13) is perhaps best appreciated after the combinatorial one. The two sides meet again in Section 4.2, where the independence property is characterized in terms of convergent subsequences by means of Rosenthal's dichotomy, and in Chapter 6, where Harrington's lemma, the fact behind the symmetry of independence, turns out to be the exchange of iterated ultrafilter limits (see Subsection 6.1.1).

1.2. The real-valued case

Once a formula is identified with a $\{0,1\}$ -valued function, stability can be read as a property of bounded real-valued functions, namely the exchangeability of iterated limits. We follow Iovino [Iov99]. Seen this way, stability meets the reflexivity of Banach spaces, through a characterization of reflexivity due to James [Jam64] that looks very much like the order property.

1.2.1. Stability and iterated limit exchange

We first state the order property for arbitrary functions of two variables, and then show that its absence is exactly the exchangeability of two iterated limits.

Let M be an L -structure and $\varphi(x, y)$ an L -formula. We identify φ with the function $\varphi: M^{|x|} \times M^{|y|} \rightarrow \{0, 1\}$ that takes the value 1 at (a, b) if and only if $M \models \varphi(a, b)$. More generally, a function $\varphi: A \times B \rightarrow \{0, 1\}$ on arbitrary sets A, B has the *order property* if there are sequences $(a_m)_{m < \omega}$ in A and $(b_n)_{n < \omega}$ in B such that

$$\varphi(a_m, b_n) = 1 \iff m \leq n.$$

Definition 1.2.1. A function $\varphi: A \times B \rightarrow \{0, 1\}$ is *unstable* if either φ or $\neg\varphi = 1 - \varphi$ has the order property, and *stable* otherwise. A formula $\varphi(x, y)$ is *stable on M* if the corresponding function on $M^{|x|} \times M^{|y|}$ is stable.



Remark 1.2.2. A formula $\varphi(x, y)$ is stable in the sense of Definition 1.1.4 if and only if φ is stable on every model of T . Indeed, if φ has the order property, then by compactness (see Remark 1.1.5) some model contains an infinite ladder, so the formula φ is unstable on that model. Conversely, if φ or $\neg\varphi$ has the order property on some model M of T , then the model M contains n -ladders for φ for every n : directly in the first case, and by Remark 1.1.5 in the second, since $\neg\varphi(a_m, b_n) \iff m \leq n$ means $\varphi(a_m, b_n) \iff m > n$.

To state the analytic form of stability, we use limits along ultrafilters. For a family $(a_i)_{i \in I}$ in a topological space X , a point $a \in X$, and an ultrafilter $\mathcal{U}$ on I , we write

$$(1.1) \quad \lim_{i, \mathcal{U}} a_i = a$$

if a is a limit of (a_i) along $\mathcal{U}$. We use freely the standard facts about such limits: limits along ultrafilters are unique in Hausdorff spaces, always exist in compact spaces, describe closures, and are preserved by continuous maps.

Along an infinite ladder, the two iterated limits of φ along a nonprincipal ultrafilter on ω already differ: for fixed m the values $\varphi(a_m, b_n)$ are eventually 1, and for fixed n the values $\varphi(a_m, b_n)$ are eventually 0, so one iterated limit is 1 and the other is 0. This is the phenomenon that the analytic form of stability forbids. Ultrafilters on arbitrary index sets are convenient, but ladders are indexed by ω . The next proposition, which we use several times, bridges the gap: from iterated limits taken along arbitrary families it extracts sequences on which the two iterated limits are already visible, one above the diagonal and the other below.

Proposition 1.2.3. *Let $\varphi: A \times B \rightarrow [0, 1]$ be a function, let $(a_i)_{i \in I}$ be a family in A and $(b_j)_{j \in J}$ a family in B , and let $\mathcal{U}, \mathcal{V}$ be ultrafilters on I, J with*

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \alpha, \quad \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j) = \beta.$$

Then there are $i_0, i_1, \dots \in I$ and $j_0, j_1, \dots \in J$ such that

$$|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{1-m} \text{ if } m < n, \quad |\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{1-n} \text{ if } n < m.$$

Proof. By definition of the limits, for every $\varepsilon > 0$ there are $U_\varepsilon \in \mathcal{U}$ and $V_\varepsilon \in \mathcal{V}$ such that

$$i \in U_\varepsilon \implies \left| \lim_{j, \mathcal{V}} \varphi(a_i, b_j) - \alpha \right| \leq \varepsilon,$$

$$j \in V_\varepsilon \implies \left| \lim_{i, \mathcal{U}} \varphi(a_i, b_j) - \beta \right| \leq \varepsilon.$$

Moreover, for all $\hat{i} \in I, \hat{j} \in J$ and $\varepsilon > 0$ there are $V_\varepsilon^{\hat{i}} \in \mathcal{V}$ and $U_\varepsilon^{\hat{j}} \in \mathcal{U}$ such that

$$j \in V_\varepsilon^{\hat{i}} \implies \left| \varphi(a_{\hat{i}}, b_j) - \lim_{j, \mathcal{V}} \varphi(a_{\hat{i}}, b_j) \right| \leq \varepsilon,$$

$$i \in U_\varepsilon^{\hat{j}} \implies \left| \varphi(a_i, b_{\hat{j}}) - \lim_{i, \mathcal{U}} \varphi(a_i, b_{\hat{j}}) \right| \leq \varepsilon.$$

Choose recursively

$$i_N \in U_{2^{-N}} \cap \bigcap_{k < N} U_{2^{-k}}^{j_k}, \quad j_N \in V_{2^{-N}} \cap \bigcap_{k \leq N} V_{2^{-k}}^{i_k}.$$

These sets are finite intersections of members of an ultrafilter, hence nonempty. If $m < n$, then we have $j_n \in V_{2^{-n}}^{i_m}$ and $i_m \in U_{2^{-m}}$, so we have $|\varphi(a_{i_m}, b_{j_n}) - \alpha| \leq 2^{-n} + 2^{-m} \leq 2^{1-m}$. If $n < m$, then we have $i_m \in U_{2^{-m}}^{j_n}$ and $j_n \in V_{2^{-n}}$, so we have $|\varphi(a_{i_m}, b_{j_n}) - \beta| \leq 2^{-m} + 2^{-n} \leq 2^{1-n}$. ☺

Figure 1.2 shows the array produced by Proposition 1.2.3.

With Proposition 1.2.3 in hand, we can show that for $\{0, 1\}$ -valued functions the order property is exactly the failure of limit exchange.

Proposition 1.2.4. *Let $\varphi: A \times B \rightarrow \{0, 1\}$ be a function. The following are equivalent:*

(1) *The function φ is stable.*

(2) *For all sequences (a_m) in A and (b_n) in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have*

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$

Proof. (2) $\Rightarrow$ (1). Suppose that φ is unstable. If φ has the order property, there are sequences (a_m) in A and (b_n) in B with $\varphi(a_m, b_n) = 1 \iff m \leq n$. Let $\mathcal{U} = \mathcal{V}$ be a nonprincipal ultrafilter on $\mathbb{N}$. For fixed m we have $\varphi(a_m, b_n) = 1$ for all $n \geq m$, so the left-hand side of (2) is 1. For fixed n we have $\varphi(a_m, b_n) = 0$ for all $m > n$, so the right-hand side is 0. If instead $\neg\varphi$ has the order property, the two sides are 0 and 1.

(1) $\Rightarrow$ (2). Suppose that (2) fails: for some sequences $(a_m), (b_n)$ and ultrafilters $\mathcal{U}, \mathcal{V}$, the two iterated limits are $\alpha \neq \beta$. Since φ is $\{0, 1\}$ -valued, both limits lie in $\{0, 1\}$. By

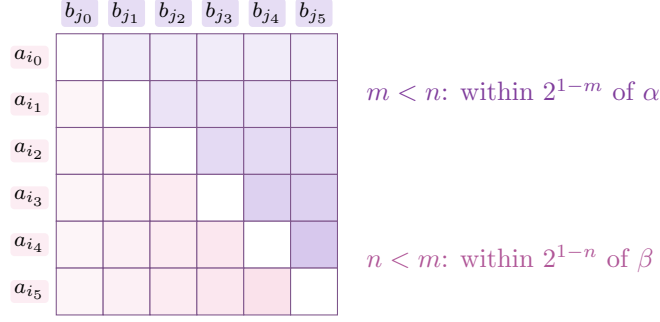


FIGURE 1.2. The array $\varphi(a_{i_m}, b_{j_n})$ of Proposition 1.2.3. Above the diagonal the values approach α , below it they approach β , and the deepening shades show the tolerances 2^{1-m} and 2^{1-n} shrinking. Nothing is claimed on the diagonal. When $\alpha \neq \beta$ the array is a ladder in the limit: compare the staircase of Figure 1.1.

Proposition 1.2.3 there are sequences (a'_m) in A and (b'_n) in B with $|\varphi(a'_m, b'_n) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|\varphi(a'_m, b'_n) - \beta| \leq 2^{1-n}$ for $n < m$. Since φ is $\{0, 1\}$ -valued, we have $\varphi(a'_m, b'_n) = \alpha$ for $2 \leq m < n$ and $\varphi(a'_m, b'_n) = \beta$ for $2 \leq n < m$. Put $c_m = a'_{2m+2}$ and $d_n = b'_{2n+3}$. Since $2m+2 < 2n+3 \iff m \leq n$, we get $\varphi(c_m, d_n) = \alpha$ if $m \leq n$ and $\varphi(c_m, d_n) = \beta$ if $m > n$. Thus the function φ has the order property if $\alpha = 1$, and the function $\neg\varphi$ does if $\alpha = 0$. ☺

Condition (2) of Proposition 1.2.4 makes sense for every bounded real-valued function, and we take this condition as the definition of stability in the real-valued setting.

Definition 1.2.5. A bounded function $\varphi: A \times B \rightarrow \mathbb{R}$ is *stable* if, for all sequences (a_m) in A and (b_n) in B and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n).$$

By Proposition 1.2.4, the new definition agrees with Definition 1.2.1 for $\{0, 1\}$ -valued functions, and by Proposition 1.2.3, the sequences in Definition 1.2.5 may be replaced by arbitrary families, as the next corollary records. For a bounded set of continuous functions on a compact space, the same condition is the hypothesis of Grothendieck's double limit criterion (see Appendix A).

Corollary 1.2.6. If $\varphi: A \times B \rightarrow [0, 1]$ is stable, then we have $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$ for all families $(a_i)_{i \in I}$, $(b_j)_{j \in J}$ and all ultrafilters $\mathcal{U}$ on I , $\mathcal{V}$ on J .

Proof. If the two limits are $\alpha \neq \beta$, take (i_m) , (j_n) as in Proposition 1.2.3 and a nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$. For fixed m , the limit $\lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n})$ is within 2^{1-m} of α , so we have $\lim_{m, \mathcal{W}} \lim_{n, \mathcal{W}} \varphi(a_{i_m}, b_{j_n}) = \alpha$. Symmetrically, the other iterated limit is β , which contradicts the stability of φ . ☺

The proof of Corollary 1.2.6 used a single nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$, in both coordinates, and this is no accident. For ultrafilters $\mathcal{U}$ on I and $\mathcal{V}$ on J , let $\mathcal{U} \otimes \mathcal{V}$ be the ultrafilter on $I \times J$ consisting of the sets X with $\{i : \{j : (i, j) \in X\} \in \mathcal{V}\} \in \mathcal{U}$. For every bounded $f: I \times J \rightarrow \mathbb{R}$ we have $\lim_{(i, j), \mathcal{U} \otimes \mathcal{V}} f(i, j) = \lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} f(i, j)$, so the two iterated limits of Definition 1.2.5 are the limits of $\varphi(a_m, b_n)$ along $\mathcal{U} \otimes \mathcal{V}$ and along the transpose of $\mathcal{V} \otimes \mathcal{U}$. These two ultrafilters can differ, because the tensor product of ultrafilters is not commutative, and the simplest witness already lives on $\mathbb{N} \times \mathbb{N}$: if $\mathcal{W}$ is a nonprincipal ultrafilter on $\mathbb{N}$, then the upper triangle $\{(m, n) : m < n\}$ belongs to $\mathcal{W} \otimes \mathcal{W}$, because each of its rows is

cofinite, while the lower triangle $\{(m, n) : m > n\}$ does not, because each of its rows is finite. A ladder is exactly a pair of sequences along which φ is the indicator function of the closed upper triangle $\{(m, n) : m \leq n\}$. Thus the order property says that φ can tell $\mathcal{W} \otimes \mathcal{W}$ from its own transpose, and the proofs of Proposition 1.2.4 and Corollary 1.2.6 show the following: a bounded function φ is stable if and only if, for one (equivalently, for every) nonprincipal ultrafilter $\mathcal{W}$ on $\mathbb{N}$ and all sequences (a_m) and (b_n) , we have

$$\lim_{m, \mathcal{W}} \lim_{n, \mathcal{W}} \varphi(a_m, b_n) = \lim_{n, \mathcal{W}} \lim_{m, \mathcal{W}} \varphi(a_m, b_n).$$

Stable functions are blind to the noncommutativity of the tensor square of a single ultrafilter.

1.2.2. Stability and continuous extendability

Exchanging iterated limits is a statement about continuity in disguise. In Proposition 1.2.8 below we show that a function $\varphi: A \times B \rightarrow [0, 1]$ is stable exactly when the function extends to a separately continuous function on pairs of types. The types in question live in compact spaces built from A and B .

Definition 1.2.7. Let $\varphi: A \times B \rightarrow [0, 1]$ be a function. For $a \in A$, the *left φ -type* of a is the function $\text{ltp}_\varphi(a): y \mapsto \varphi(a, y)$ on B . For $b \in B$, the *right φ -type* of b is the function $\text{rtp}_\varphi(b): x \mapsto \varphi(x, b)$ on A . The *space of left φ -types* $\text{LS}(\varphi)$ is the closure of $\{\text{ltp}_\varphi(a) : a \in A\}$ in $[0, 1]^B$ with the product topology, and the *space of right φ -types* $\text{RS}(\varphi)$ is the closure of $\{\text{rtp}_\varphi(b) : b \in B\}$ in $[0, 1]^A$.

By Tychonoff's theorem, the spaces $[0, 1]^A$ and $[0, 1]^B$ are compact, and so are their closed subspaces $\text{RS}(\varphi)$ and $\text{LS}(\varphi)$. Both spaces are quotients of Stone–Čech compactifications. The map $a \mapsto \text{ltp}_\varphi(a)$ from the discrete space A into the compact space $[0, 1]^B$ extends to a continuous map $\beta A \rightarrow [0, 1]^B$, namely $\mathcal{U} \mapsto \lim_{a, \mathcal{U}} \text{ltp}_\varphi(a)$, whose image is compact and contains the realized types, and is therefore $\text{LS}(\varphi)$. Thus every left φ -type is the limit of the realized types along an ultrafilter on A , the principal ultrafilters give the realized types, and the space $\text{LS}(\varphi)$ is a continuous image of βA ; likewise the space $\text{RS}(\varphi)$ is a continuous image of βB .

Proposition 1.2.8. *For $\varphi: A \times B \rightarrow [0, 1]$, the following are equivalent:*

- (1) *The function φ is stable.*
- (2) *There is a separately continuous function $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ such that*

$$\hat{\varphi}(\text{ltp}_\varphi(a), \text{rtp}_\varphi(b)) = \varphi(a, b) \quad \text{for all } (a, b) \in A \times B.$$

Proof. (1) $\Rightarrow$ (2). Let $p \in \text{LS}(\varphi)$ and $q \in \text{RS}(\varphi)$ be given. There are families $(a_i)_{i \in I}$ in A , $(b_j)_{j \in J}$ in B and ultrafilters $\mathcal{U}$ on I , $\mathcal{V}$ on J with

$$p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i), \quad q = \lim_{j, \mathcal{V}} \text{rtp}_\varphi(b_j).$$

By Corollary 1.2.6, we have $\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{j, \mathcal{V}} \lim_{i, \mathcal{U}} \varphi(a_i, b_j)$. Define $\hat{\varphi}(p, q)$ to be this common value. Since limits in the product topology are computed coordinatewise, this common value satisfies

$$(1.2) \quad \hat{\varphi}(p, q) = \lim_{i, \mathcal{U}} q(a_i) = \lim_{j, \mathcal{V}} p(b_j).$$

The map $\hat{\varphi}$ is well defined. Suppose also that $p = \lim_{i, \mathcal{U}'} \text{ltp}_\varphi(a'_i)$ and $q = \lim_{j, \mathcal{V}'} \text{rtp}_\varphi(b'_j)$. Then we have $\lim_{j, \mathcal{V}} \varphi(x, b_j) = q(x) = \lim_{j, \mathcal{V}'} \varphi(x, b'_j)$ for every $x \in A$, and $\lim_{i, \mathcal{U}} \varphi(a_i, y) =$

$p(y) = \lim_{i, \mathcal{U}} \varphi(a'_i, y)$ for every $y \in B$. Hence, using Corollary 1.2.6 for the middle step, we have

$$\lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}} \varphi(a_i, b_j) = \lim_{i, \mathcal{U}} \lim_{j, \mathcal{V}'} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}} \varphi(a_i, b'_j) = \lim_{j, \mathcal{V}'} \lim_{i, \mathcal{U}'} \varphi(a'_i, b'_j).$$

The map $\hat{\varphi}$ extends φ . Take constant families.

The map $\hat{\varphi}$ is separately continuous. Fix p , and let $q = \lim_{k, \mathcal{W}} q_k$ in $\text{RS}(\varphi)$, where $\mathcal{W}$ is an ultrafilter on a set K . Write $q_k = \lim_{j, \mathcal{V}_k} \text{rtp}_\varphi(b_{k,j})$ with j ranging over a set J_k . On $J' = \{(k, j) : k \in K, j \in J_k\}$ let $\mathcal{V}'$ be the ultrafilter of all $X \subseteq J'$ with $\{k : \{j : (k, j) \in X\} \in \mathcal{V}_k\} \in \mathcal{W}$. For every family $(z_{k,j})$ in a compact Hausdorff space we have $\lim_{(k,j), \mathcal{V}'} z_{k,j} = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} z_{k,j}$. It follows that $q = \lim_{(k,j), \mathcal{V}'} \text{rtp}_\varphi(b_{k,j})$, and by (1.2), we have

$$\hat{\varphi}(p, q) = \lim_{(k,j), \mathcal{V}'} p(b_{k,j}) = \lim_{k, \mathcal{W}} \lim_{j, \mathcal{V}_k} p(b_{k,j}) = \lim_{k, \mathcal{W}} \hat{\varphi}(p, q_k).$$

Thus the function $\hat{\varphi}(p, \cdot)$ is continuous. Continuity of $\hat{\varphi}(\cdot, q)$ is proved symmetrically.

(2) $\Rightarrow$ (1). Let (a_m) , (b_n) be sequences in A , B , let $\mathcal{U}, \mathcal{V}$ be ultrafilters on $\mathbb{N}$, and put $p = \lim_{m, \mathcal{U}} \text{ltp}_\varphi(a_m)$ and $q = \lim_{n, \mathcal{V}} \text{rtp}_\varphi(b_n)$. Then the pair (p, q) lies in $\text{LS}(\varphi) \times \text{RS}(\varphi)$, and by separate continuity

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} \varphi(a_m, b_n) = \lim_{m, \mathcal{U}} \hat{\varphi}(\text{ltp}_\varphi(a_m), q) = \hat{\varphi}(p, q) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} \varphi(a_m, b_n). \quad \text{☺}$$

1.2.3. Stability and reflexivity

The pairing between a reflexive Banach space and its dual, restricted to the product of the two unit balls with their weak topologies, is a separately continuous function on a product of compact spaces. In Theorem 1.2.9 below we state Iovino's theorem that this example is universal: the stable functions are exactly the restrictions of the pairing between the unit balls of a reflexive Banach space E and of E^* .

For a Banach space E with dual E^* , we write B_E for the closed unit ball of E and $\langle x, y \rangle$ for $y(x)$, where $x \in E$ and $y \in E^*$. Recall that E is *reflexive* if the canonical embedding of E into E^{**} is surjective.

Theorem 1.2.9 (Iovino). *For $\varphi : A \times B \rightarrow [0, 1]$, the following are equivalent:*

- (1) *The function φ is stable.*
- (2) *There are a reflexive Banach space E and maps $u : A \rightarrow B_E$ and $v : B \rightarrow B_{E^*}$ such that*

$$\varphi(x, y) = \langle u(x), v(y) \rangle \quad \text{for all } (x, y) \in A \times B.$$

Proof. See [Iov99, Theorem 3.3]. For (1) $\Rightarrow$ (2), stability is used to show that the operator $T : \ell^\infty(A)^* \rightarrow \ell^\infty(B)$ given by $T\mu(y) = \int \varphi(x, y) d\mu(x)$ is weakly compact; this step goes back to Krivine and Maurey [KM81] and Raynaud [Ray83]. The theorem of Davis, Figiel, Johnson and Pełczyński [DFJ+74] then factors T through a reflexive space. The implication (2) $\Rightarrow$ (1) follows from the weak compactness of B_E and the weak* compactness of B_{E^*} . ☺

Read back into model theory, the theorem says the following: by Remark 1.2.2 and Proposition 1.2.4, a formula $\varphi(x, y)$ is stable in the sense of Definition 1.1.4 if and only if, in every model M of T , the function φ on M is the restriction of the pairing map $B_E \times B_{E^*} \rightarrow \mathbb{R}$ for some reflexive Banach space E (depending on M), composed with maps $M^{|x|} \rightarrow B_E$ and $M^{|y|} \rightarrow B_{E^*}$.

1.2.4. Stability and polynomial approximation

By Theorem 1.1.6, if φ does not have the order property, every φ -type over a model of T is defined by a Boolean combination of instances $\varphi(a, y)$. On $\{0, 1\}$ the Boolean connectives are polynomials, namely $\neg t = 1 - t$ and $t \wedge s = ts$. The real-valued analogue of a definition is therefore a *uniform limit of polynomials in instances of φ* , and in Theorem 1.2.11 below we use the Stone–Weierstrass theorem to show that stability is exactly what is needed for such approximations to exist. Continuous logic treats definability in the same way, and Ben Yaacov used polynomial approximation to derive definability of types from Grothendieck’s double limit criterion (see the historical remarks at the end of this chapter, and Appendix A). We use the Stone–Weierstrass theorem in the following form.

Fact 1.2.10 (Stone–Weierstrass). *Let K be a compact Hausdorff space and let $\mathcal{F} \subseteq C(K)$ be a family of continuous real-valued functions that separates the points of K . Then every $f \in C(K)$ is a uniform limit on K of functions of the form $P(g_1, \dots, g_k)$, where $g_1, \dots, g_k \in \mathcal{F}$ and P is a real polynomial in k variables.*

To apply the Stone–Weierstrass theorem we need continuous functions on the type spaces that separate points, and the instances of φ supply them. Fix $\varphi: A \times B \rightarrow [0, 1]$. For $a \in A$, let $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$ be the evaluation map $e_a(q) = q(a)$, and for $b \in B$, let $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$ be $e_b(p) = p(b)$. These maps are continuous, because $\text{RS}(\varphi)$ and $\text{LS}(\varphi)$ carry the product topology, and on realized types they are instances of φ , since $e_a(\text{rtp}_\varphi(b)) = \varphi(a, b) = e_b(\text{ltp}_\varphi(a))$. The family $\{e_a : a \in A\}$ separates the points of $\text{RS}(\varphi)$, since two distinct functions $q \neq q'$ in $[0, 1]^A$ differ at some $a \in A$, and likewise the family $\{e_b : b \in B\}$ separates the points of $\text{LS}(\varphi)$.

Theorem 1.2.11. *For $\varphi: A \times B \rightarrow [0, 1]$, the following are equivalent:*

- (1) *The function φ is stable.*
- (2) *There is a function $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ with $\hat{\varphi}(\text{ltp}_\varphi(a), \text{rtp}_\varphi(b)) = \varphi(a, b)$ for all $(a, b) \in A \times B$, such that for every $\varepsilon > 0$:*
 - (a) *for every $p \in \text{LS}(\varphi)$ there are $a_1, \dots, a_k \in A$ and a real polynomial P with*

$$|\hat{\varphi}(p, q) - P(q(a_1), \dots, q(a_k))| \leq \varepsilon \quad \text{for all } q \in \text{RS}(\varphi);$$

- (b) *for every $q \in \text{RS}(\varphi)$ there are $b_1, \dots, b_l \in B$ and a real polynomial Q with*

$$|\hat{\varphi}(p, q) - Q(p(b_1), \dots, p(b_l))| \leq \varepsilon \quad \text{for all } p \in \text{LS}(\varphi).$$

Proof. (1) $\Rightarrow$ (2). Let $\hat{\varphi}$ be the separately continuous function given by Proposition 1.2.8. For fixed p , the function $\hat{\varphi}(p, \cdot)$ is continuous on the compact Hausdorff space $\text{RS}(\varphi)$. Applying Fact 1.2.10 with $\mathcal{F} = \{e_a : a \in A\}$ gives (a). Clause (b) follows in the same way, using the continuity of $\hat{\varphi}(\cdot, q)$ on $\text{LS}(\varphi)$ and $\mathcal{F} = \{e_b : b \in B\}$.

(2) $\Rightarrow$ (1). Each function $q \mapsto P(q(a_1), \dots, q(a_k))$ is continuous on $\text{RS}(\varphi)$. By (a), the function $\hat{\varphi}(p, \cdot)$ is a uniform limit of such functions, hence continuous. By (b), the function $\hat{\varphi}(\cdot, q)$ is continuous for the same reason. Hence the function $\hat{\varphi}$ is separately continuous, and the function φ is stable by Proposition 1.2.8. 

Evaluated at realized right types, clause (a) says that left types are approximated by polynomials in instances. Suppose that φ is stable, and let $\hat{\varphi}$ be the separately continuous extension given by Proposition 1.2.8, as in the proof of (1) $\Rightarrow$ (2). Then we have $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = p(b)$ for $p \in \text{LS}(\varphi)$ and $b \in B$: if $p = \lim_{i \in \mathbb{N}} \text{ltp}_\varphi(a_i)$, continuity in the first

variable gives $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = \lim_{i, \mathcal{U}} \varphi(a_i, b) = p(b)$. Taking $q = \text{rtp}_\varphi(b)$ in (a) therefore gives

$$|p(b) - P(\varphi(a_1, b), \dots, \varphi(a_k, b))| \leq \varepsilon \quad \text{for all } b \in B.$$

Thus every left φ -type is a uniform limit on B of polynomials in instances $\varphi(a, y)$ with $a \in A$, and symmetrically for right types.

The approximation in Theorem 1.2.11 is required on all of $\text{RS}(\varphi)$ (respectively $\text{LS}(\varphi)$), not only at realized types, and this requirement cannot be weakened: the next example shows that approximation on B alone does not imply stability, since there every type is trivially approximable on B and yet the function has the order property.

Example 1.2.12. Let $A = B = \mathbb{N}$, and define $\varphi(m, n) = 1$ if $m \leq n$ and $\varphi(m, n) = 0$ otherwise. Then the function φ has the order property and is therefore unstable. Here the left type $\text{ltp}_\varphi(m)$ is the indicator function of $[m, \infty)$, and the space $\text{RS}(\varphi)$ consists of the indicator functions of the intervals $[0, n]$ together with the constant function $\mathbf{1}$. Similarly, the space $\text{LS}(\varphi)$ consists of the functions $\text{ltp}_\varphi(m)$ together with the constant function $\mathbf{0}$. Thus every $p \in \text{LS}(\varphi)$, viewed as a function on B , is either an instance $\varphi(m, \cdot)$ or the constant 0, which is trivially a polynomial in instances. Symmetrically, every $q \in \text{RS}(\varphi)$ is an instance or the constant 1. The failure appears at the unrealized pair $(\mathbf{0}, \mathbf{1})$. If $\hat{\varphi}$ were separately continuous and extended φ as in Theorem 1.2.11, then we would have $\hat{\varphi}(\mathbf{0}, \text{rtp}_\varphi(n)) = \mathbf{0}(n) = 0$ for all n , and since $\text{rtp}_\varphi(n) \rightarrow \mathbf{1}$, we would get $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 0$. On the other hand, we would also have $\hat{\varphi}(\text{ltp}_\varphi(m), \mathbf{1}) = \mathbf{1}(m) = 1$ for all m , and since $\text{ltp}_\varphi(m) \rightarrow \mathbf{0}$, we would get $\hat{\varphi}(\mathbf{0}, \mathbf{1}) = 1$.

We now return to formulas, whose left types are their φ -types in another guise. Let M be a model of T , and identify each $p \in S_\varphi(M)$ with the characteristic function of D_p on $M^{|y|}$. With this identification we have $\text{LS}(\varphi) = S_\varphi(M)$. Indeed, for $a \in M$ the left type $\text{ltp}_\varphi(a)$ is $\text{tp}_\varphi(a/M)$, and by Lemma 1.1.2, every $p \in S_\varphi(M)$ agrees on each finite set of parameters with some realized type, so the type p is a pointwise limit of realized types. Conversely, a pointwise limit of realized types determines a set of φ -formulas over M that is finitely satisfiable in M , hence consistent with the elementary diagram of M , and so a complete φ -type.

In this two-valued setting, the quotient map $\beta M^{|x|} \rightarrow S_\varphi(M)$ is transparent. Recall from Section 1.1 that $S_\varphi(M)$ is the Stone space of the Boolean algebra $\mathcal{B}_\varphi(M)$ generated by the sets $\varphi(M, b)$ with $b \in M^{|y|}$. An ultrafilter $\mathcal{U}$ on $M^{|x|}$ goes to its trace on $\mathcal{B}_\varphi(M)$, that is, to the type containing $\varphi(x, b)$ exactly when $\varphi(M, b) \in \mathcal{U}$. The realized types are the traces of principal ultrafilters, and Lemma 1.1.2 says that every type is the trace of some ultrafilter.

For $\{0, 1\}$ -valued types, an approximation within $\frac{1}{3}$ can be rounded, and polynomials in instances become Boolean combinations of instances.

Corollary 1.2.13. *Let M be an L -structure and let $\varphi(x, y)$ be a formula that is stable on M . Then every $p \in \text{LS}(\varphi)$ is defined by a Boolean combination of instances: there are $a_1, \dots, a_k \in M^{|x|}$ and a Boolean combination $\psi(y)$ of $\varphi(a_1, y), \dots, \varphi(a_k, y)$ such that, for all $b \in M^{|y|}$, we have*

$$p(b) = 1 \iff M \models \psi(b).$$

Proof. Apply Theorem 1.2.11 with $\varepsilon = \frac{1}{3}$ to get $a_1, \dots, a_k$ and P . Both $p(b)$ and $\varphi(a_i, b)$ lie in $\{0, 1\}$, so we have $p(b) = 1$ if and only if the tuple $t = (\varphi(a_1, b), \dots, \varphi(a_k, b))$ belongs to $S = \{t \in \{0, 1\}^k : P(t) > \frac{1}{2}\}$. Take $\psi(y) = \bigvee_{t \in S} \bigwedge_{i \leq k} \varphi(a_i, y)^{t_i}$ (recall that $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$). ☺

Since a formula without the order property is stable on every model of T (see Remark 1.2.2), Corollary 1.2.13 already gives (1) $\Rightarrow$ (3) of Theorem 1.1.6, by analytic means. The price is positivity: the definitions obtained this way are Boolean combinations of instances rather than positive ones, and their size depends on p .



Remark 1.2.14(1) *Lattice version.* The lattice form of the Stone–Weierstrass theorem replaces polynomials by the operations max, min and affine maps. Continuous logic uses this form to show that a small set of connectives generates all continuous connectives up to uniform approximation [BU10, Proposition 1.4].

- (2) *Convex combinations.* Stability gives more than a polynomial approximation. In Proposition A.1.7 below we show, using Grothendieck’s criterion, that stability of φ implies that $\{e_a : a \in A\}$ is relatively weakly compact in $C(\text{RS}(\varphi))$. In that case the equality $p = \lim_{i, \mathcal{U}} \text{ltp}_\varphi(a_i)$ implies that $\hat{\varphi}(p, \cdot)$ is the weak limit of the functions e_{a_i} along $\mathcal{U}$. Mazur’s lemma then shows that $\hat{\varphi}(p, \cdot)$ is a uniform limit of convex combinations $\sum_j \lambda_j e_{a_{i_j}}$. Thus the left type p is a uniform limit on B of averages of instances $\varphi(a, y)$. Ben Yaacov [Ben14] uses this convex approximation to recover the familiar forms of the defining formulas.

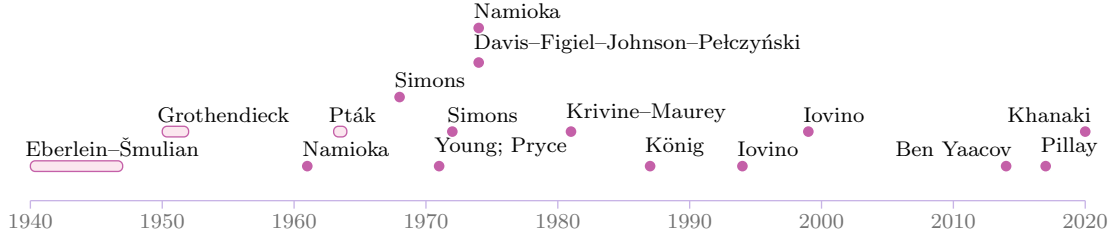
1.3. Historical remarks



Section 1.1. Counting types begins with Morley’s proof of Łoś’s conjecture [Mor65]. There he introduced totally transcendental theories, defined by a restriction on the topology of the type spaces of those theories, and showed that for countable theories total transcendence is equivalent to a restriction on the number of types: only countably many types over countable sets, the condition now called ω -stability. The general notion arose soon after. Setting out to extend Morley’s categoricity theorem to uncountable theories, Rowbottom generalized ω -stability to λ -stability: a theory is λ -stable if there are at most λ types over sets of size λ [Row64]. Shelah then called a theory *stable* when λ -stable for some λ , and began the study of stable theories in general in [She69]. In [She71] he moved from theories to single formulas, defining when a formula $\varphi(x; y)$ is *stable in λ* : the formula φ has at most λ φ -types over sets of size λ [She71, Definition 2.1]. He also defined the *order property* [She71, Definition 2.2] and proved that stability is equivalent to the absence of the order property. The ladder formulation of Definition 1.1.4 is a finitary restatement of Shelah’s condition. Definable types were introduced by Gaifman in his study of end extensions of models of Peano arithmetic [Gai76], and Shelah arrived at them independently. The theorem that φ -types are definable when φ is stable is [She71, Theorem 3.1]; see also [She90, Chapter II]. Today this theorem is often called the fundamental theorem of stability theory. Ben Yaacov [Ben14] observed that the same result follows from Grothendieck’s 1952 work on weak compactness in function spaces [Gro52] (see Appendix A below). This is the viewpoint of Section 1.2.



Section 1.2.1. Interchanging iterated limits is an old theme. For double sequences of reals, the classical sufficient condition is uniform convergence in one variable (the Moore–Osgood theorem). The condition of Definition 1.2.5 requires the iterated limits to agree along *all* pairs of sequences. This condition comes from functional analysis, and reached model theory through Banach space geometry.

**1940–1947**

Eberlein–Šmulian. Šmulian (1940) and Eberlein [Ebe47] (1947) prove that in a Banach space weak compactness and weak sequential compactness coincide, so that weak compactness can be tested along sequences.

1950–1952

Grothendieck. Grothendieck proves the *double limit criterion*, announced in [Gro50] and proved in [Gro52]. For compact K , a bounded set $H \subseteq C(K)$ is relatively weakly compact if and only if $\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$ for all sequences (f_n) in H and (x_m) in K for which both iterated limits exist. For an arbitrary topological space, the same condition decides when pointwise limits of functions in H are continuous. See Appendix A.

1961

Namioka. Namioka proves a combinatorial “substitute for Lebesgue’s bounded convergence theorem” [Nam61]. From this result he obtains, without measure theory, Krein’s theorem: in a Banach space, the closed convex hull of a weakly compact set is weakly compact.

1963–1964

Pták. Pták [Ptá63] finds a purely combinatorial lemma about averages, from which both the double limit criterion and Krein’s theorem follow. In [Ptá64] he uses an interchange of iterated limits to decide which bounded functions on a product $X \times Y$, continuous in each variable separately, extend to bilinear forms with the same kind of continuity.

1968

Simons. Simons [Sim68] introduces the *Fubini characterization* of the interchange of double limits. A bounded function on a product satisfies the iterated limit condition if and only if the two iterated integrals of the function agree, for every pair of measures on the factors. This characterization plays an important role in the proof of the factorization theorem of Davis, Figiel, Johnson and Pelczyński [DFJ+74].

1971

Young; Pryce. N. J. Young [You71] gives topological proofs of Pták’s double-limit theorems, based on Grothendieck’s work rather than combinatorics, and generalizes Pták’s bilinear extension theorem. Pryce [Pry71] uses a trick of Whitley’s to simplify proofs about pointwise compact sets of continuous functions.

1972

Simons. In three consecutive papers, Simons gives an elementary proof of Pták’s lemma [Sim72c]. He also proves an inequality about suprema and upper limits, now known as *Simons’ inequality* [Sim72a], and relates minimax theorems, which say when $\max \min = \min \max$, to James’s characterization of reflexivity [Sim72b].

- 1974** • **Davis–Figiel–Johnson–Pełczyński.** Davis, Figiel, Johnson and Pełczyński [DFJ+74] prove that every weakly compact operator factors through a reflexive Banach space. This is the key step in the proof of Theorem 1.2.9.
- 1974** • **Namioka.** Namioka [Nam74] studies when separately continuous functions on products are jointly continuous at many points. This is the question behind Proposition 1.2.8.
- 1981** • **Krivine–Maurey.** Krivine and Maurey [KM81] apply the interchange of limits to the norm: a Banach space is *stable* if $\lim_m \lim_n \|x_m + y_n\| = \lim_n \lim_m \|x_m + y_n\|$ for all bounded sequences for which both limits exist. They prove that every stable Banach space contains, for some p , copies of ℓ_p that are as close to isometric as desired, which generalizes a theorem of Aldous about subspaces of L_1 .
- 1987** • **König.** König [Kön87] studies the double limit relation in angelic spaces, where compactness can be tested by sequences.
- 1994** • **Iovino.** Iovino’s Ph.D. thesis [Iov94] introduces stability for real-valued logic.
- 1999** • **Iovino.** Iovino [Iov99] observes that Shelah’s order property is the 2-valued case of the same condition, and proves Theorem 1.2.9. Propositions 1.2.4 and 1.2.3 are taken from his paper.
- 2014** • **Ben Yaacov.** Ben Yaacov [Ben14] derives the definability of types in stable theories directly from Grothendieck’s criterion.
- 2017** • **Pillay.** Pillay [Pil17] comments on the model-theoretic reading of Grothendieck’s criterion, and derives from the criterion the positive form of the definitions.
- 2020** • **Khanaki.** Khanaki [Kha20] studies stability, NIP (see Chapter 4), and related properties systematically, through topological properties of function spaces.

⌘ **Section 1.2.2.** Spaces of left and right φ -types (see Definition 1.2.7) are the analytic counterpart of Shelah’s spaces of φ -types. The use of *types* in Banach space theory, as limits of functions $y \mapsto \|x_n + y\|$, is due to Krivine and Maurey [KM81]. Proposition 1.2.8, which characterizes stability by the existence of a separately continuous extension to pairs of types, is Proposition 2.6 of Iovino [Iov99]. Separately continuous functions on products of compact spaces, and their relation to iterated limits, are the subject of Grothendieck’s paper [Gro52].

⌘ **Section 1.2.3.** James [Jam64] characterized reflexivity by the absence of sequences (a_m) in B_E and (f_n) in B_{E^*} whose pairing takes one value for $m < n$ and another for $m \geq n$. This is a Banach-space form of the order property. In the early 1980s, Krivine and Maurey [KM81] and Raynaud [Ray83] showed that stable functions give rise to weakly compact operators. The factorization theorem of Davis, Figiel, Johnson and Pełczyński [DFJ+74] (1974) says that every weakly compact operator factors through a reflexive space. Iovino [Iov99] combined these results to prove Theorem 1.2.9: stable functions are exactly the pairings between the unit balls of a reflexive space E and of E^* .

⌘ **Section 1.2.4.** Weierstrass proved in 1885 that every continuous function on a compact interval is a uniform limit of polynomials [Wei85]. Stone replaced the interval by an arbitrary compact Hausdorff space and the coordinate function by any family of functions that separates points [Sto37, Sto48]. Ben Yaacov and Usvyatsov [BU10] used Stone's theorem to lay the foundations of continuous first-order logic. There Stone's theorem shows that finitely many connectives suffice and that continuous functions on type spaces are exactly the definable predicates. Ben Yaacov [Ben14] combined Grothendieck's criterion with Mazur's lemma [Maz33] (a weakly convergent sequence has convex combinations converging in norm), and so obtained the defining formulas of stable types as uniform limits of averages of instances; see Remark 1.2.14.

Chapter 2

Stability, trees, and definability of types

This chapter proves the definability theorem, Theorem 1.1.6, which Chapter 1 stated without proof. The argument passes through binary trees, and along the way we show that a formula is stable exactly when it cannot encode tall trees. As in Section 1.1, we fix a first-order theory T in a language L and an L -formula $\varphi(x, y)$ consistent with T .

2.1. The equivalence proved in this chapter

Four conditions on φ turn out to coincide: instability, the order property, the presence of tall binary trees, and the existence of a non-definable φ -type. The notions of φ -type, definable type, ladder, order property and stability come from Section 1.1; the newcomer, a binary tree for φ , needs a little notation for finite 0–1 sequences.

Notation 2.1.1. Recall that $\varphi^1 = \varphi$ and $\varphi^0 = \neg\varphi$. For an ordinal μ , we write $2^{<\mu}$ for the set of 0–1 sequences of length $<\mu$, and we write $2^{\leq n} = 2^{<n} \cup 2^n$. We write $\sigma \sqsubseteq \tau$ if σ is an initial segment of τ , $\sigma \sqsubset \tau$ if moreover $\sigma \neq \tau$, $|\sigma|$ for the length of σ , $\sigma \frown \langle u \rangle$ for the extension of σ by one entry u , and $\eta \upharpoonright k$ for the initial segment of length k of a sequence η of length $>k$.

Definition 2.1.2. A *tree of height n for φ in a model M of T* consists of families $(b_\sigma : \sigma \in 2^{<n})$ and $(a_\eta : \eta \in 2^n)$ of tuples of M such that

$$\sigma \frown \langle u \rangle \sqsubseteq \eta \implies M \models \varphi(a_\eta, b_\sigma)^u.$$

In other words, the leaf a_η satisfies $\varphi(x, b_\sigma)$ exactly when the branch η turns right ($u = 1$) at the node σ . We say that φ *encodes binary trees of arbitrarily large finite height* if for every n some model of T contains a tree of height n for φ .

Figure 2.1 shows a tree of height 3. Each node b_σ splits the leaves above it into two halves: the leaves on the right satisfy $\varphi(x, b_\sigma)$, and the leaves on the left satisfy $\neg\varphi(x, b_\sigma)$. The branch leading to a leaf a_η is thus a record, node by node, of which of the formulas $\varphi(x, b_\sigma)$ the leaf satisfies, and a tree of height n realizes all 2^n such records.

Like ladders, trees can be shortened. A tree of height $n + 1$ restricts to one of height n : keep the nodes b_σ with $\sigma \in 2^{<n}$, and for each $\eta \in 2^n$ keep the leaf $a_{\eta \frown \langle 0 \rangle}$.

Theorem 2.1.3. *Let $\varphi(x, y)$ be an L -formula. The following are equivalent:*

- (1) *The formula φ is unstable (see Definition 1.1.4).*
- (2) *The formula φ has the order property (see Definition 1.1.4).*
- (3) *The formula φ encodes binary trees of arbitrarily large finite height.*
- (4) *Some φ -type over some model of T is not definable (see Definitions 1.1.1 and 1.1.3).*

The proof goes around a cycle of implications. The equivalence (1) $\Leftrightarrow$ (2) is the definition of stability (see Definition 1.1.4). In Section 2.3 we turn long ladders into tall trees, which gives (2) $\Rightarrow$ (3) (see Proposition 2.3.1). In Section 2.4 we use tall trees to produce more φ -types than there are formulas to define them, which gives (3) $\Rightarrow$ (4) (see Proposition 2.4.2).

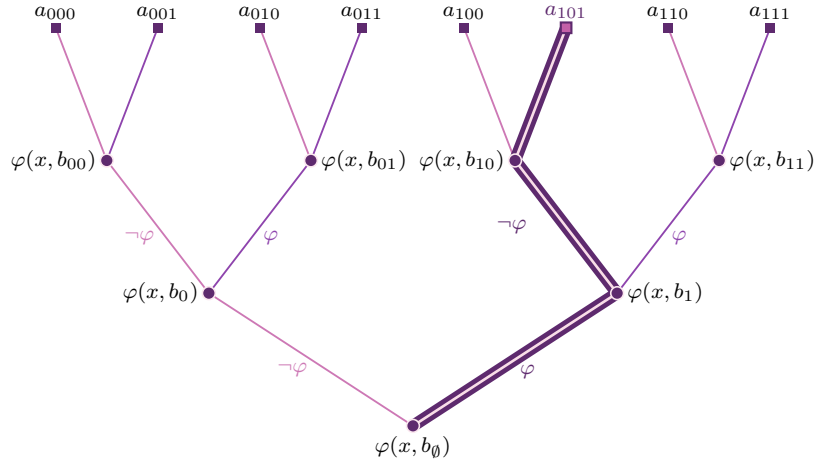


FIGURE 2.1. A tree of height 3 for φ . Each round node is labeled by the formula $\varphi(x, b_\sigma)$ given by its parameter b_σ , and the squares a_η are the leaves. An edge going up and to the right from the node $\varphi(x, b_\sigma)$ is labeled φ (purple), and an edge going up and to the left is labeled $\neg\varphi$ (pink): every leaf above a right edge out of that node satisfies $\varphi(x, b_\sigma)$, and every leaf above a left edge satisfies $\neg\varphi(x, b_\sigma)$. The edges into the leaves are not labeled, but their colors follow the same rule. For example, the thick branch leading up to a_{101} goes right, left, right, so that $M \models \varphi(a_{101}, b_\emptyset) \wedge \neg\varphi(a_{101}, b_1) \wedge \varphi(a_{101}, b_{10})$. The formula φ is not required to say anything about a_η and b_σ when σ is not an initial segment of η , such as a_{101} and b_0 .

The cycle closes with the definability theorem, which we prove first, in Section 2.2, and whose contrapositive is (4) $\Rightarrow$ (2) (see Theorem 2.2.1 and Corollary 2.2.2).

Chapter 3 gives alternative proofs of two of these implications, each of which yields information that the proofs in this chapter do not provide. Here the way from tall trees back to the order property passes through non-definable types, and so requires counting types over models of large cardinality. In Section 3.1 we prove (3) $\Rightarrow$ (2) directly instead, by finite combinatorics based on Ramsey's theorem (see Proposition 3.1.5 and Corollary 3.1.6). This direct proof stays inside a single model and is quantitative: a tree of height $2^{\binom{2n}{n}} - 1$ in a model M already yields an n -ladder in M . In Section 3.2 the classical rank argument proves (4) $\Rightarrow$ (3), again in contrapositive form: Proposition 3.2.3 shows that the absence of tall trees alone makes every φ -type definable, without going through the order property, and gives the definition explicitly in terms of a rank. Unlike the definitions of Section 2.2, however, these definitions are not positive Boolean combinations of instances of φ .

2.2. From the absence of the order property to type definability

We begin with the implication of the definability theorem that carries the weight, (1) $\Rightarrow$ (2) of Theorem 1.1.6: if φ does not have the order property, then every φ -type over a model has a definition built from instances of φ using only $\wedge$ and $\vee$. The candidate definitions are easy to write down, and the work lies in choosing the finitely many parameters that they use.

Let M be a model of T , let $p \in S_\varphi(M)$ be given, and recall that $D_p = \{b \in M : \varphi(x, b) \in p\}$. For a finite set $F \subseteq M$, we shall write

$$\mathcal{W}_F = \{\{a \in F : M \models \varphi(a, b)\} : b \in D_p\} \quad \text{and} \quad d_F\varphi(y) = \bigvee_{W \in \mathcal{W}_F} \bigwedge_{a \in W} \varphi(a, y).$$

The family $\mathcal{W}_F$ consists of subsets of the finite set F , so the family is finite and the formula $d_F\varphi$ is a positive Boolean combination of instances of φ over M . In words, the formula $d_F\varphi(y)$ says that, for some $b' \in D_p$, every $a \in F$ with $M \models \varphi(a, b')$ also satisfies $\varphi(a, y)$. Every parameter $b \in D_p$ satisfies $d_F\varphi$, whatever F is (take $b' = b$). The task is to choose F so that no parameter outside D_p satisfies $d_F\varphi$, and this is where the absence of the order property is used.

Theorem 2.2.1. *Assume that $\varphi(x, y)$ does not have the order property, let M be a model of T , and let $p \in S_\varphi(M)$ be given. Then there is a finite set $F \subseteq M$ such that $d_F\varphi$ is a φ -definition of p ; that is, for all $b \in M$ we have $\varphi(x, b) \in p$ if and only if $M \models d_F\varphi(b)$.*

Proof. For $b \in M$ we shall write $S_b = \{a \in M : M \models \varphi(a, b)\}$. The only property of p we use is that p is finitely satisfiable in M (see Lemma 1.1.2).

© **Claim.** There is a finite $F \subseteq M$ such that $S_b \cap F \not\subseteq S_{b'}$ for all $b \in D_p$ and $b' \in M \setminus D_p$.

Granting the claim, we check that $d_F\varphi$ defines p . Let $b \in M$ be given.

- (a) If $b \in D_p$, then the set $W := S_b \cap F$ belongs to $\mathcal{W}_F$ and satisfies $W \subseteq S_b$, so we have $M \models \bigwedge_{a \in W} \varphi(a, b)$. Hence $M \models d_F\varphi(b)$.
- (b) If $b \notin D_p$ and $W \in \mathcal{W}_F$, say $W = S_{b_0} \cap F$ with $b_0 \in D_p$, then we have $W \not\subseteq S_b$ by the claim. Thus some $a \in W$ has $M \models \neg\varphi(a, b)$, and hence we have $M \models \neg\bigwedge_{a \in W} \varphi(a, b)$. As this holds for every $W \in \mathcal{W}_F$, we conclude that $M \models \neg d_F\varphi(b)$.

© **Proof of the claim.** Suppose the claim fails. Then for every finite $F \subseteq M$ there are $b \in D_p$ and $b' \in M \setminus D_p$ with $S_b \cap F \subseteq S_{b'}$, i.e. with $M \models \varphi(a, b) \rightarrow \varphi(a, b')$ for all $a \in F$. Letting the failure of the claim and the finite satisfiability of p take turns, we build a configuration in which Ramsey's theorem will find a ladder. We choose recursively, for $i < \omega$, elements $b_i \in D_p$, $b'_i \in M \setminus D_p$ and $a_i \in M$ as follows.

- (a) Apply the failure of the claim to $F = \{a_j : j < i\}$ to get $b_i \in D_p$ and $b'_i \in M \setminus D_p$ with

$$(2.1) \quad M \models \varphi(a_j, b_i) \rightarrow \varphi(a_j, b'_i) \quad \text{for all } j < i.$$

- (b) Since $\varphi(x, b_k) \in p$ and $\neg\varphi(x, b'_k) \in p$ for $k \leq i$, Lemma 1.1.2 gives $a_i \in M$ with

$$(2.2) \quad M \models \varphi(a_i, b_k) \wedge \neg\varphi(a_i, b'_k) \quad \text{for all } k \leq i.$$

Color each pair $j < i$ by the pair of truth values $(\varphi(a_j, b_i), \varphi(a_j, b'_i)) \in \{0, 1\}^2$. By (2.1), the color $(1, 0)$ never occurs. By Ramsey's theorem there is an infinite $H = \{h_0 < h_1 < \dots\} \subseteq \omega$ on which all pairs have the same color c . Let $s, t < \omega$ be given.

© **Case $c \in \{(0, 1), (1, 1)\}$.** If $s < t$, then we have $M \models \varphi(a_{h_s}, b'_{h_t})$ by homogeneity. If $s \geq t$, then we have $M \models \neg\varphi(a_{h_s}, b'_{h_t})$ by (2.2). Thus the formula $\varphi(a_{h_s}, b'_{h_t})$ holds if and only if $s < t$, and with $\beta_t = b'_{h_{t+1}}$ we get $M \models \varphi(a_{h_s}, \beta_t) \iff s \leq t$.

© **Case $c = (0, 0)$.** If $s < t$, then we have $M \models \neg\varphi(a_{h_s}, b_{h_t})$ by homogeneity. If $s \geq t$, then we have $M \models \varphi(a_{h_s}, b_{h_t})$ by (2.2). Thus the formula $\varphi(a_{h_s}, b_{h_t})$ holds if and only if $s \geq t$. For a given n , reversing the first n indices by putting $\alpha_s = a_{h_{n-1-s}}$ and $\beta_t = b_{h_{n-1-t}}$ gives $M \models \varphi(\alpha_s, \beta_t) \iff s \leq t$ for $s, t < n$.

In both cases the model M contains an n -ladder for every n , so the formula φ has the order property, a contradiction. This proves the claim and the theorem. ☺

Corollary 2.2.2. *If φ does not have the order property, then for every model M of T , every $p \in S_\varphi(M)$ is definable by a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$. In particular, if φ is quantifier-free, then the type p has a quantifier-free φ -definition.*

The proof of Theorem 2.2.1 applies Ramsey's theorem for pairs to an infinite sequence, inside an argument by contradiction, and so says nothing about sizes. A finite Ramsey-type argument, adapted to binary trees rather than to sequences, reappears in Section 3.1, where it extracts ladders of prescribed length directly from trees of explicit height.

2.3. From the order property to tall trees

To prove (3) $\Rightarrow$ (1) of Theorem 1.1.6, we need a supply of many φ -types over a small model, and binary trees (see Definition 2.1.2) are the natural source. The first step, taken in this section, is to grow tall trees out of long ladders, which is (2) $\Rightarrow$ (3) of Theorem 2.1.3. The idea is to place the nodes and the leaves of a binary tree on a line, in such a way that the leaves above the right edge out of each node lie to the left of that node and the leaves above the left edge lie to the right of the node. A ladder read along this line then has exactly the pattern of a tree.

Proposition 2.3.1. *If a model M of T contains an N -ladder for φ , where $N = 2^{n+1} - 1$, then it contains a tree of height n . Hence, if φ has the order property, then it encodes binary trees of arbitrarily large finite height.*

Proof. For $\sigma \in 2^{\leq n}$ put

$$d(\sigma) = \sum_{k < |\sigma|} (1 - \sigma(k)) 2^{-(k+1)} + 2^{-(|\sigma|+1)} \in (0, 1).$$

The binary expansion of $d(\sigma)$ is $(1 - \sigma(0), \dots, 1 - \sigma(|\sigma| - 1), 1)$, which ends in 1. Thus the map d is injective on $2^{\leq n}$, and the image V of d has exactly $|2^{\leq n}| = 2^{n+1} - 1 = N$ elements.

© **Claim.** If $\sigma \smallfrown \langle u \rangle \sqsubseteq \eta$ with $\eta \in 2^n$, then we have $d(\eta) < d(\sigma) \iff u = 1$.

Indeed, let $k = |\sigma|$. Since $\eta \restriction k = \sigma$ and $\eta(k) = u$, we have

$$d(\eta) - d(\sigma) = -u 2^{-(k+1)} + R, \quad R = \sum_{k < j < n} (1 - \eta(j)) 2^{-(j+1)} + 2^{-(n+1)}.$$

Here we have $0 < R \leq (2^{-(k+1)} - 2^{-n}) + 2^{-(n+1)} < 2^{-(k+1)}$. Hence the difference $d(\eta) - d(\sigma)$ is negative if $u = 1$ and positive if $u = 0$, which proves the claim.

Let $a_1, \dots, a_N, b_1, \dots, b_N \in M$ satisfy $\varphi(a_i, b_j) \iff i \leq j$, and let $e: V \rightarrow \{1, \dots, N\}$ be the order isomorphism. Put $b_\sigma := b_{e(d(\sigma))}$ for $\sigma \in 2^{\leq n}$ and $a_\eta := a_{e(d(\eta))}$ for $\eta \in 2^n$. If $\sigma \smallfrown \langle u \rangle \sqsubseteq \eta$, then we have $d(\eta) \neq d(\sigma)$, and so we have

$$M \models \varphi(a_\eta, b_\sigma) \iff e(d(\eta)) \leq e(d(\sigma)) \iff d(\eta) < d(\sigma) \iff u = 1.$$

Thus the families (b_σ) and (a_η) form a tree of height n in M . ☺

Figure 2.2 shows the construction for $n = 2$.

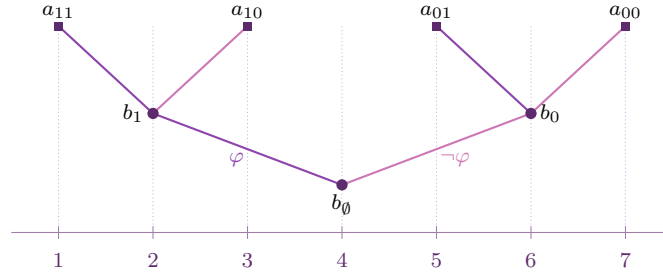


FIGURE 2.2. The proof of Proposition 2.3.1 for $n = 2$. The seven sequences of length at most 2 are placed at the positions $e(d(\sigma)) = 1, \dots, 7$ of a ladder of length 7, and the tree grows upward. In a ladder, the formula $\varphi(a_i, b_j)$ holds when a_i lies to the left of b_j (that is, $i \leq j$), so here the φ -edges (purple) go up and to the left and the $\neg\varphi$ -edges (pink) go up and to the right. The bound $R < 2^{-(k+1)}$ in the claim is what keeps every leaf on the correct side of each node below it: the leaf a_{10} , for instance, lies to the left of b_0 and to the right of b_1 .

2.4. Tall trees produce non-definable types

Tall trees produce more φ -types over a small model than there are formulas to define them, and this proves (3) $\Rightarrow$ (4) of Theorem 2.1.3. The count is done in Proposition 2.4.2 below. To prepare for it, in Lemma 2.4.1 below we use compactness to turn finite trees into a tree of infinite height, whose branches will give the types. When L is countable, the count is easiest to see over a countable model, and this case is the picture to keep in mind: a tree of height ω has $2^{\aleph_0}$ branches, each branch gives a φ -type over a countable model, and only countably many formulas are available to define them. In Section 2.5 the same tree of height ω reappears as a copy of the Cantor space inside $S_\varphi(M)$.

Lemma 2.4.1 (compactness). *Suppose that φ encodes binary trees of arbitrarily large finite height, and let μ be an infinite cardinal. Then there are a model M' of T and tuples $(b_s)_{s \in 2^{<\mu}}$ in M' such that for every $\eta \in 2^\mu$ the set*

$$\Sigma_\eta(x) = \{\varphi(x, b_{\eta|_\alpha})^{\eta(\alpha)} : \alpha < \mu\}$$

is finitely satisfiable in M' .

Proof. Add new tuples of constants c_s ($s \in 2^{<\mu}$) and let Γ be T together with the sentences

$$\gamma_{\eta, F} := \exists x \bigwedge_{\alpha \in F} \varphi(x, c_{\eta|_\alpha})^{\eta(\alpha)} \quad (\eta \in 2^\mu, F \subseteq \mu \text{ finite}).$$

By compactness, we need only show that every finite $\Gamma_0 \subseteq \Gamma$ has a model.

Let γ_{η_i, F_i} ($i < m$) be the sentences of this form in Γ_0 (if $m = 0$ there is nothing to do). Only finitely many nodes of $2^{<\mu}$ are mentioned in these sentences, and the plan is to embed them into a finite tree, preserving the direction in which each branch leaves each node. Let S be the finite set of all $\eta_i|_\alpha$ and $\eta_i|(\alpha + 1)$ with $i < m$ and $\alpha \in F_i$. Note that $\alpha + 1 < \mu$ because μ is a limit ordinal. Let $S' = \{s \wedge s' : s, s' \in S\}$, where $s \wedge s'$ is the longest common initial segment. The set S' is finite and closed under $\wedge$, because in a tree the meet $s \wedge s' \wedge s''$ equals one of the pairwise meets. Thus the set S' has a least element r , and every $s \in S' \setminus \{r\}$ has a longest predecessor $s^- \in S'$ with $s^- \sqsubset s$.

Define $f: S' \rightarrow 2^{<N}$, where $N = |S'| + 1$, by recursion on length, setting $f(r) = \emptyset$ and $f(s) = f(s^-) \hat{\ } \langle s(|s^-|) \rangle$. Then we have $|f(s)| < N$.

☺ **Claim.** If $s, s' \in S'$ and $s \sqsubset s'$, then we have $f(s) \hat{\ } \langle s'(|s|) \rangle \sqsubseteq f(s')$.

The proof is by induction on $|s'|$. If $s'^- = s$, this is the definition of f . Otherwise we have $s \sqsubset s'^- \sqsubset s'$ by maximality of s'^- . By induction, we have $f(s) \hat{\ } \langle s'^-(|s|) \rangle \sqsubseteq f(s'^-) \sqsubseteq f(s')$, and finally we have $s'^-(|s|) = s'(|s|)$.

By hypothesis, take a model M_0 of T containing a tree $(b_\sigma)_{\sigma \in 2^{<N}}$, $(a_\nu)_{\nu \in 2^N}$ of height N . Interpret c_s as $b_{f(s)}$ for $s \in S'$, and arbitrarily otherwise. Fix $i < m$, let $\alpha^* = \max F_i$ and $s^* = \eta_i(\alpha^* + 1) \in S'$. Pick $\nu \in 2^N$ extending $f(s^*)$. For $\alpha \in F_i$ we have $\eta_i|\alpha \sqsubset s^*$ and $s^*(\alpha) = \eta_i(\alpha)$. By the Claim, we have $f(\eta_i|\alpha) \hat{\ } \langle \eta_i(\alpha) \rangle \sqsubseteq \nu$, which gives $M_0 \models \varphi(a_\nu, b_{f(\eta_i|\alpha)})^{\eta_i(\alpha)}$. Hence the tuple a_ν witnesses γ_{η_i, F_i} , and the structure M_0 with this interpretation is a model of Γ_0 .

By compactness, the theory Γ has a model. The L -reduct M' of that model, with $b_s = c_s^{M'}$, is as required. ☺

In the next proposition we write $\kappa = |L| + \aleph_0$, which is the number of L -formulas.

Proposition 2.4.2. *If φ encodes binary trees of arbitrarily large finite height, then for every cardinal $\lambda \geq \kappa$ there is a model M of T with $|M| \leq \lambda$ and $|S_\varphi(M)| > \lambda$. In particular, there are a model M of T with $|M| \leq \kappa$ and a type $p \in S_\varphi(M)$ such that p is not definable.*

Proof. Let μ be the least cardinal with $2^\mu > \lambda$. Then the cardinal μ is infinite and satisfies $\mu \leq \lambda$. Since $2^\nu \leq \lambda$ for every $\nu < \mu$, we have $|2^{<\mu}| = \sum_{\nu < \mu} 2^\nu \leq \mu \cdot \lambda = \lambda$.

Take M' and (b_s) as in Lemma 2.4.1. By the downward Löwenheim–Skolem theorem, there is $M \preceq M'$ with $|M| \leq \lambda$ containing all b_s . Each finite part of Σ_η is satisfiable in M' , hence in M , since the formula $\exists x \bigwedge \dots$ is an $L(M)$ -sentence. Thus the set Σ_η is consistent with the elementary diagram of M and extends to some $p_\eta \in S_\varphi(M)$.

If $\eta \neq \eta'$, let α be least with $\eta(\alpha) \neq \eta'(\alpha)$. Put $s = \eta|\alpha = \eta'|\alpha$; then the types p_η and $p_{\eta'}$ disagree on $\varphi(x, b_s)$. Hence $|S_\varphi(M)| \geq 2^\mu > \lambda$.

For the last statement take $\lambda = \kappa$. A definable type is determined by any φ -definition of the type, and there are only $|L(M)| \leq \kappa$ formulas over M . Thus at most κ types in $S_\varphi(M)$ are definable, and some $p \in S_\varphi(M)$ is not. ☺

Proof of (3) $\Rightarrow$ (1) in Theorem 1.1.6. If φ has the order property, then it encodes binary trees of arbitrarily large finite height by Proposition 2.3.1, and so some φ -type over some model of T is not definable by Proposition 2.4.2. ☺

This completes the proof of Theorem 1.1.6, and with it that of Theorem 2.1.3. The equivalence of (2) and (3) in Theorem 2.1.3 is the form in which trees are used later (see Section 2.5), so we record it separately.

Theorem 2.4.3. *The formula φ has the order property if and only if it encodes binary trees of arbitrarily large finite height.*

Proof. The forward direction is Proposition 2.3.1. Conversely, if φ encodes tall trees, then some φ -type is not definable (see Proposition 2.4.2), so the formula φ has the order property by (1) $\Rightarrow$ (3) of Theorem 1.1.6. ☺

2.5. Types over countable models: a perfect set dichotomy

Section 2.4 measured instability by counting types; topology measures instability by shape, and over countable models the two measurements agree. Throughout this section the

language L is countable. For a model M of T , the space $S_\varphi(M)$ is the Stone space of the Boolean algebra $\mathcal{B}_\varphi(M)$ (see Section 1.1). If M is countable, the algebra $\mathcal{B}_\varphi(M)$ is countable, so its Stone space $S_\varphi(M)$ is a compact metrizable zero-dimensional space. By the Cantor–Bendixson theorem, such a space is either countable, in which case the space is scattered, or contains a homeomorphic copy of the Cantor space 2^ω . Stability decides which alternative occurs.

Theorem 2.5.1. *Assume that L is countable. The following are equivalent:*

- (1) *The formula φ is stable.*
 - (2) *For every countable model M of T , the space $S_\varphi(M)$ is countable.*
 - (3) *For every countable model M of T , the space $S_\varphi(M)$ is scattered.*
- Moreover, if φ is unstable, then some countable model M of T has a closed subset of $S_\varphi(M)$ homeomorphic to 2^ω .*

Proof. (1) $\Rightarrow$ (2). By Theorem 1.1.6, every $p \in S_\varphi(M)$ has a φ -definition, which is an $L(M)$ -formula, and the definition determines p . Since there are only countably many $L(M)$ -formulas, the space $S_\varphi(M)$ is countable.

(2) $\Leftrightarrow$ (3). This is the Cantor–Bendixson theorem for compact metrizable spaces.

(2) $\Rightarrow$ (1) and the last statement. Suppose that φ is unstable. By Theorem 2.4.3, the formula φ encodes binary trees of arbitrarily large finite height, so Lemma 2.4.1 with $\mu = \omega$ gives a model M' of T and tuples $(b_\sigma)_{\sigma \in 2^{<\omega}}$ in M' such that every set $\Sigma_\eta(x)$, for $\eta \in 2^\omega$, is finitely satisfiable in M' . By the downward Löwenheim–Skolem theorem, there is a countable model $M \preceq M'$ containing every b_σ , and by elementarity every $\Sigma_\eta(x)$ is finitely satisfiable in M . For $\sigma \in 2^{<\omega}$, let C_σ be the clopen set of types $p \in S_\varphi(M)$ that contain $\varphi(x, b_{\sigma|_\alpha})^{\sigma(\alpha)}$ for every $\alpha < |\sigma|$. Each C_σ is nonempty, because the formulas in question form a finitely satisfiable subset of Σ_η for any η extending σ , and the set C_σ is the disjoint union of $C_{\sigma \smallfrown \langle 0 \rangle}$ and $C_{\sigma \smallfrown \langle 1 \rangle}$, which disagree on $\varphi(x, b_\sigma)$. Thus the sets C_σ form a Cantor scheme. The closed set $D = \bigcap_n \bigcup_{|\sigma|=n} C_\sigma$ maps continuously onto 2^ω , by sending p to the unique η with $p \in C_{\eta|_n}$ for every n ; the map is onto because each $\bigcap_n C_{\eta|_n}$ is a decreasing intersection of nonempty closed sets in a compact space. Hence the space $S_\varphi(M)$ is uncountable, so (2) fails, and by the Cantor–Bendixson theorem the uncountable compact metrizable space $S_\varphi(M)$ contains a closed copy of 2^ω . ☺

A tree of infinite height is thus nothing but a Cantor scheme of clopen sets in a type space (see Figure 2.3), and stability says that type spaces over countable models are scattered. Morley used the same topology to define the totally transcendental theories, whose type spaces over countable sets are scattered, and whose Morley rank is the Cantor–Bendixson rank on type spaces over $\aleph_0$ -saturated models. The rank R introduced in Section 3.2 below is a binary splitting rank on the clopen sets $[\theta]$, a finite combinatorial cousin of the Cantor–Bendixson rank: Lemma 3.2.2(iii) says that splitting a clopen set into two pieces of rank at least m raises the rank.

2.6. Historical remarks

⌘ **Section 2.2.** Definability of φ -types over models for formulas without the order property is due to Shelah [She71, She90]. The sharper form proved here, in which the definition is a positive Boolean combination of instances of φ itself, goes back to Hrushovski and Pillay [HP94], as Pillay points out in [Pil96, Chapter 1]. Pillay [Pil17, Remark 2.4] derives

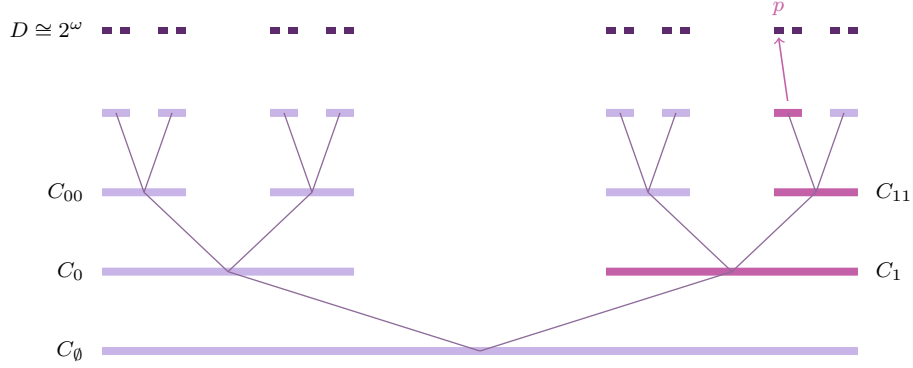


FIGURE 2.3. The Cantor scheme in the proof of Theorem 2.5.1, growing upward. Each bar is a nonempty clopen set C_σ of φ -types, and C_σ splits into $C_{\sigma \smallfrown \langle 0 \rangle}$ (left) and $C_{\sigma \smallfrown \langle 1 \rangle}$ (right), which disagree on $\varphi(x, b_\sigma)$. The pink bars $C_1 \supseteq C_{11} \supseteq C_{110} \supseteq \dots$ follow one branch η , and their intersection contains a type p with $p \in C_{\eta|n}$ for every n . Distinct branches give distinct types, so the types along the $2^{\aleph_0}$ branches form the closed set D , which maps onto 2^ω .

the positive form directly from Grothendieck's double limit criterion (see Theorem A.1.3 below) by a compactness argument. Positivity matters in applications: for example, positivity gives quantifier-free definitions for quantifier-free formulas.

- ⌘ **Section 2.3.** Binary trees enter stability theory through Shelah's local rank $R(-, \varphi, 2)$. In [She71, Lemma 2.7] the rank of a type is computed by the consistency of the family of formulas indexed by the branches of a binary tree of height n . That is, the formulas of infinite 2-rank are exactly those that encode binary trees of arbitrarily large finite height (see Definition 2.1.2). Shelah showed that this happens exactly when φ has the order property [She71, She90]. The two directions behave very differently quantitatively. A ladder of length exponential in n yields a tree of height n (see Proposition 2.3.1). In the other direction, a tree yields only a ladder of length logarithmic in the height of the tree; see Section 3.1.
- ⌘ **Section 2.4.** The construction of this section is the local form of Shelah's counting argument [She71]. A tree of infinite height gives $2^{\aleph_0}$ φ -types over a countable set of parameters, which is incompatible with definability of all φ -types. This is also the origin of the *stability spectrum*: a theory with the order property has 2^λ types over some set of size λ , for every infinite λ [She90].

Chapter 3

More stability, trees, and definability

Chapter 2 established the equivalences of Theorem 2.1.3 by the most direct route to the definability theorem. In that route, the way from tall trees back to the order property goes through non-definable types and a count of types over large models. This chapter develops the quantitative side of the theory, which that route leaves out. In Section 3.1 we show by a direct combinatorial argument, based on Ramsey’s theorem and carried out inside a single model, how tall a tree must be to contain a ladder of a given length. In Section 3.2 we present the classical proof of definability of types, which measures types by the height of the trees that can be grown above them and produces the definitions explicitly from this rank, without passing through the order property. The conventions are those of Chapter 2.

3.1. Trees give the order property: a quantitative proof by Ramsey’s theorem

Theorem 2.4.3 was derived from the definability theorem; here we give a direct and quantitative proof of the implication “tall trees $\Rightarrow$ order property”. The proof is pure finite combinatorics, the Boolean case of a Ramsey-type argument due to Hodges, and we follow the presentation of Conant and Terry [CT26]. Finding a ladder inside a tree requires passing to smaller trees embedded in it, and the first definition says what such an embedding is.

Definition 3.1.1. Let $s, t \geq 1$ be given.

- (a) A *node embedding* is a map $\theta: 2^{<s} \rightarrow 2^{<t}$ such that $\rho \frown \langle u \rangle \sqsubseteq \tau$ implies $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$.
- (b) A *full embedding* is a map $\theta: 2^{\leq s} \rightarrow 2^{\leq t}$ sending $2^{<s}$ into $2^{<t}$ and 2^s into 2^t , with the same property for all $\rho, \tau \in 2^{\leq s}$.

Lemma 3.1.2. *Every node embedding $\theta: 2^{<s} \rightarrow 2^{<t}$ extends to a full embedding. If (b_σ) , (a_η) is a tree of height t in M and θ is a full embedding $2^{\leq s} \rightarrow 2^{\leq t}$, then the families $(b_{\theta(\sigma)} : \sigma \in 2^{<s})$ and $(a_{\theta(\eta)} : \eta \in 2^s)$ form a tree of height s in M .*

Proof. For $\eta \in 2^s$, let $\theta(\eta)$ be any element of 2^t extending $\theta(\eta|(s-1)) \frown \langle \eta(s-1) \rangle$. Suppose that $\rho \frown \langle u \rangle \sqsubseteq \eta$. If $\rho = \eta|(s-1)$, the requirement holds by construction. Otherwise we have $\rho \frown \langle u \rangle \sqsubseteq \eta|(s-1)$, and therefore $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\eta|(s-1)) \sqsubseteq \theta(\eta)$. The second statement is immediate, since whether $\varphi(a_\eta, b_\sigma)$ holds depends only on the direction u with $\sigma \frown \langle u \rangle \sqsubseteq \eta$, and full embeddings preserve this direction. $\odot$

To find a ladder, the proof of Proposition 3.1.5 below splits the nodes of a tree into two classes and needs a tall tree embedded in one of them. The next lemma, a Ramsey theorem for node embeddings, provides such an embedding; Figure 3.1 illustrates it for $t_0 = t_1 = 2$. In its proof and in the rest of this chapter, for $u \in \{0, 1\}$ and a finite 0–1 sequence σ , we write $\langle u \rangle \frown \sigma$ for the sequence obtained by putting u in front of σ .

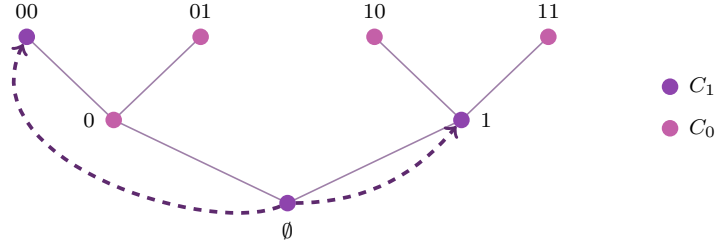


FIGURE 3.1. The nodes $2^{<3}$, growing upward, coloured with two colours C_0 (pink) and C_1 (purple). The dashed arrows show a node embedding $\theta: 2^{<2} \rightarrow 2^{<3}$ into C_1 , with $\theta(\emptyset) = \emptyset$, $\theta(0) = 00$ and $\theta(1) = 1$: the image of the left child lies above the image of the root in the left direction, and the image of the right child in the right direction, but the map θ need not send children to children. This is the conclusion of Lemma 3.1.3 for $t_0 = t_1 = 2$ and $t = 3$. The colour C_0 admits no such embedding here.

Lemma 3.1.3 (Ramsey lemma for trees). *Let $t_0, t_1 \geq 1$ be given, let $t = t_0 + t_1 - 1$, and suppose that $2^{<t} = C_0 \cup C_1$. Then for some $u \in \{0, 1\}$ there is a node embedding $\theta: 2^{<t_u} \rightarrow 2^{<t}$ with image contained in C_u .*

Proof. Induction on t . If $t = 1$, then we have $t_0 = t_1 = 1$, and the map $\emptyset \mapsto \emptyset$ works for the u with $\emptyset \in C_u$. Now suppose that $t > 1$, and choose w with $\emptyset \in C_w$. If $t_w = 1$, the map $\emptyset \mapsto \emptyset$ works, so assume that $t_w \geq 2$. For $v \in \{0, 1\}$ and $i \in \{0, 1\}$ let $C_i^v = \{\sigma \in 2^{<t-1} : \langle v \rangle \frown \sigma \in C_i\}$. Then the union $C_0^v \cup C_1^v$ covers $2^{<t-1}$, and we have $(t_w - 1) + t_{1-w} - 1 = t - 1$. By induction, for each v either

- (i) there is a node embedding $\chi: 2^{<t_{1-w}} \rightarrow C_{1-w}^v$, in which case the map $\sigma \mapsto \langle v \rangle \frown \chi(\sigma)$ is a node embedding $2^{<t_{1-w}} \rightarrow 2^{<t}$ with image in C_{1-w} , and we are done; or
- (ii) there is a node embedding $\chi_v: 2^{<t_w-1} \rightarrow C_w^v$.

If (ii) holds for both v , define $\theta: 2^{<t_w} \rightarrow 2^{<t}$ by $\theta(\emptyset) = \emptyset$ and $\theta(\langle v \rangle \frown \sigma) = \langle v \rangle \frown \chi_v(\sigma)$. The image of θ lies in C_w . Suppose that $\rho \frown \langle u \rangle \sqsubseteq \tau$. If $\rho = \emptyset$, then we have $\tau = \langle u \rangle \frown \sigma$ and $\emptyset \frown \langle u \rangle \sqsubseteq \theta(\tau)$. If $\rho = \langle v \rangle \frown \rho'$, then we have $\tau = \langle v \rangle \frown \tau'$ with $\rho' \frown \langle u \rangle \sqsubseteq \tau'$, and therefore $\chi_v(\rho') \frown \langle u \rangle \sqsubseteq \chi_v(\tau')$. In both cases we have $\theta(\rho) \frown \langle u \rangle \sqsubseteq \theta(\tau)$, so the map θ is a node embedding. ☺

The ladders found inside a tree come in two orientations, according to the side of the split on which the Ramsey lemma lands, and the induction keeps track of both.

Definition 3.1.4. Let $(b_\sigma)_{\sigma \in 2^{<t}}$, $(a_\eta)_{\eta \in 2^t}$ be a tree in M . An *up-ladder of length k* in this tree consists of nodes $\sigma_1, \dots, \sigma_k$ and leaves $\eta_1, \dots, \eta_k$ such that $M \models \varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$. A *down-ladder of length ℓ* is defined in the same way with “ $p > q$ ” in place of “ $p \leq q$ ”.

The height needed to force a ladder is governed by a familiar recursion. Define $B(k, \ell)$ for $k, \ell \geq 1$ by $B(k, 1) = B(1, \ell) = 1$ and $B(k, \ell) = B(k-1, \ell) + B(k, \ell-1) + 1$ for $k, \ell \geq 2$. By induction on $k + \ell$, using Pascal’s rule $\binom{k+\ell-3}{k-2} + \binom{k+\ell-3}{k-1} = \binom{k+\ell-2}{k-1}$, we have $B(k, \ell) = 2\binom{k+\ell-2}{k-1} - 1$. The recursion is the Erdős–Szekeres recursion for the Ramsey numbers $R(k, \ell)$, and the bound mirrors $R(k, \ell) \leq \binom{k+\ell-2}{k-1}$. The proof below follows the Erdős–Szekeres pattern as well: we fix a leaf η_* and split the nodes according to whether a_{η_*} satisfies the corresponding instance of φ . Lemma 3.1.3 finds a large subtree whose nodes all lie on one

side of the split, induction finds a ladder inside that subtree, and the pair formed by η_* and the root of the subtree extends the ladder by one rung.

Proposition 3.1.5. *For all $k, \ell \geq 1$, every tree of height $B(k, \ell)$ for φ in a model of T contains an up-ladder of length k or a down-ladder of length ℓ .*

Proof. Induction on $k + \ell$. In a tree of height 1, the root $\emptyset$ with the leaf $\langle 1 \rangle$ is an up-ladder of length 1, and the root with the leaf $\langle 0 \rangle$ is a down-ladder of length 1. This settles the case $k = 1$ or $\ell = 1$.

Let $k, \ell \geq 2$ be given, let $t = B(k, \ell)$, and let $(b_\sigma), (a_\eta)$ be a tree of height t in M . Fix a leaf $\eta_* \in 2^t$ and put

$$P^+ = \{\sigma \in 2^{<t} : M \models \varphi(a_{\eta_*}, b_\sigma)\}, \quad P^- = 2^{<t} \setminus P^+.$$

Let $t^+ = B(k-1, \ell) + 1$ and $t^- = B(k, \ell-1) + 1$, so that $t^+ + t^- - 1 = t$. By Lemma 3.1.3, one of the two cases below occurs.

Case +: there is a node embedding $\theta: 2^{<t^+} \rightarrow 2^{<t}$ with image in P^+ . Extend θ to a full embedding $\hat{\theta}$ (see Lemma 3.1.2) and set $\psi(\tau) = \hat{\theta}(\langle 0 \rangle \frown \tau)$ for $\tau \in 2^{\leq t^+ - 1}$. Then the map ψ is a full embedding of height $B(k-1, \ell)$. Put $\sigma_0 = \theta(\emptyset)$. Since $\emptyset \frown \langle 0 \rangle \subseteq \langle 0 \rangle \frown \tau$, we have $\sigma_0 \frown \langle 0 \rangle \subseteq \psi(\tau)$ for every τ . By Lemma 3.1.2, the embedding ψ induces a tree of height $B(k-1, \ell)$ in M . By induction, the induced tree contains a down-ladder of length ℓ , which is then a down-ladder in the original tree, or an up-ladder $(\sigma'_p, \eta'_p)_{p=1}^{k-1}$. In the latter case, let

$$(\sigma_1, \eta_1) = (\sigma_0, \eta_*), \quad (\sigma_{p+1}, \eta_{p+1}) = (\psi(\sigma'_p), \psi(\eta'_p)) \quad (1 \leq p < k).$$

We check that $\varphi(a_{\eta_p}, b_{\sigma_q}) \iff p \leq q$.

- (a) For $p = 1$, every σ_q is in the image of θ , which lies in P^+ , so the formula $\varphi(a_{\eta_*}, b_{\sigma_q})$ holds, and indeed $1 \leq q$.
- (b) For $p \geq 2$ and $q = 1$, we have $\sigma_0 \frown \langle 0 \rangle \subseteq \eta_p$, so the formula $\neg \varphi(a_{\eta_p}, b_{\sigma_0})$ holds, and indeed $p > 1$.
- (c) For $p, q \geq 2$: this is the up-ladder property of the smaller tree.

This is an up-ladder of length k .

Case -: there is a node embedding $\theta: 2^{<t^-} \rightarrow 2^{<t}$ with image in P^- . Proceed symmetrically, with $\psi(\tau) = \hat{\theta}(\langle 1 \rangle \frown \tau)$, a full embedding of height $B(k, \ell-1)$ whose leaves all extend $\sigma_0 \frown \langle 1 \rangle$. By induction the induced tree contains an up-ladder of length k , and we are done, or a down-ladder $(\sigma'_p, \eta'_p)_{p=1}^{\ell-1}$. Prepend (σ_0, η_*) as before.

- (a) For $p = 1$, the formula $\neg \varphi(a_{\eta_*}, b_{\sigma_q})$ holds for all q , because the nodes lie in P^- , and indeed $1 \not\leq q$.
- (b) For $p \geq 2$ and $q = 1$, the formula $\varphi(a_{\eta_p}, b_{\sigma_0})$ holds, because $\sigma_0 \frown \langle 1 \rangle \subseteq \eta_p$, and indeed $p > 1$.

This is a down-ladder of length ℓ . ☺

Corollary 3.1.6. *If a model M of T contains a tree of height $B(n+1, n+1)$ for φ , then it contains an n -ladder for φ .*

Proof. By Proposition 3.1.5, the model M contains an up-ladder or a down-ladder of length $n+1$. An up-ladder of length $n+1$ is an $(n+1)$ -ladder, and so contains an n -ladder. A down-ladder of length $n+1$ yields an n -ladder by Remark 1.1.5. ☺

3.2. The classical proof of definability, via ranks

In Proposition 3.2.3 below we give the classical argument that, in the absence of tall trees, every φ -type is definable. Together with Proposition 2.3.1, the argument gives another route to (1) $\Rightarrow$ (3) in Theorem 1.1.6, at the cost of positivity: the definitions obtained in this section are not positive Boolean combinations of instances of φ , since they contain quantifiers and negations that do not occur in φ (see Lemma 3.2.2(iv)). The idea is to measure each finite piece of a type by the height of the tallest tree that can be grown above the piece, and to define the type by a piece of minimal height.

Until the end of the proof of Proposition 3.2.3, we assume that there is n such that no model of T contains a tree of height n , and we fix a model M of T . By a φ -conjunction over M we mean a finite conjunction $\theta(x)$ of formulas $\varphi(x, b)^u$ with $b \in M$.

Definition 3.2.1. For a φ -conjunction θ over M and $m < \omega$, we write $R(\theta) \geq m$ if there are $b_\sigma \in M$ ($\sigma \in 2^{<m}$) such that for every $\eta \in 2^m$, we have

$$M \models \exists x \left(\theta(x) \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k})^{\eta(k)} \right).$$

Thus $R(\theta) \geq 0$ if and only if $M \models \exists x \theta(x)$. When θ is satisfiable in M , we write $R(\theta)$ for the largest m with $R(\theta) \geq m$.

In Lemma 3.2.2(i) below we show that this largest m exists.

Lemma 3.2.2.

- (i) *The inequality $R(\theta) < n$ holds for every θ ; in particular, the rank $R(\theta)$ is well defined whenever θ is satisfiable in M .*
- (ii) *If $\theta' = \theta \wedge \chi$, then we have $R(\theta') \leq R(\theta)$.*
- (iii) *If $b \in M$ satisfies $R(\theta \wedge \varphi(x, b)^u) \geq m$ for both $u \in \{0, 1\}$, then we have $R(\theta) \geq m + 1$.*
- (iv) *For each θ and m there is an $L(M)$ -formula $\psi_{\theta, m}(y)$ such that for all $b \in M$ we have $R(\theta \wedge \varphi(x, b)) \geq m \iff M \models \psi_{\theta, m}(b)$.*

Proof. (i) If $R(\theta) \geq n$, witnessed by (b_σ) , then choosing a witness $a_\eta \in M$ for each $\eta \in 2^n$ gives a tree of height n in M , which is impossible.

(ii) A witness for $R(\theta') \geq m$ is a witness for $R(\theta) \geq m$.

(iii) Let $(b_\sigma^u)_{\sigma \in 2^{<m}}$ witness $R(\theta \wedge \varphi(x, b)^u) \geq m$. Put $b_\emptyset = b$ and $b_{\langle u \rangle \frown \sigma} = b_\sigma^u$. A branch of length $m + 1$ has the form $\langle u \rangle \frown \eta$ with $\eta \in 2^m$, and the corresponding formula is $\theta \wedge \varphi(x, b)^u \wedge \bigwedge_{k < m} \varphi(x, b_{\eta|k}^u)^{\eta(k)}$, which is satisfiable in M .

(iv) Take

$$\psi_{\theta, m}(y) := \exists (z_\sigma)_{\sigma \in 2^{<m}} \bigwedge_{\eta \in 2^m} \exists x \left(\theta(x) \wedge \varphi(x, y) \wedge \bigwedge_{k < m} \varphi(x, z_{\eta|k})^{\eta(k)} \right),$$

whose parameters are those of θ . ☺

Proposition 3.2.3. *Let M be a model of T . If for some n no model of T contains a tree of height n , then every $p \in S_\varphi(M)$ is definable.*

Proof. Let Φ_p be the set of finite conjunctions of members of p . Each $\theta \in \Phi_p$ is satisfiable in M , because the sentence $\exists x \theta$ is an $L(M)$ -sentence consistent with the elementary diagram of M , hence true in M . By Lemma 3.2.2(i), choose $\theta \in \Phi_p$ with $m = R(\theta)$ minimal. We show that $\psi_{\theta, m}$ is a φ -definition of p , i.e. that for $b \in M$, we have

$$\varphi(x, b) \in p \iff R(\theta \wedge \varphi(x, b)) \geq m.$$

If $\varphi(x, b) \in p$, then the conjunction $\theta \wedge \varphi(x, b)$ belongs to Φ_p , so we have $R(\theta \wedge \varphi(x, b)) \geq m$ by minimality. Conversely, suppose that $\varphi(x, b) \notin p$. Then the formula $\neg\varphi(x, b)$ belongs to p , so we have $R(\theta \wedge \neg\varphi(x, b)) \geq m$ by minimality. If also $R(\theta \wedge \varphi(x, b)) \geq m$, then Lemma 3.2.2(iii) would give $R(\theta) \geq m + 1$, a contradiction. The equivalence together with Lemma 3.2.2(iv) shows that p is definable. $\odot$



Remark 3.2.4.

- (a) The passages between ladders and trees are effective. An up- or down-ladder of length $n + 1$ appears in any tree of height $B(n + 1, n + 1) = 2^{\binom{2n}{n}} - 1$. Conversely, a ladder of length $2^{n+1} - 1$ yields a tree of height n .
- (b) Theorem 2.2.1 uses only that p is finitely satisfiable in M (see Lemma 1.1.2). Consequently, the theorem applies to any φ -type over a set A that is finitely satisfiable in A (see Definition 3.2.5(a) below). The definition given by the theorem has its parameters in the finite set F , whereas the rank argument of Proposition 3.2.3 yields a definition whose parameters are those of the formula θ of minimal rank.
- (c) Without finite satisfiability, definitions over the parameter set cannot be expected in general, and the restriction to types over models is essential. Definition 3.2.5 and Example 3.2.6 below make this precise.

Definition 3.2.5. Let N be a model of T , and let $A \subseteq N$ and $p \in S_\varphi(N)$ be given.

- (a) A *complete φ -type over A* is defined as in Definition 1.1.1, with b ranging over A instead of M ; such a type is *finitely satisfiable in A* if every finite subset of the type is realized by an element of A .
- (b) The type p is *definable over A* if p has a φ -definition that is an $L(A)$ -formula.
- (c) The type p is *definable almost over A* if p has a φ -definition of the form $\psi(y, c)$, with $\psi(y, z)$ an L -formula and c a tuple of N , such that only finitely many distinct subsets of N are of the form $\psi(N, c')$ with $c' \in N$ realizing the same complete type over A as c . In other words, the definition of p has only finitely many conjugates over A .

Definable over A implies definable almost over A : take c to enumerate the parameters from A , so that $c' = c$ is the only possibility. The converse fails, as the next example shows with two equivalence classes.

Example 3.2.6. Let T be the theory of an equivalence relation E with exactly two classes, both infinite, let $\varphi(x, y)$ be $E(x, y)$, and let $A = \emptyset$. Let N be a model of T with classes C_1 and C_2 , and let $p \in S_\varphi(N)$ be the φ -type saying that x lies in C_1 , i.e. the type that contains $E(x, b)$ if and only if $b \in C_1$. For any $c \in C_1$, the formula $E(y, c)$ is a φ -definition of p . All elements of N have the same complete type over $\emptyset$, so the sets $E(N, c')$ are exactly C_1 and C_2 . Hence the type p is definable almost over $\emptyset$. The type p is not definable over $\emptyset$: since N has an automorphism swapping C_1 and C_2 , every L -formula $\psi(y)$ without parameters defines $\emptyset$ or N , never C_1 . Note that p is not finitely satisfiable in $\emptyset$.

3.3. Historical remarks



Section 3.1. The Ramsey-type argument that extracts ladders from trees is due to Hodges, and appears in his book [Hod93]. Hodges's argument gives a direct combinatorial proof that tall trees force the order property, without going through definability of types. Conant

and Terry [CT26] recently revisited the argument with explicit bounds, and extended the argument from $\{0,1\}$ -valued formulas to real-valued functions. Their version is the one followed here. Such quantitative bounds are what is needed for applications of stability to finite combinatorics, such as the regularity lemma for stable graphs.

✂ **Section 3.2.** The rank $R(\theta)$ is Shelah's local rank $R(\theta, \varphi, 2)$ [She71, She90]. The proof that a type of minimal rank is definable is the classical route to definability of types, the one found in most textbook treatments, for instance Pillay [Pil96] and Tent–Ziegler [TZ12], and predates the combinatorial and analytic proofs given elsewhere in these notes. As noted above, the resulting definitions involve quantifiers, whereas those of Section 2.2 do not.

Chapter 4

From stability to NIP

In a stable theory no formula can encode arbitrarily long ladders. Many of the best understood structures in mathematics fail this test for a simple reason: they carry a definable linear order. Dense linear orders, the ordered field of real numbers and, more generally, o-minimal structures are all unstable, and yet their definable sets are remarkably well behaved. To account for this kind of tameness we introduce the *independence property*, a configuration much wilder than a ladder, and study the formulas and theories that avoid the independence property, called *NIP*. Since a ladder is one of the patterns that the independence property demands, every stable formula is NIP (see Proposition 4.1.2); the orders just mentioned are NIP as well, so that NIP generalizes both stability and the tameness of order. Section 4.1 treats formulas, with the conventions of Section 1.1. Section 4.2 treats bounded functions, with the conventions of Section 1.2; there NIP becomes the existence of pointwise convergent subsequences, and the independence property a copy of $\beta\mathbb{N}$.

4.1. The independence property

A ladder (see Definition 1.1.4) is a special pattern: the element a_i satisfies $\varphi(x, b_j)$ exactly for $j \geq i$. The independence property asks instead that *every* pattern be realized.

Definition 4.1.1. Let M be a model of T , and let $n \geq 1$ be given.

- (a) Tuples $b_1, \dots, b_n$ of M are *shattered by φ in M* if for every $S \subseteq \{1, \dots, n\}$ there is a tuple a_S of M such that, for all $1 \leq i \leq n$, we have

$$M \models \varphi(a_S, b_i) \iff i \in S.$$

- (b) The formula φ has the *independence property* if for every n some model of T contains n tuples shattered by φ . Otherwise the formula φ is *NIP* (“not the independence property”); NIP formulas are also called *dependent*.
- (c) The *VC dimension* of φ is the largest n such that some model of T contains n tuples shattered by φ , and ∞ if there is no largest such n .

A theory is *NIP* if every formula is NIP.

Thus the formula φ is NIP if and only if the VC dimension of φ is finite. The terminology comes from learning theory, where shattering is a property of a family of sets. For $a \in M$ we write $\varphi(a, M) = \{b \in M : M \models \varphi(a, b)\}$. If $b_1, \dots, b_n$ are shattered by φ in M , then every subset of $\{b_1, \dots, b_n\}$ is cut out by one of the sets $\varphi(a, M)$, so that the family $\{\varphi(a, M) : a \in M\}$ shatters $\{b_1, \dots, b_n\}$ in the sense of learning theory.

As for ladders, an $(n+1)$ -shattered family contains an n -shattered one, so by compactness the formula φ has the independence property if and only if some model of T contains an infinite sequence $(b_i)_{i < \omega}$ such that for every $S \subseteq \omega$ there is a_S with $\varphi(a_S, b_i)$ if and only if $i \in S$.

Proposition 4.1.2. *If φ does not have the order property, then the formula φ is NIP.*

Proof. We prove the contrapositive, by finding a ladder among the patterns that shattering provides. Suppose that $b_1, \dots, b_n$ are shattered by φ in a model M of T , witnessed by the tuples a_S . For $1 \leq i \leq n$ put $S_i = \{j : i \leq j \leq n\}$ and $a_i = a_{S_i}$. We then have $M \models \varphi(a_i, b_j)$ if and only if $j \in S_i$, that is, if and only if $i \leq j$. Thus the suffixes S_i cut out a staircase pattern, and the tuples $a_1, \dots, a_n$ and $b_1, \dots, b_n$ form an n -ladder. Since n is arbitrary, the formula φ has the order property whenever it has the independence property. $\odot$

Figure 4.1 shows the proof for $n = 3$.

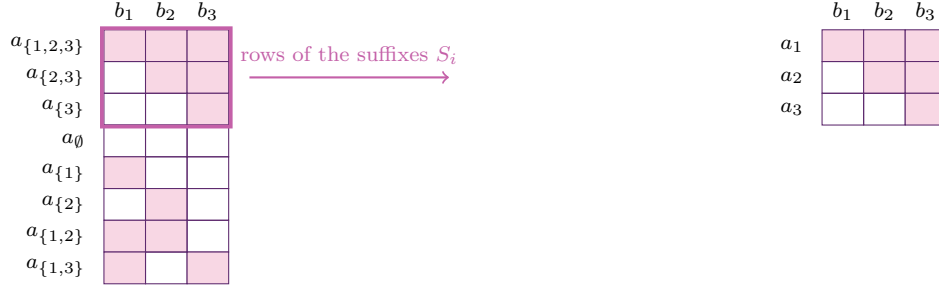


FIGURE 4.1. The proof of Proposition 4.1.2 for $n = 3$. Left: shattered tuples b_1, b_2, b_3 , with one row a_S for each $S \subseteq \{1, 2, 3\}$ and a pink cell where $\varphi(a_S, b_j)$ holds. Right: the three rows indexed by the suffixes $S_1 = \{1, 2, 3\}$, $S_2 = \{2, 3\}$ and $S_3 = \{3\}$ (framed) form the staircase of a 3-ladder (compare Figure 1.1). A ladder uses only n of the 2^n rows, which is why the order property does not give the independence property (see Example 4.1.3).

The converse of Proposition 4.1.2 fails, and the first example of the chapter, the order of the rationals, shows why NIP is the right notion for ordered structures.

Example 4.1.3. Let T be the theory of $(\mathbb{Q}, <)$ and let $\varphi(x, y)$ be $x < y$. Then the elements $a_i = i$ and $b_j = j + \frac{1}{2}$ satisfy $\varphi(a_i, b_j)$ if and only if $i \leq j$, so the formula φ has the order property and is unstable. However, the formula φ is NIP, with VC dimension 1. Indeed, let b_1, b_2 be elements of a model, say $b_1 \leq b_2$. Shattering requires some a with $a < b_1$ and $a \not< b_2$, that is, some a with $b_2 \leq a < b_1$, which is impossible. Hence no two elements are shattered. A single element b is shattered, by any $a < b$ and any $a \geq b$.

The random graph lies at the other extreme: its edge relation realizes every pattern on every finite set of vertices.

Example 4.1.4. Let T be the theory of the random graph, whose models are the graphs in which, for all disjoint finite sets U, V of vertices, some vertex outside $U \cup V$ is adjacent to every vertex of U and to no vertex of V . Let $\varphi(x, y)$ be the edge relation $E(x, y)$. Any n distinct vertices $b_1, \dots, b_n$ are shattered: for $S \subseteq \{1, \dots, n\}$ apply the extension property to $U = \{b_i : i \in S\}$ and $V = \{b_i : i \notin S\}$. Thus the relation E has the independence property.

The independence property of $\varphi(x, y)$ refers to shattering tuples on the y -side. In Proposition 4.1.5 below we show that the choice of side does not matter, at the price of an exponential

loss in the number of shattered tuples. We write $\varphi^{\text{opp}}(y, x)$ for $\varphi(x, y)$ with the roles of the variables exchanged.

Proposition 4.1.5. *The formula φ is NIP if and only if φ^{opp} is NIP. More precisely, if some model contains 2^n tuples shattered by φ , then that model contains n tuples shattered by φ^{opp} .*

Proof. The idea is to index the shattered tuples by sets, so that membership itself becomes the pattern, and then to read the pattern from the other side. Let M be a model of T containing 2^n tuples shattered by φ , index them as b_T with $T \subseteq \{1, \dots, n\}$, and let the tuples a_S with $S \subseteq \mathcal{P}(\{1, \dots, n\})$ witness that they are shattered by φ . For $1 \leq i \leq n$ put $S_i = \{T : i \in T\}$ and $a_i = a_{S_i}$. We then have $M \models \varphi(a_i, b_T)$ if and only if $T \in S_i$, that is, if and only if $i \in T$. Thus for every $T \subseteq \{1, \dots, n\}$ the tuple b_T satisfies $\varphi^{\text{opp}}(b_T, a_i)$ exactly for $i \in T$, and the tuples $a_1, \dots, a_n$ are shattered by φ^{opp} . Applying this in both directions gives the equivalence. ☺



Remark 4.1.6. Examples 4.1.3 and 4.1.4 are unstable for different reasons, and a theorem of Shelah says that there are no others. A formula $\psi(x, y)$ has the *strict order property* if for every n some model M of T contains tuples $b_1, \dots, b_n$ with $\psi(M, b_1) \subsetneq \psi(M, b_2) \subsetneq \dots \subsetneq \psi(M, b_n)$, where $\psi(M, b) = \{a \in M : M \models \psi(a, b)\}$. The formula $x < y$ of Example 4.1.3 is the basic case. Shelah's theorem states that a theory is unstable if and only if some formula has the independence property or some formula has the strict order property [She90, Chapter II]. Thus an unstable theory either encodes arbitrary subsets, as the random graph does, or encodes an infinite order; NIP theories are those in which only the second reason can occur.

4.2. The real-valued case

As in Section 1.2, formulas give way to bounded functions on arbitrary sets, and the goal is an analytic form of NIP: in Theorem 4.2.4 below we show that a function is NIP exactly when every sequence of its evaluation maps has a pointwise convergent subsequence. The definition must first be adapted. On arbitrary sets compactness is not available, so shattering is required only for finite patterns, with a uniform gap between the values on the two sides.

Definition 4.2.1. Let A, B be sets and $\varphi: A \times B \rightarrow \mathbb{R}$ a bounded function. A sequence $(b_n)_{n < \omega}$ in B is *independent* for φ if there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $a \in A$ with

$$\varphi(a, b_n) \geq s \text{ for } n \in P \quad \text{and} \quad \varphi(a, b_n) \leq r \text{ for } n \in Q.$$

The function φ has the *independence property* if some sequence in B is independent for φ , and is *NIP* otherwise.

For a $\{0, 1\}$ -valued function, any $0 < r < s < 1$ may be used, and a sequence is independent exactly when each finite part of the sequence is shattered in the sense of Definition 4.1.1. By compactness, a formula $\varphi(x, y)$ is NIP in the sense of Definition 4.1.1 if and only if the corresponding function on $M^{|x|} \times M^{|y|}$ is NIP for every model M of T .

As in the $\{0, 1\}$ -valued case (see Proposition 4.1.2), stability implies NIP. The proof there turned a shattered set into a ladder, a staircase being one of the patterns that shattering provides. Here the ladder appears among the types: ultrafilter limits of witnesses give types p_m that form a ladder against the b_n , and the separate continuity of the extension $\hat{\varphi}$ of Proposition 1.2.8 rules the ladder out.

Proposition 4.2.2. *Every stable function $\varphi: A \times B \rightarrow [0, 1]$ is NIP.*

Proof. Let $\hat{\varphi}: \text{LS}(\varphi) \times \text{RS}(\varphi) \rightarrow [0, 1]$ be the separately continuous function of Proposition 1.2.8. Taking a constant family in (1.2) gives $\hat{\varphi}(p, \text{rtp}_\varphi(b)) = p(b)$ for all $p \in \text{LS}(\varphi)$ and $b \in B$.

Suppose, towards a contradiction, that (b_n) is independent for φ , with reals $r < s$. For $m < N$ choose $a_{m,N} \in A$ with

$$\varphi(a_{m,N}, b_n) \geq s \text{ for } m \leq n < N \quad \text{and} \quad \varphi(a_{m,N}, b_n) \leq r \text{ for } n < m,$$

and let $a_{m,N}$ be arbitrary for $N \leq m$. Fix a nonprincipal ultrafilter $\mathcal{U}$ on ω and put

$$p_m = \lim_{N, \mathcal{U}} \text{ltp}_\varphi(a_{m,N}) \in \text{LS}(\varphi).$$

For fixed n the displayed inequalities hold for all $N > \max(m, n)$, so we have

$$(4.1) \quad p_m(b_n) \geq s \text{ if } n \geq m, \quad p_m(b_n) \leq r \text{ if } n < m.$$

Now put $q = \lim_{n, \mathcal{U}} \text{rtp}_\varphi(b_n) \in \text{RS}(\varphi)$ and $p = \lim_{m, \mathcal{U}} p_m \in \text{LS}(\varphi)$. We compute $\hat{\varphi}(p, q)$ in two ways.

First, by continuity of $\hat{\varphi}(p_m, \cdot)$ and (4.1), we have $\hat{\varphi}(p_m, q) = \lim_{n, \mathcal{U}} p_m(b_n) \geq s$ for every m , since $p_m(b_n) \geq s$ for all $n \geq m$. By continuity of $\hat{\varphi}(\cdot, q)$, we obtain $\hat{\varphi}(p, q) = \lim_{m, \mathcal{U}} \hat{\varphi}(p_m, q) \geq s$.

Second, limits in $\text{LS}(\varphi)$ are computed coordinatewise. Since $p_m(b_n) \leq r$ for all $m > n$, we have

$$p(b_n) = \lim_{m, \mathcal{U}} p_m(b_n) \leq r \text{ for every } n.$$

By continuity of $\hat{\varphi}(p, \cdot)$, we obtain

$$\hat{\varphi}(p, q) = \lim_{n, \mathcal{U}} \hat{\varphi}(p, \text{rtp}_\varphi(b_n)) = \lim_{n, \mathcal{U}} p(b_n) \leq r.$$

Hence we have $s \leq \hat{\varphi}(p, q) \leq r$, contradicting $r < s$. ☹

Stability was characterized in Section 1.2 by the exchange of iterated limits. NIP, the weaker condition by Proposition 4.2.2, is characterized by the evaluation maps $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$, given by $e_b(p) = p(b)$ for $b \in B$ as in Subsection 1.2.4: every sequence of them has a pointwise convergent subsequence. The analytic input is Rosenthal's dichotomy: a bounded sequence of functions has either a pointwise convergent subsequence or a subsequence that behaves like an independent sequence.

Fact 4.2.3 (Rosenthal). *Let X be a set and (f_n) a uniformly bounded sequence of real-valued functions on X . Then the sequence (f_n) has a subsequence that converges pointwise on X , or a subsequence (f_{n_k}) for which there are reals $r < s$ such that for all disjoint finite sets $P, Q \subseteq \omega$ there is $x \in X$ with $f_{n_k}(x) \geq s$ for $k \in P$ and $f_{n_k}(x) \leq r$ for $k \in Q$.*

Theorem 4.2.4. *For $\varphi: A \times B \rightarrow [0, 1]$, the following are equivalent:*

- (1) *The function φ is NIP.*
- (2) *For every sequence (b_n) in B , the sequence of functions (e_{b_n}) has a subsequence that converges pointwise on $\text{LS}(\varphi)$.*

Proof. (2) $\Rightarrow$ (1). Suppose that (b_n) is independent for φ , with reals $r < s$, and let (b_{n_k}) be a subsequence. For each $K < \omega$ choose $a_K \in A$ with $\varphi(a_K, b_{n_k}) \geq s$ for even $k < K$ and $\varphi(a_K, b_{n_k}) \leq r$ for odd $k < K$. Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω , and put $p = \lim_{K, \mathcal{U}} \text{ltp}_\varphi(a_K)$, a point of $\text{LS}(\varphi)$. For each fixed k the inequality for k holds for all $K > k$, so the value $e_{b_{n_k}}(p) = p(b_{n_k})$ is at least s for even k and at most r for odd k . Hence

the sequence $(e_{b_{n_k}}(p))_k$ does not converge. Since the subsequence was arbitrary, condition (2) fails.

(1) $\Rightarrow$ (2). Suppose that condition (2) fails for (b_n) . By Fact 4.2.3 applied to $X = \text{LS}(\varphi)$ and $f_n = e_{b_n}$, there are a subsequence (b_{n_k}) and reals $r < s$ such that for all disjoint finite $P, Q \subseteq \omega$ there is $p \in \text{LS}(\varphi)$ with $p(b_{n_k}) \geq s$ for $k \in P$ and $p(b_{n_k}) \leq r$ for $k \in Q$. Fix r' and s' with $r < r' < s' < s$. Since p is a pointwise limit of left types $\text{ltp}_\varphi(a)$ and only finitely many coordinates are involved, some $a \in A$ satisfies $\varphi(a, b_{n_k}) > s'$ for $k \in P$ and $\varphi(a, b_{n_k}) < r'$ for $k \in Q$. Thus the sequence (b_{n_k}) is independent for φ , with $r' < s'$, and the function φ is not NIP. ☺

The failure of condition (2) has a striking topological form: an independent sequence produces a copy of $\beta\mathbb{N}$.

Proposition 4.2.5. *If (b_n) is independent for φ , then the map $\mathcal{U} \mapsto \lim_{n, \mathcal{U}} e_{b_n}$ is a homeomorphism from $\beta\mathbb{N}$ onto a closed subset of the pointwise closure of $\{e_b : b \in B\}$ in $[0, 1]^{\text{LS}(\varphi)}$.*

Proof. The map $n \mapsto e_{b_n}$ from $\mathbb{N}$ into the compact space $[0, 1]^{\text{LS}(\varphi)}$ extends to the continuous map $\mathcal{U} \mapsto \lim_{n, \mathcal{U}} e_{b_n}$ on $\beta\mathbb{N}$, whose image lies in the pointwise closure of $\{e_b : b \in B\}$. Since $\beta\mathbb{N}$ is compact and $[0, 1]^{\text{LS}(\varphi)}$ is Hausdorff, it remains to be shown that the map is injective. Let $r < s$ witness the independence of (b_n) , let $\mathcal{U} \neq \mathcal{V}$ be ultrafilters on $\mathbb{N}$, and choose $X \in \mathcal{U}$ with $\mathbb{N} \setminus X \in \mathcal{V}$. For each finite $F \subseteq \mathbb{N}$ there is $a_F \in A$ with $\varphi(a_F, b_n) \geq s$ for $n \in F \cap X$ and $\varphi(a_F, b_n) \leq r$ for $n \in F \setminus X$. Let $\mathcal{D}$ be an ultrafilter on the finite subsets of $\mathbb{N}$ that contains $\{G : G \supseteq F\}$ for every finite F , and put $p = \lim_{F, \mathcal{D}} \text{ltp}_\varphi(a_F) \in \text{LS}(\varphi)$. Then we have $p(b_n) \geq s$ for $n \in X$ and $p(b_n) \leq r$ for $n \notin X$. Hence the limit of $e_{b_n}(p)$ along $\mathcal{U}$ is at least s , the limit along $\mathcal{V}$ is at most r , and the two limits of (e_{b_n}) differ at p . ☺

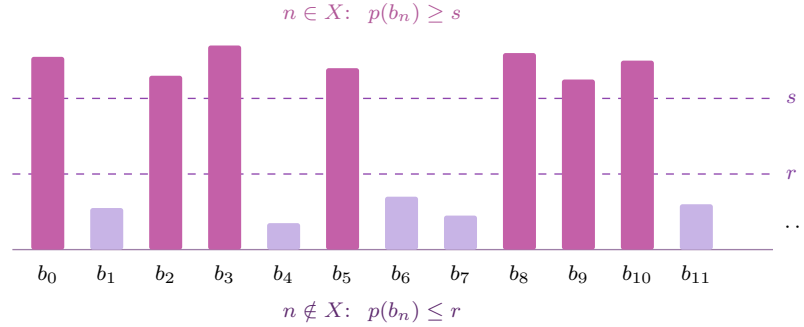


FIGURE 4.2. The proof of Proposition 4.2.5. Independence of (b_n) lets a single point p of $\text{LS}(\varphi)$ cut out any prescribed set $X \subseteq \mathbb{N}$ (pink bars): the values $p(b_n)$ lie above s on X and below r off X . Two distinct ultrafilters $\mathcal{U} \ni X$ and $\mathcal{V} \ni \mathbb{N} \setminus X$ therefore give limits of (e_{b_n}) that differ at p , and $\beta\mathbb{N}$ embeds in the closure of the instances.

Thus the instances of φ , viewed as continuous functions on $\text{LS}(\varphi)$, face a dichotomy. If φ is NIP, every sequence of instances has a pointwise convergent subsequence (see Theorem 4.2.4); if φ has the independence property, the closure of the instances contains $\beta\mathbb{N}$, a compact space in which no sequence of distinct points converges. The dichotomy is the topological face of Rosenthal's ℓ_1 dichotomy. In the theory of Bourgain, Fremlin and Talagrand [BFT78], the same dichotomy separates the *Rosenthal compacta*, the compact spaces of Baire class 1 functions on Polish spaces, from compact spaces of functions that contain

$\beta\mathbb{N}$, and Simon [Sim15b] develops the NIP side of this picture for spaces of types. Stable functions sit at the rigid end of the spectrum: their type spaces are Eberlein compacta (see Corollary A.1.8 below).



Remark 4.2.6. Proposition 4.2.2 can also be derived, with heavier machinery, from Theorem 4.2.4. By Grothendieck’s double limit criterion, in the form of Proposition A.1.7, stability of φ makes $\{e_b : b \in B\}$ relatively weakly compact in $C(\text{LS}(\varphi))$. By Corollary A.1.6, which rests on the Eberlein–Šmulian theorem (see Fact A.1.5 below), every sequence (e_{b_n}) then has a pointwise convergent subsequence. Thus condition (2) of Theorem 4.2.4 holds, and the function φ is NIP by the implication (2) $\Rightarrow$ (1), which does not use Rosenthal’s dichotomy. The direct proof of Proposition 4.2.2 avoids both Grothendieck’s criterion and the Eberlein–Šmulian theorem.



Remark 4.2.7. Theorem 1.2.9 represents stable functions by the pairing of a reflexive Banach space E with E^* . Khanaki [Kha15] proved that φ is stable, NIP, or NSOP (that is, without the strict order property of Remark 4.1.6) according as a Banach space naturally associated with φ is reflexive, contains no copy of ℓ_1 , or is weakly sequentially complete (every weakly Cauchy sequence converges weakly).

4.3. Historical remarks



Section 4.1. The independence property was introduced by Shelah [She71], in the same paper as the order property. His theorem that an unstable theory has the independence property or the strict order property is in [She90, Chapter II]. The combinatorial notion of shattering, and the VC dimension, were introduced independently in probability and learning theory by Vapnik and Chervonenkis [VC71]; the polynomial bound on the number of traces (the subsets cut out on an n -element set) of a family of finite VC dimension was found independently by Sauer [Sau72] and by Shelah [She72]. Laskowski [Las92] observed that a formula is NIP exactly when the family of sets defined by the formula has finite VC dimension, connecting model theory with learning theory.



Section 4.2. Rosenthal’s dichotomy comes from his characterization of Banach spaces containing ℓ_1 [Ros74], and the systematic study of pointwise compact sets of Baire class 1 functions is due to Rosenthal [Ros77] and to Bourgain, Fremlin and Talagrand [BFT78]. Their work entered model theory much later. The development is summarized in the following timeline.



1971 • **Shelah; Vapnik–Chervonenkis.** Shelah [She71] introduces the independence property. Independently, Vapnik and Chervonenkis [VC71] introduce shattering and what is now called VC dimension, in the study of uniform convergence of empirical frequencies.

1972 • **Sauer; Shelah.** Sauer [Sau72] and Shelah [She72] independently prove the polynomial bound on the number of traces of a set system of finite VC dimension.

- 1974–1978

Rosenthal; Bourgain–Fremlin–Talagrand. Rosenthal proves the ℓ_1 dichotomy [Ros74] and studies pointwise compact sets of Baire class 1 functions [Ros77]. Bourgain, Fremlin and Talagrand [BFT78] develop the theory of these “Rosenthal compacta”.
- 1981

• **Poizat.** Poizat [Poi81] characterizes countable NIP theories by a bound on the number of types, and develops the theory of unstable theories without the independence property.
- 1986

• **Pillay–Steinhorn.** Pillay and Steinhorn [PS86] develop the theory of o-minimal structures, following van den Dries. These structures are NIP and are the main source of unstable NIP theories.
- 1992

• **Laskowski.** Laskowski [Las92] identifies NIP formulas with formulas defining families of finite VC dimension.
- 2008–2011

Hrushovski–Peterzil–Pillay; Hrushovski–Pillay. The study of invariant and generically stable measures in NIP theories [HPP08, HP11], motivated by Pillay’s conjecture on definable groups in o-minimal structures, turns NIP into a central topic of model theory.
- 2009

• **Ben Yaacov.** Ben Yaacov [Ben09] extends VC classes to $[0, 1]$ -valued functions, the continuous analogue of NIP used in Section 4.2.
- 2015

• **Simon.** Simon [Sim15b] applies the theory of Rosenthal compacta to spaces of φ -types of NIP formulas, and publishes the monograph [Sim15a] on NIP theories.
- 2015–2020

• **Khanaki.** Khanaki [Kha20, Kha15] relates stability, NIP and NSOP to three properties of Banach spaces (see Remark 4.2.7).

Chapter 5

From stability to simplicity

A stable formula has definable types, and a definition extends its type canonically to every larger set of parameters (see Theorem 1.1.6 and the remarks following Definition 1.1.3). A stable theory therefore has, over a model, a preferred way of extending a type without adding information, and in Chapter 6 we show that this way is unique. The random graph of Example 4.1.4 is not stable, since its edge relation has the independence property, and yet it carries an evident notion of independence. Over a set C , a vertex a is independent from a set B when a is not one of the vertices of B outside C ; once this holds, the extension axioms leave a free to be joined to the vertices of B in any way whatever. Uniqueness is lost precisely here: a new vertex may be joined to a given vertex b or not, and neither choice tells us more about b than the other. What survives is amalgamation. Suppose that c_1 and c_2 have the same type over C , that c_1 is independent from b_1 and c_2 from b_2 , and that b_1 and b_2 are independent from each other. Then a single vertex realizes the type of c_1 over $C \cup \{b_1\}$ and the type of c_2 over $C \cup \{b_2\}$ at once. This is the *independence theorem*. It holds in the *simple* theories of Shelah, and, together with the other properties of independence, it characterizes them. The theme of this chapter fits in one line: from stability to simplicity, uniqueness gives way to amalgamation.

Simplicity has a combinatorial definition in the spirit of the order property and of the independence property: a theory is simple if no formula has the *tree property* (see Definition 5.2.7), a configuration assembled from *dividing*, the way in which a formula can say something essentially new about a set. Section 5.1 fixes conventions, and Section 5.2 defines both notions. Section 5.3 places simplicity in the classification: every stable theory is simple, a theory is stable if and only if it is simple and NIP (see Theorem 5.3.4), and in a simple theory every type is nondividing over a small part of its domain. Section 5.4 computes dividing in the random graph from the extension axioms alone, proves that the random graph is simple, and proves the independence theorem there. Section 5.5 asks what these proofs used, and finds a single property of the finite graphs, *disjoint 3-amalgamation*. Every Fraïssé limit with this property is simple (see Theorem 5.5.9), and two classical Fraïssé limits without it, the rational order and the Henson graph, are not. Section 5.6 passes to real-valued functions, as Chapters 1 and 4 did, and shows by counting approximate definitions that stable functions have no tree property.

5.1. Conventions

This section fixes the setting for the rest of the notes. From now on T is a *complete* theory in the language L , and $\kappa = |T| = |L| + \aleph_0$. We work inside a monster model $\mathbb{M}$ of T : a $\bar{\kappa}$ -saturated and strongly $\bar{\kappa}$ -homogeneous model for some very large cardinal $\bar{\kappa}$ (larger than 2^{2^κ} , say). A set is *small* if of cardinality $< \bar{\kappa}$. Letters A, B, C denote small subsets of $\mathbb{M}$; tuples may be infinite, but are always small. We freely identify a set with an enumerating

tuple. A *model* is a small elementary substructure $M \preceq \mathbb{M}$. Every model of T of size $< \bar{\kappa}$ embeds elementarily in $\mathbb{M}$, and every type over a small set is realized in $\mathbb{M}$.

We write $\text{tp}(a/A)$ for the complete type of a over A , $S_x(A)$ for the set of complete types over A in the variables x , and $a \equiv_A a'$ for $\text{tp}(a/A) = \text{tp}(a'/A)$. We write MB for $M \cup B$. By strong homogeneity, we have $a \equiv_A a'$ if and only if $\sigma(a) = a'$ for some $\sigma \in \text{Aut}(\mathbb{M}/A)$.

Until now, stability has been a property of single formulas. We record the corresponding notion for the theory T , together with a sufficient condition by counting types, which Section 5.5 uses to recognize stable Fraïssé limits.

Definition 5.1.1.

- (a) The theory T is *stable* if no L -formula $\varphi(x, y)$ has the order property (see Definition 1.1.4).
- (b) For an infinite cardinal λ , the theory T is λ -*stable* if $|S_x(A)| \leq \lambda$ whenever $|A| \leq \lambda$ and x is a finite tuple of variables.

Lemma 5.1.2. *If T is λ -stable for some $\lambda \geq \kappa$, then T is stable.*

Proof. Suppose that $\varphi(x, y)$ has the order property. By Proposition 2.3.1, the formula φ encodes binary trees of arbitrarily large finite height. By Proposition 2.4.2 there is a model N of T with $|N| \leq \lambda$ and $|S_\varphi(N)| > \lambda$. Since T is complete, we may assume $N \preceq \mathbb{M}$. Every φ -type over N is consistent with the elementary diagram of N , so each φ -type over N extends to a complete type in $S_x(N)$, and distinct φ -types have distinct extensions. Hence $|S_x(N)| > \lambda \geq |N|$, and T is not λ -stable. $\odot$

5.2. Dividing and the tree property

The goal of this section is to define dividing and the tree property, and with them the simple theories. A small set of formulas with parameters in $\mathbb{M}$ is *consistent* if it is realized in $\mathbb{M}$. We often identify a tuple with the set of its coordinates; thus, for tuples a, b and a set C , the inclusion $a \cap b \subseteq C$ says that every element that is a coordinate of both a and b belongs to C .

A formula $\varphi(x, b)$ may say about x something that a set C already knows, or something new. Two examples show the difference. In a model of the theory of $(\mathbb{Q}, <)$, the formula $b_1 < x < b_2$ is one of infinitely many formulas $b_1^i < x < b_2^i$, with $b_2^i < b_1^{i+1}$, which look alike over $\emptyset$ and are pairwise contradictory. In the random graph, the formula $E(x, b)$, where b is a vertex, looks like $E(x, b')$ for every other vertex b' , but any finite number of these formulas hold together, because the extension axioms provide a common neighbour. The first formula *divides* over $\emptyset$, and the second does not. Indiscernible sequences make “look alike” precise.

Definition 5.2.1. Let C be a small set. A sequence $(b_i)_{i < \omega}$ of tuples of the same length is *indiscernible over C* if for every n and all $i_1 < \dots < i_n$ and $j_1 < \dots < j_n$ in ω , we have

$$(b_{i_1}, \dots, b_{i_n}) \equiv_C (b_{j_1}, \dots, b_{j_n}).$$

Every infinite subsequence of a sequence indiscernible over C is indiscernible over C ; a sequence indiscernible over C is indiscernible over every subset of C ; and, taking $n = 1$, all its terms have the same type over C .

Indiscernible sequences are plentiful. Every sequence leaves a trace of the formulas that hold of all its increasing tuples, and Ramsey’s theorem, together with compactness, produces an indiscernible sequence with the same trace. For a sequence $(b_i)_{i < \omega}$ and a small set C , the *Ehrenfeucht–Mostowski type* of (b_i) over C is the set of $L(C)$ -formulas $\psi(y_1, \dots, y_n)$ such that $\mathbb{M} \models \psi(b_{i_1}, \dots, b_{i_n})$ for all $i_1 < \dots < i_n$ in ω .

Lemma 5.2.2 (Extraction of indiscernible sequences). *Let C be a small set, and let $(b_i)_{i < \omega}$ be tuples of the same length. Then there is a sequence $(b'_i)_{i < \omega}$, indiscernible over C , that realizes the Ehrenfeucht–Mostowski type of (b_i) over C : for every formula $\psi(y_1, \dots, y_n)$ of that type and all $j_1 < \dots < j_n$, we have $\mathbb{M} \models \psi(b'_{j_1}, \dots, b'_{j_n})$. In particular, if all the b_i have the same type over C , then $b'_j \equiv_C b_0$ for every j .*

Proof. Let Σ be the set of $L(C)$ -formulas in variables y_i ($i < \omega$) consisting of

- (a) $\psi(y_{j_1}, \dots, y_{j_n})$, for every ψ in the Ehrenfeucht–Mostowski type of (b_i) over C and all $j_1 < \dots < j_n$, and
- (b) $\theta(y_{i_1}, \dots, y_{i_n}) \leftrightarrow \theta(y_{j_1}, \dots, y_{j_n})$, for every $L(C)$ -formula θ and all $i_1 < \dots < i_n$ and $j_1 < \dots < j_n$.

A realization of Σ is a sequence as required: (b) says that it is indiscernible over C , and (a) that it realizes the Ehrenfeucht–Mostowski type. Since Σ has at most $|C| + \aleph_0$ parameters, by saturation it suffices to show that every finite $\Sigma_0 \subseteq \Sigma$ is satisfiable.

Let $\theta_1, \dots, \theta_m$ be the formulas that occur in the formulas of type (b) in Σ_0 , and let n be the largest number of variables among them (if there are none, take $H = \omega$ below). Colour each set $\{i_1 < \dots < i_n\} \subseteq \omega$ by the sequence of truth values of $\theta_l(b_{i_1}, \dots, b_{i_{n_l}})$, $l = 1, \dots, m$, where $n_l \leq n$ is the number of variables of θ_l . There are at most 2^m colours, so Ramsey's theorem gives an infinite set $H \subseteq \omega$ all of whose n -element subsets have the same colour. Let $h_0 < h_1 < \dots$ enumerate H , and assign b_{h_i} to y_i . Every increasing n_l -tuple from H is the beginning of an increasing n -tuple from H , because H is infinite, so the truth value of θ_l on increasing n_l -tuples from H does not depend on the tuple; hence the formulas of type (b) in Σ_0 hold. The formulas of type (a) hold because they hold of all increasing tuples of the original sequence. For the last assertion, every formula of $\text{tp}(b_0/C)$, in one variable y_1 , belongs to the Ehrenfeucht–Mostowski type. ☺

In the Rado graph and in Fraïssé limits, quantifier elimination gives a sharper conclusion over finite sets: an infinite subsequence of the given sequence is itself indiscernible (see Lemmas 5.4.4 and 5.5.7).

Definition 5.2.3. Let C be a small set, let $\varphi(x, y)$ be an L -formula, and let b be a tuple of the length of y .

- (a) The formula $\varphi(x, b)$ *divides over C* if there is a sequence $(b_i)_{i < \omega}$, indiscernible over C , with $b_0 = b$, such that the set $\{\varphi(x, b_i) : i < \omega\}$ is inconsistent.
- (b) For $k \geq 2$, a set of formulas is *k -inconsistent* if each of its subsets with k elements is inconsistent, and $\varphi(x, b)$ *k -divides over C* if the sequence in (a) can be chosen so that $\{\varphi(x, b_i) : i < \omega\}$ is k -inconsistent.
- (c) Let $B \supseteq C$ be small. A type $p \in S_x(B)$ *divides over C* if some formula of p divides over C .

Every $L(B)$ -formula has the form $\varphi(x, b)$ with $\varphi \in L$ and b a finite tuple from B , so clause (c) makes sense. The next lemma shows that dividing is k -dividing for a suitable k , so that the two notions of (a) and (b) agree.

Lemma 5.2.4. *Let $(b_i)_{i < \omega}$ be indiscernible over C , and suppose that $\{\varphi(x, b_i) : i < \omega\}$ is inconsistent. Then there is $k \geq 2$ such that $\{\varphi(x, b_i) : i < \omega\}$ is k -inconsistent. In particular, a formula divides over C if and only if it k -divides over C for some $k \geq 2$.*

Proof. By compactness and saturation, some finite subset $\{\varphi(x, b_{i_1}), \dots, \varphi(x, b_{i_k})\}$, with $i_1 < \dots < i_k$, is inconsistent; adding formulas keeps it inconsistent, so we may assume $k \geq 2$.

Thus $\mathbb{M} \models \neg \exists x \bigwedge_{l=1}^k \varphi(x, b_{i_l})$. For $j_1 < \dots < j_k$, indiscernibility gives $(b_{j_1}, \dots, b_{j_k}) \equiv_C (b_{i_1}, \dots, b_{i_k})$, and hence $\mathbb{M} \models \neg \exists x \bigwedge_{l=1}^k \varphi(x, b_{j_l})$. $\odot$



Remark 5.2.5. If $C \subseteq C'$ and $\varphi(x, b)$ k -divides over C' , then $\varphi(x, b)$ k -divides over C , because a sequence indiscernible over C' is indiscernible over C . Consequently, if $C \subseteq C' \subseteq B$ and $p \in S_x(B)$ divides over C' , then p divides over C . The terms of the sequence in Definition 5.2.3 all have the type of b over C , so dividing is a strengthening of Shelah's original condition, which asks only for infinitely many realizations of $\text{tp}(b/C)$ whose formulas are k -inconsistent; in fact, the two conditions are equivalent (see [TZ12, Chapter 7]), but we shall not need this.

Example 5.2.6. Let T be the theory of $(\mathbb{Q}, <)$, let $b_1 < b_2$ be elements of $\mathbb{M}$, and let $\theta(x; y_1, y_2)$ be the formula $y_1 < x < y_2$. Choose pairs $b^i = (b_1^i, b_2^i)$ with $b^0 = (b_1, b_2)$, with $b_1^i < b_2^i$, and with $b_2^i < b_1^{i+1}$ for every i . By quantifier elimination for dense linear orders without endpoints (see Fact 5.5.2(b) and Example 5.5.3(2) below, or [Hod93, Theorem 7.4.1]), the type over $\emptyset$ of a tuple is determined by the order between its coordinates, and any two increasing n -tuples of the sequence (b^i) are ordered alike. Hence the sequence (b^i) is indiscernible over $\emptyset$. The intervals (b_1^i, b_2^i) are pairwise disjoint, so $\{\theta(x, b^i) : i < \omega\}$ is 2-inconsistent, and $\theta(x, b_1, b_2)$ 2-divides over $\emptyset$.

In Example 5.2.6 we can go on: inside the interval (b_1, b_2) there are infinitely many disjoint subintervals, inside each of these infinitely many more, and so on. Choosing one interval at each stage, we obtain a consistent descending sequence of formulas; at each stage, the available choices are pairwise inconsistent. The tree property abstracts this picture. We use the notation of Chapter 2 for sequences, with ω in place of 2. Thus the set $\omega^{<\omega}$ consists of the finite sequences of natural numbers, ω^ω is the set of infinite ones, $\sigma \frown \langle i \rangle$ is the extension of σ by the entry i , and $\eta \upharpoonright n$ is the initial segment of η of length n .

Definition 5.2.7. Let $\varphi(x, y)$ be an L -formula, and let $k \geq 2$ be given. The formula φ has the *k -tree property* if there are tuples b_σ , for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, such that the following hold:

- (a) for every $\eta \in \omega^\omega$, the set $\{\varphi(x, b_{\eta \upharpoonright n}) : 1 \leq n < \omega\}$ is consistent;
- (b) for every $\sigma \in \omega^{<\omega}$, the set $\{\varphi(x, b_{\sigma \frown \langle i \rangle}) : i < \omega\}$ is k -inconsistent.

The formula φ has the *tree property* if it has the k -tree property for some $k \geq 2$. The theory T is *simple* if no L -formula has the tree property.

Along each branch the formulas are consistent; among the immediate successors of a node, called *siblings*, they are k -inconsistent. Each step up a branch therefore chooses one of infinitely many incompatible options, and a simple theory is one that cannot accommodate an ω -sequence of such choices made by a single formula. The trees of Chapter 2 were binary, with parameters at the nodes and elements at the leaves; the trees here branch infinitely, carry parameters at every node, and have no leaves. Figure 5.1 shows the configuration.

By compactness, trees of finite height suffice.



Remark 5.2.8. Let $\varphi(x, y)$ be an L -formula, and let $k \geq 2$ be given. Suppose that for every $n \geq 1$ there are tuples b_σ , for $\sigma \in \omega^{\leq n}$ with $\sigma \neq \emptyset$, such that condition (a) of Definition 5.2.7 holds for the branches of length n (the set $\{\varphi(x, b_{\eta \upharpoonright m}) : 1 \leq m \leq n\}$ is consistent for every $\eta \in \omega^n$) and condition (b) holds for every $\sigma \in \omega^{<n}$. Then φ has the k -tree property. Indeed, by compactness a branch $\{\varphi(x, b_{\eta \upharpoonright m}) : 1 \leq m < \omega\}$ is consistent if and only if each of its finite initial segments is. Hence the k -tree property for a family of variables (y_σ) is expressed

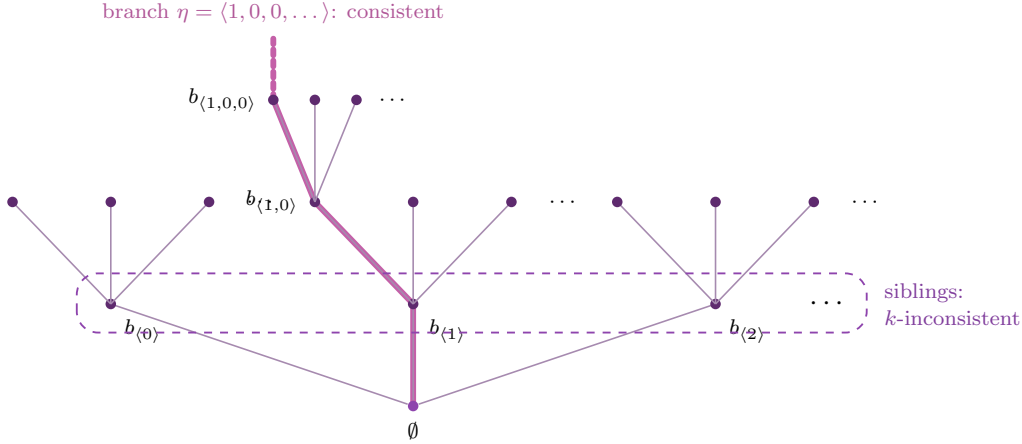


FIGURE 5.1. The k -tree property (see Definition 5.2.7). The tree grows upward; the node σ carries the parameter b_σ . Along every branch (pink) the formulas $\varphi(x, b_{\eta|n})$ have a common solution, while the formulas attached to the siblings of any node (framed for the successors of the root) are k -inconsistent: no k of them hold together.

by the set of formulas

$$\exists x \bigwedge_{m=1}^n \varphi(x, y_{\eta|m}) \quad (\eta \in \omega^\omega, n \geq 1), \quad \neg \exists x \bigwedge_{l=1}^k \varphi(x, y_{\sigma \smallfrown \langle i_l \rangle}) \quad (\sigma \in \omega^{<\omega}, i_1 < \dots < i_k).$$

A finite subset of this set mentions only nodes of length at most some n , and the given family of height n satisfies it. By compactness and saturation, the whole set is realized in $\mathbb{M}$.

5.3. Simplicity between stability and NIP

The goal of this section is to place simplicity in the classification. We show that every stable theory is simple, that a theory with the strict order property is not, and that the simple NIP theories are exactly the stable ones. We close with the first structural consequence of simplicity: every type is nondividing over a subset of its domain of size at most κ .

Proposition 5.3.1. *If T is stable, then T is simple.*

Proof. The idea is to count branches. A tree with λ successors at every node has $\lambda^{\aleph_0}$ branches, whereas a single φ -type can follow at most $2^{\aleph_0}$ of them, because at each node it accepts fewer than k successors; a stable formula has too few types to follow all the branches.

Suppose that $\varphi(x, y)$ has the k -tree property. Put $\lambda_0 = 2^\kappa$, $\lambda_{n+1} = 2^{\lambda_n}$, and $\lambda = \sup_n \lambda_n$. Then $\lambda \geq 2^\kappa$, the set $\lambda^{<\omega}$ has cardinality λ , and λ has cofinality ω , so that $\lambda^{\aleph_0} > \lambda$ by König's theorem.

© **Step 1: a wide tree.** There are tuples b_σ , for $\sigma \in \lambda^{<\omega}$ with $\sigma \neq \emptyset$, satisfying the two conditions of Definition 5.2.7 with λ in place of ω . Indeed, the conditions are expressed by formulas as in Remark 5.2.8, and a finite set of them mentions finitely many nodes. Let $J \subseteq \lambda$ be the finite set of entries of these nodes, and let $e: J \rightarrow \omega$ be injective. The map sending σ to the sequence $e \circ \sigma$ preserves initial segments and the relation of being siblings, so the tuples $b_{e \circ \sigma}$ of the given tree satisfy the finite set. By compactness and saturation, the wide

tree exists in $\mathbb{M}$. By the downward Löwenheim–Skolem theorem, there is a model $M \preceq \mathbb{M}$ with $|M| = \lambda$ that contains all the tuples b_σ .

⊙ **Step 2: few types.** By Theorem 1.1.6, every $p \in S_\varphi(M)$ has a φ -definition that is a positive Boolean combination of instances $\varphi(a, y)$ with $a \in M$. The type p is determined by the set D_p , hence by its definition, and there are at most $|M| + \aleph_0 = \lambda$ such combinations. Therefore $|S_\varphi(M)| \leq \lambda$.

⊙ **Step 3: many branches.** For each $\eta \in \lambda^\omega$, choose a_η realizing $\{\varphi(x, b_{\eta|n}) : n \geq 1\}$, and put $p_\eta = \text{tp}_\varphi(a_\eta/M)$. Fix $p \in S_\varphi(M)$. For each node σ , the set $\text{Acc}_\sigma = \{\alpha < \lambda : \varphi(x, b_{\sigma \smallfrown \langle \alpha \rangle}) \in p\}$ of successors accepted by p has fewer than k elements: otherwise p would contain k siblings, whose conjunction is inconsistent in $\mathbb{M}$ and hence in M , whereas p is consistent with the elementary diagram of M . If $p_\eta = p$, then $\eta(n) \in \text{Acc}_{\eta|n}$ for every n . Hence the map sending such an η to the sequence of positions of $\eta(n)$ in $\text{Acc}_{\eta|n}$ is injective into $(k-1)^\omega$, and at most $2^{\aleph_0}$ branches η satisfy $p_\eta = p$. It follows that

$$\lambda^{\aleph_0} = |\lambda^\omega| \leq |S_\varphi(M)| \cdot 2^{\aleph_0} \leq \lambda,$$

because $\lambda \geq 2^\kappa \geq 2^{\aleph_0}$; this contradicts $\lambda^{\aleph_0} > \lambda$. ☹

Simplicity is thus weaker than stability. A definable infinite order is enough to destroy it: the strict order property (see Remark 4.1.6) produces the picture that follows Example 5.2.6, nested intervals with infinitely many disjoint subintervals inside each interval.

Proposition 5.3.2. *If some formula has the strict order property, then T is not simple.*

Proof. Let $\psi(x, y)$ have the strict order property, and write $\psi(\mathbb{M}, c)$ for the set of solutions of $\psi(x, c)$ in $\mathbb{M}$. By compactness and saturation, there are tuples c_q , for $q \in \mathbb{Q}$, with $\psi(\mathbb{M}, c_q) \subsetneq \psi(\mathbb{M}, c_r)$ whenever $q < r$: a finite set of these conditions mentions finitely many rationals $q_1 < \dots < q_n$, and is satisfied by a strictly increasing chain of length n , which exists in some model of T and hence in $\mathbb{M}$. Let $\theta(x; y_1, y_2)$ be the formula $\psi(x, y_2) \wedge \neg\psi(x, y_1)$, and for rationals $q < r$ put

$$\Delta(q, r) = \psi(\mathbb{M}, c_r) \setminus \psi(\mathbb{M}, c_q),$$

the set of solutions of $\theta(x; c_q, c_r)$; it is nonempty. Two properties of these sets follow from monotonicity of the chain:

- (i) if $q \leq q' < r' \leq r$, then $\Delta(q', r') \subseteq \Delta(q, r)$;
- (ii) if $r \leq q' < r'$, then $\Delta(q, r) \cap \Delta(q', r') = \emptyset$, since $\Delta(q, r) \subseteq \psi(\mathbb{M}, c_r) \subseteq \psi(\mathbb{M}, c_{q'})$ and $\Delta(q', r')$ is disjoint from $\psi(\mathbb{M}, c_{q'})$.

Assign rationals $q_\sigma < r_\sigma$ to the nodes $\sigma \in \omega^{<\omega}$ by recursion, starting from $(q_\emptyset, r_\emptyset) = (0, 1)$ and putting, with $d = r_\sigma - q_\sigma$,

$$q_{\sigma \smallfrown \langle i \rangle} = q_\sigma + d(1 - 2^{-i}), \quad r_{\sigma \smallfrown \langle i \rangle} = q_\sigma + d(1 - 2^{-i-1}).$$

Then $q_\sigma \leq q_{\sigma \smallfrown \langle i \rangle} < r_{\sigma \smallfrown \langle i \rangle} \leq r_\sigma$, and $r_{\sigma \smallfrown \langle i \rangle} \leq q_{\sigma \smallfrown \langle j \rangle}$ for $i < j$. Put $b_\sigma = (c_{q_\sigma}, c_{r_\sigma})$ for $\sigma \neq \emptyset$. For a branch η and $n \geq 1$, property (i) gives $\Delta(q_{\eta|n}, r_{\eta|n}) \subseteq \Delta(q_{\eta|m}, r_{\eta|m})$ for $m \leq n$, so any element of the nonempty set $\Delta(q_{\eta|n}, r_{\eta|n})$ satisfies $\theta(x, b_{\eta|m})$ for all $m \leq n$. By property (ii), the sets attached to two siblings are disjoint. Thus θ has the 2-tree property. ☹

The two propositions combine with Shelah's theorem, quoted in Remark 4.1.6, to characterize the stable theories as the simple NIP ones.

Fact 5.3.3 (Shelah [She90, Chapter II]). *If T is unstable, then some formula has the independence property or some formula has the strict order property.*



FIGURE 5.2. The proof of Proposition 5.3.2. The node σ carries the rational interval $(q_\sigma, r_\sigma]$, drawn at the height of the length of σ ; the successors of a node carry disjoint subintervals of its interval. The formula $\theta(x, b_\sigma)$ defines $\Delta(q_\sigma, r_\sigma) = \psi(\mathbb{M}, c_{r_\sigma}) \setminus \psi(\mathbb{M}, c_{q_\sigma})$, a band of the chain $(\psi(\mathbb{M}, c_q))_{q \in \mathbb{Q}}$. Nested intervals give nested bands, so a branch (pink) is consistent; disjoint intervals give disjoint bands, so siblings are 2-inconsistent.

Theorem 5.3.4. *A complete theory is stable if and only if it is simple and NIP.*

Proof. If T is stable, then T is simple by Proposition 5.3.1, and every formula is NIP by Proposition 4.1.2. Conversely, if T is unstable, then by Fact 5.3.3 some formula has the independence property, so that T is not NIP, or some formula has the strict order property, so that T is not simple by Proposition 5.3.2. ☺

Thus simplicity and NIP are the two halves of stability: an unstable theory encodes either arbitrary subsets or an infinite order, simplicity forbids the order, and NIP forbids the subsets. Neither half contains the other. Figure 5.3 places the Fraïssé limits studied in Section 5.5 in the four regions that this division determines.

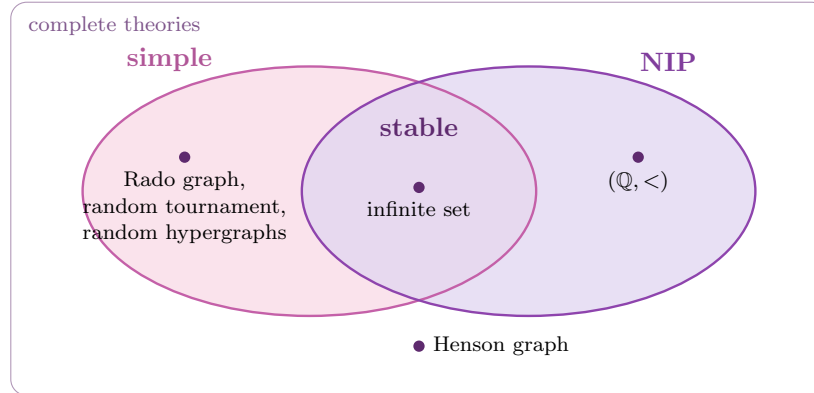


FIGURE 5.3. Theorem 5.3.4: the stable theories are the simple NIP theories. The pure infinite set is stable (see Example 5.5.3). The Rado graph is simple (see Theorem 5.4.6) and has the independence property (see Example 4.1.4); the random tournament and the random hypergraphs are simple (see Theorem 5.5.9), and their extension properties shatter every finite set, as in Example 4.1.4. The rational order is NIP, being o-minimal [PS86], and is not simple (see Proposition 5.3.2). The Henson graph is neither (see Example 5.5.10).

The first consequence of simplicity concerns dividing. In a stable theory a type over a model is determined by its restriction to a submodel of size at most κ , as Chapter 6 shows; in a simple theory a type is nondividing over a subset of size at most κ . The proof runs the construction of the tree property backwards: a long sequence of dividing formulas is turned into a tree. The tree is built one level at a time, using automorphisms to copy a sequence that witnesses dividing above each leaf.

Lemma 5.3.5. *Let $\varphi(x, y)$ be an L -formula, and let $k \geq 2$ be given.*

- (1) *Let $n \geq 1$, and let $a_0, \dots, a_{n-1}$ be tuples such that $\{\varphi(x, a_m) : m < n\}$ is consistent and, for every $m < n$, the formula $\varphi(x, a_m)$ k -divides over $A_m = \{a_l : l < m\}$. Then there are tuples b_σ , for $\sigma \in \omega^{\leq n}$ with $\sigma \neq \emptyset$, such that*
 - (i) *for every $\eta \in \omega^n$, we have $(b_{\eta|1}, \dots, b_{\eta|n}) \equiv (a_0, \dots, a_{n-1})$, and*
 - (ii) *for every $\sigma \in \omega^{<n}$, the set $\{\varphi(x, b_{\sigma \smallfrown \langle i \rangle}) : i < \omega\}$ is k -inconsistent.**In particular, these tuples form a tree of height n as in Remark 5.2.8.*
- (2) *If for every $n \geq 1$ there are tuples $a_0, \dots, a_{n-1}$ as in (1), possibly depending on n , then φ has the k -tree property. This holds in particular if $(a_n)_{n < \omega}$ is a sequence such that $\{\varphi(x, a_n) : n < \omega\}$ is consistent and, for every n , the formula $\varphi(x, a_n)$ k -divides over $A_n = \{a_m : m < n\}$.*

Proof. (1) For each $m < n$, fix a sequence $(a_m^i)_{i < \omega}$, indiscernible over A_m , with $a_m^0 = a_m$ and $\{\varphi(x, a_m^i) : i < \omega\}$ k -inconsistent. We build the tuples level by level: for $p = 1, \dots, n$ we find b_σ for $\sigma \in \omega^{\leq p}$, $\sigma \neq \emptyset$, satisfying (i) and (ii) with p in place of n and $(a_0, \dots, a_{p-1})$ in place of $(a_0, \dots, a_{n-1})$. Condition (i) gives condition (a) of Definition 5.2.7 for the branches of length p , because $\{\varphi(x, a_m) : m < p\}$ is consistent. For $p = 1$, put $b_{\langle i \rangle} = a_0^i$; all terms of an indiscernible sequence have the same type, so (i) holds, and (ii) holds by the choice of (a_0^i) .

Suppose that the tuples have been built for some $p < n$. For each $\eta \in \omega^p$, by (i) and strong homogeneity there is $f_\eta \in \text{Aut}(\mathbb{M})$ with $f_\eta(a_0, \dots, a_{p-1}) = (b_{\eta|1}, \dots, b_{\eta|p})$. Keep the tuples already built, and put $b_{\eta \smallfrown \langle i \rangle} = f_\eta(a_p^i)$ for $i < \omega$. Since $a_p^i \equiv_{A_p} a_p$, we have

$$(b_{\eta|1}, \dots, b_{\eta|p}, b_{\eta \smallfrown \langle i \rangle}) = f_\eta(a_0, \dots, a_{p-1}, a_p^i) \equiv (a_0, \dots, a_{p-1}, a_p^i) \equiv (a_0, \dots, a_p),$$

which is (i) for $p+1$. The set $\{\varphi(x, b_{\eta \smallfrown \langle i \rangle}) : i < \omega\}$ is the image under f_η of a k -inconsistent set, and hence is k -inconsistent; this is (ii) at the new level, and (ii) at the old levels is unchanged.

(2) By (1), for every $n \geq 1$ there is a tree of height n whose branches are consistent and whose siblings are k -inconsistent, and Remark 5.2.8 gives the k -tree property. For the last assertion, apply (1) to the initial segments $a_0, \dots, a_{n-1}$ of the sequence. ☺

Part (2) gives the local character of dividing at once. The finite form (1) is used in Chapter 7, where the tuples $a_0, \dots, a_{n-1}$ are produced afresh for each n .

Theorem 5.3.6 (Local character of dividing). *Assume that T is simple, and let a be a finite tuple and B a small set. Then there is $A \subseteq B$ with $|A| \leq \kappa$ such that $\text{tp}(a/B)$ does not divide over A .*

Proof. Suppose not. We choose, by recursion on $\alpha < \kappa^+$, finite tuples b_α from B , formulas $\varphi_\alpha(x, y) \in L$ and integers $k_\alpha \geq 2$. Put $B_\alpha = \bigcup_{\beta < \alpha} b_\beta$, a subset of B of size at most κ . By assumption $\text{tp}(a/B)$ divides over B_α , so some formula of $\text{tp}(a/B)$, say $\varphi_\alpha(x, b_\alpha)$ with $\varphi_\alpha \in L$ and b_α from B , divides over B_α , and by Lemma 5.2.4 it k_α -divides over B_α for some $k_\alpha \geq 2$.

There are only κ pairs (φ, k) with $\varphi \in L$ and $k < \omega$, and κ^+ is regular, so some pair (φ, k) equals $(\varphi_\alpha, k_\alpha)$ for unboundedly many α . Choose such $\alpha_0 < \alpha_1 < \dots$, and put $a_n = b_{\alpha_n}$. The set $A_n = \{a_m : m < n\}$ is contained in B_{α_n} , so $\varphi(x, a_n)$ k -divides over A_n by Remark 5.2.5. The formulas $\varphi(x, a_n)$ all belong to $\text{tp}(a/B)$, and so they form a consistent set. By Lemma 5.3.5(2), the formula φ has the k -tree property, contradicting simplicity. $\odot$



Remark 5.3.7. The converse of Theorem 5.3.6 holds as well: if every type over a small set is nondividing over a subset of size at most κ , then T is simple (Shelah [She80]; see [TZ12, Chapter 7] or Casanovas [Cas11]). Kim [Kim98] and Kim and Pillay [KP97] characterized the simple theories by an abstract relation of independence, as Chapter 6 does for the stable theories; their theorem is the subject of Chapter 7. In Sections 5.4 and 5.5 we prove the corresponding statements directly, from the axioms, for the random graph and for a class of Fraïssé limits.

5.4. The Rado graph

The goal of this section is to compute dividing in the random graph, from the extension axioms alone, and to derive from the computation that the random graph is simple and that its independence relation satisfies the independence theorem but not uniqueness. Let T_{RG} be the theory of Example 4.1.4: the axioms say that E is irreflexive and symmetric and, for all m, n , that for all disjoint finite sets U, V of vertices with $|U| = m$ and $|V| = n$ some vertex outside $U \cup V$ is adjacent to every vertex of U and to no vertex of V . These are the *extension axioms*. A countable model is easy to write down.

Example 5.4.1. Let the vertices be the natural numbers, and for $i < j$ join i and j if the i -th binary digit of j is 1 (digits counted from the right, starting at 0). This graph satisfies the extension axioms. Let U, V be disjoint finite sets of natural numbers, choose N larger than every element of $U \cup V$, and put

$$z = 2^N + \sum_{u \in U} 2^u.$$

Then $z \geq 2^N > N$, so $z \notin U \cup V$ and z is larger than every element of $U \cup V$. The binary digits of z at the positions $u \in U$ are 1 and at the positions $v \in V$ are 0, so z is adjacent to every vertex of U and to no vertex of V . The graph is called the *Rado graph* $\mathcal{R}$ (see Figure 5.4).

Fact 5.4.2. *The theory T_{RG} is complete, $\aleph_0$ -categorical, and admits elimination of quantifiers; its countable model is the Rado graph (see [Hod93, Theorem 7.4.1], or Fact 5.5.2 below).*

From now until the end of this section, the theory T is T_{RG} . By quantifier elimination, the type of a tuple over a set is determined by the equalities and edges that it satisfies with parameters from the set. We use this in the following form: for small tuples a, a' and a small set C ,

$$(5.1) \quad a \equiv_C a' \iff \begin{array}{l} \text{the map that fixes } C \text{ pointwise and sends each} \\ \text{coordinate of } a \text{ to the corresponding coordinate of } a' \text{ is} \\ \text{a well defined bijection from } C \cup a \text{ onto } C \cup a' \text{ that} \\ \text{preserves edges and non-edges.} \end{array}$$

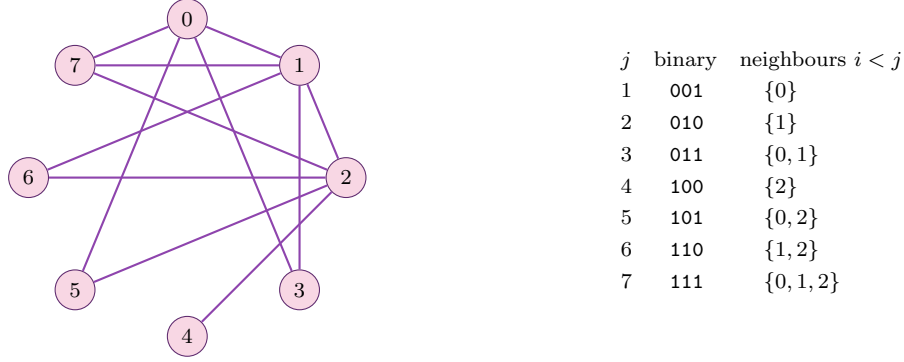


FIGURE 5.4. The vertices $0, \dots, 7$ of the Rado graph of Example 5.4.1. For $i < j$, the vertex j is joined to i exactly when the binary expansion of j has digit 1 in position i ; the table lists the expansions. The new neighbour of U that avoids V is obtained by writing 1 in the positions of U , 0 in the positions of V , and a 1 far to the left.

An injective map between graphs that preserves edges and non-edges is an *embedding*. The computation of dividing rests on two tools. The first is the extension axioms, which let us embed into $\mathbb{M}$ any graph that agrees with $\mathbb{M}$ on a given small set.

Lemma 5.4.3. *Let C be a small set of vertices, and let X be a graph whose vertex set is the disjoint union of C and a small set X_0 , and whose restriction to C is the graph induced by $\mathbb{M}$ on C . Then there is an embedding $g: X \rightarrow \mathbb{M}$ with $g(c) = c$ for every $c \in C$.*

Proof. Introduce a variable x_v for each $v \in X_0$, and let Σ be the set of formulas $x_v \neq x_w$, $x_v \neq c$, $E(x_v, x_w)^u$ and $E(x_v, c)^u$ (for $v \neq w$ in X_0 and $c \in C$) that describe X , where $u \in \{0, 1\}$ records whether the edge is present in X . A realization of Σ is an embedding of the required kind, and Σ is a small set of formulas over C , so by saturation it suffices to show that every finite subset of Σ is satisfiable. A finite subset mentions finite sets $C_0 \subseteq C$ and $\{v_1, \dots, v_m\} \subseteq X_0$. Choose images $w_1, \dots, w_m \in \mathbb{M}$ one at a time: if $w_1, \dots, w_{l-1}$ have been chosen, apply the extension axiom to U , the set of elements of $C_0 \cup \{w_1, \dots, w_{l-1}\}$ that must be adjacent to w_l , and V , the set of the others. The vertex w_l so obtained lies outside $C_0 \cup \{w_1, \dots, w_{l-1}\}$, so the new vertices are distinct from each other and from C_0 . ☺

The second tool is Ramsey's theorem, in the sharper form announced after Lemma 5.2.2: over a finite set, an infinite subsequence of the given sequence is itself indiscernible.

Lemma 5.4.4. *Let D be a finite set of vertices, and let $(b_i)_{i < \omega}$ be tuples of the same finite length. Then there is an infinite set $H \subseteq \omega$ such that $(b_i)_{i \in H}$ is indiscernible over D .*

Proof. Call the *diagram* over D of a tuple the set of equalities and edges, and of their negations, that hold among its coordinates and elements of D . Since D is finite and the tuples have fixed length, there are finitely many possible diagrams over D of one tuple b_i , and of a pair (b_i, b_j) . By the pigeonhole principle, all b_i with i in some infinite set have the same diagram over D ; colouring each pair $i < j$ of this set by the diagram of (b_i, b_j) over D , Ramsey's theorem gives an infinite homogeneous set H . Each equality or edge involves at most two vertices, hence at most two of the tuples. Hence, for $i_1 < \dots < i_n$ in H , the diagram of $(b_{i_1}, \dots, b_{i_n})$ over D is determined by the diagrams of the b_{i_l} and of the pairs (b_{i_l}, b_{i_m}) with $l < m$, which do not depend on the choice of $i_1 < \dots < i_n$ in H . By (5.1), the sequence $(b_i)_{i \in H}$ is indiscernible over D . ☺

We can now compute dividing. The answer is the one suggested by the introduction to this chapter: a type over $C \cup B$ divides over C exactly when it puts one of the variables equal to an element of B outside C .

Theorem 5.4.5 (Dividing in the Rado graph). *Let a be a finite tuple, and let C and B be small sets. Then $\text{tp}(a/C \cup B)$ divides over C if and only if $a \cap B \not\subseteq C$.*

Proof. Suppose first that some coordinate a_j of a lies in $B \setminus C$, and write $d = a_j$. The formula $x_j = d$ belongs to $\text{tp}(a/C \cup B)$. By Lemma 5.4.3, applied to the graph on $C \cup \{d'_i : i < \omega\}$ in which each new vertex d'_i is adjacent to $c \in C$ exactly when d is, and no two new vertices are adjacent, there are distinct vertices $d''_i \notin C$ with these adjacencies. By (5.1), any two increasing n -tuples of (d''_i) have the same type over C , and $d''_0 \equiv_C d$. Choose $f \in \text{Aut}(\mathbb{M}/C)$ with $f(d''_0) = d$. The sequence $(f(d''_i))_{i < \omega}$ is indiscernible over C , begins with d , and consists of distinct vertices, so $\{x_j = f(d''_i) : i < \omega\}$ is 2-inconsistent. Hence $\text{tp}(a/C \cup B)$ divides over C .

Conversely, suppose that $a \cap B \subseteq C$. Let $\varphi(x, b)$ be a formula of $\text{tp}(a/C \cup B)$, with $\varphi \in L$ and b a finite tuple from $C \cup B$, and let $(b_i)_{i < \omega}$ be indiscernible over C with $b_0 = b$. We find a single tuple a'' with $a''b_i \equiv_C ab$ for all i ; then a'' satisfies every $\varphi(x, b_i)$, so $\varphi(x, b)$ does not divide over C . The idea is to join a fresh copy of a to each b_i exactly as a is joined to b . For this to be well defined, coordinates shared by different terms of the sequence must play the same role in b . Write $e[p]$ for the p -th coordinate of a tuple e .

© **Claim.** If $i \neq j$ and $b_i[p] = b_j[q]$, then $b[p] = b[q]$, and all terms of the sequence have the same p -th coordinate and the same q -th coordinate.

Exchanging the roles of (i, p) and (j, q) if necessary, we may assume $i < j$. Choose $l > j$. Since $(b_i, b_j) \equiv_C (b_i, b_l)$, we have $b_i[p] = b_l[q]$, and since $(b_i, b_j) \equiv_C (b_j, b_l)$, we have $b_j[p] = b_l[q]$. Hence $b_j[p] = b_i[p] = b_j[q]$. As $b_j \equiv_C b$, this gives $b[p] = b[q]$. Moreover, since $b_i[p] = b_j[p]$ with $i < j$, indiscernibility gives $b_{i'}[p] = b_{j'}[p]$ for all $i' < j'$; thus the p -th coordinate is constant. Finally, $b_{i'} \equiv_C b$ and $b[p] = b[q]$ give $b_{i'}[p] = b_{i'}[q]$ for every i' , so the q -th coordinate is constant as well. This proves the claim (see Figure 5.5).

	[1]	[2]	[3]	[4]	
b_0	u_0	e	v_0	e'	e, e' : constant columns, the only elements shared by two different terms
b_1	u_1	e	v_1	e'	
b_2	u_2	e	v_2	e'	
b_3	u_3	e	v_3	e'	u_i, v_i : new in each term
b_4	u_4	e	v_4	e'	

FIGURE 5.5. The claim in the proof of Theorem 5.4.5. The rows are the terms of a sequence indiscernible over C , and the columns are their coordinates. An element outside C that occurs in two different terms occupies a whole column (pink): it is the same coordinate of every term. The fresh copy of a can therefore be joined to e and e' without ambiguity, exactly as a is joined to the corresponding coordinates of b .

With the claim in hand, we build the fresh copy of a . Let I be the set of indices v with $a_v \notin C$. For $v \in I$ we have $a_v \notin b$, because $b \subseteq C \cup B$ and $a \cap B \subseteq C$. Let $V = \bigcup_i b_i \setminus C$, and let X be the graph on $C \cup V$ together with fresh vertices a'_v for $v \in I$, where $a'_v = a'_w$ exactly when $a_v = a_w$, with the following edges:

- (i) on $C \cup V$, the edges of $\mathbb{M}$;
- (ii) the vertex a'_v is adjacent to $c \in C$ if and only if a_v is adjacent to c , and a'_v is adjacent to a'_w if and only if a_v is adjacent to a_w ;
- (iii) if $e = b_i[p] \in V$, then a'_v is adjacent to e if and only if a_v is adjacent to $b[p]$.

Clause (iii) does not depend on the choice of i and p with $e = b_i[p]$: if also $e = b_j[q]$, then $b[p] = b[q]$, by the claim when $i \neq j$, and because $b_i \equiv_C b$ when $i = j$. By Lemma 5.4.3, there is an embedding g of X into $\mathbb{M}$ fixing $C \cup V$. Let a'' be the tuple with $a''_v = g(a'_v)$ for $v \in I$ and $a''_v = a_v$ for $v \notin I$. Fix i . The map that fixes C , sends a to a'' and b to b_i coordinatewise is a well defined bijection from $C \cup a \cup b$ onto $C \cup a'' \cup b_i$: on $C \cup b$ it is the bijection given by $b \equiv_C b_i$, the coordinates a_v with $v \in I$ lie outside $C \cup b$, and their images lie outside $C \cup V$. By (i)–(iii) this map preserves edges and non-edges, so $a''b_i \equiv_C ab$ by (5.1). ☺

A dividing step therefore uses up a coordinate of a , and a tuple of length n can divide at most n times. This is the heart of the proof that the random graph is simple.

Theorem 5.4.6. *Let a be a tuple of length n , and let $D_0 \subseteq D_1 \subseteq \dots \subseteq D_m$ be small sets such that $\text{tp}(a/D_{j+1})$ divides over D_j for every $j < m$. Then $m \leq n$. The theory T_{RG} is simple.*

Proof. By Theorem 5.4.5, applied with $C = D_j$ and $B = D_{j+1}$, the tuple a meets D_{j+1} in a coordinate outside D_j . Thus the sets $a \cap D_0 \subsetneq a \cap D_1 \subsetneq \dots \subsetneq a \cap D_m$ increase strictly, and a has at least m distinct coordinates, so $m \leq n$.

Suppose now that $\varphi(x, y)$ has the k -tree property, witnessed by tuples b_σ , and let n be the length of x . The idea is to find, along a branch, $n + 1$ dividing steps, which the first part forbids. For this we prune the tree so that siblings become indiscernible over the nodes below them. By recursion, define a set $P \subseteq \omega^{\leq n+1}$ of nodes: the empty sequence belongs to P , and if $\sigma \in P$ has length at most n , let $D_\sigma = \bigcup_{1 \leq l \leq |\sigma|} b_{\sigma|l}$, a finite set (empty when $\sigma = \emptyset$), choose by Lemma 5.4.4 an infinite set $H_\sigma \subseteq \omega$ such that $(b_{\sigma \smallfrown \langle i \rangle})_{i \in H_\sigma}$ is indiscernible over D_σ , and put $\sigma \smallfrown \langle i \rangle \in P$ for $i \in H_\sigma$. Choose $\eta \in \omega^{n+1}$ with $\eta|l \in P$ for every l , let a realize $\{\varphi(x, b_{\eta|l}) : 1 \leq l \leq n + 1\}$, and put $D_l = D_{\eta|l}$ for $l \leq n + 1$, where $D_{\eta|n+1}$ is defined by the same formula.

Let $l \leq n$ and $\sigma = \eta|l$. The terms $b_{\sigma \smallfrown \langle i \rangle}$ with $i \in H_\sigma$ and $i \geq \eta(l)$ form a sequence indiscernible over D_l that begins with $b_{\eta|l+1}$, and the corresponding formulas are k -inconsistent. Hence $\varphi(x, b_{\eta|l+1})$ divides over D_l , and, since this formula belongs to $\text{tp}(a/D_{l+1})$, the type $\text{tp}(a/D_{l+1})$ divides over D_l . The chain $D_0 \subseteq \dots \subseteq D_{n+1}$ has $n + 1$ dividing steps, contradicting the first part of the theorem (see Figure 5.6). ☺

Simplicity settled, we turn to the second aim of this section, the independence relation of the random graph. An independence relation over a model M is a relation $A \perp_M B$, read “ A is independent from B over M ”, and the following axioms are the yardstick for such relations. In Chapter 6 we show that a theory is stable exactly when it carries a relation satisfying all of them (see Theorem 6.2.2); the random graph carries a relation satisfying all of them but one.

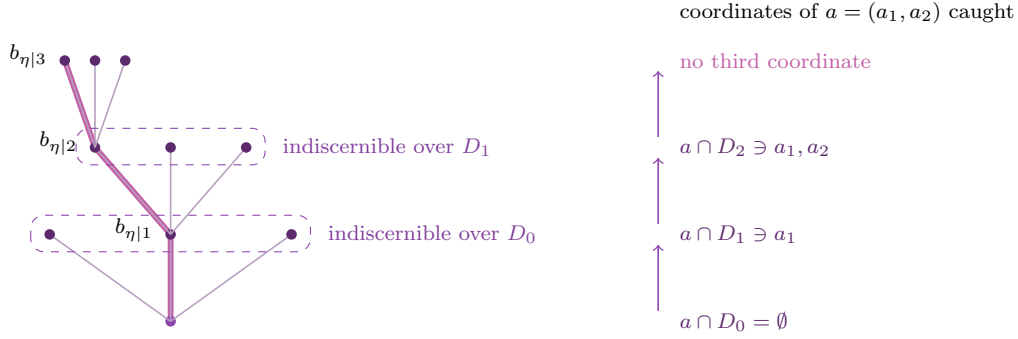


FIGURE 5.6. The proof of Theorem 5.4.6 for a pair $x = (x_1, x_2)$, so $n = 2$. The tree is pruned so that the siblings above each node are indiscernible over the parameters below it (framed). Each step up the branch (pink) is a dividing step, and by Theorem 5.4.5 it catches a new coordinate of the realization a in the parameters D_l collected so far. After two steps both coordinates are caught, and a third dividing step is impossible; a tree of height 3 cannot exist.

Definition 5.4.7. An *independence relation (over models)* is a relation $A \downarrow_M B$ between small sets A, B and models M satisfying the following axioms, for all small A, A', B, B' and all models M, N .

- (INV) *Invariance.* If $A \downarrow_M B$ and $\sigma \in \text{Aut}(\mathbb{M})$, then $\sigma(A) \downarrow_{\sigma(M)} \sigma(B)$.
- (MON) *Monotonicity.* If $A \downarrow_M B$, $A' \subseteq A$ and $B' \subseteq B$, then $A' \downarrow_M B'$.
- (BMON) *Base monotonicity.* If $A \downarrow_M B$ and N is a model with $M \subseteq N \subseteq B$, then $A \downarrow_N B$.
- (TR) *Transitivity.* If $M \subseteq N$, $A \downarrow_M N$ and $A \downarrow_N B$, then $A \downarrow_M NB$.
- (FC) *Finite character.* The relation $A \downarrow_M B$ holds if and only if $a \downarrow_M b$ for all finite tuples a from A and b from B .
- (SYM) *Symmetry.* The relation $A \downarrow_M B$ holds if and only if $B \downarrow_M A$.
- (EX) *Existence (extension).* For all A, M, B there is $A' \equiv_M A$ with $A' \downarrow_M B$.
- (UN) *Uniqueness (stationarity).* If $A \equiv_M A'$, $A \downarrow_M B$ and $A' \downarrow_M B$, then $A \equiv_{MB} A'$.
- (LOC) *Local character.* For every finite tuple a and every model N there is a model $M \subseteq N$ with $|M| \leq \kappa$ and $a \downarrow_M N$.

The relation is suggested by the computation of Theorem 5.4.5: a tuple should be independent from a set when it shares with it nothing outside the base.

Definition 5.4.8. For small sets A, B, C , write $A \downarrow_C^t B$ if $A \cap B \subseteq C$. We call $\downarrow^t$ *trivial independence*; the superscript distinguishes it from the abstract relations $\downarrow$ of Definition 5.4.7.

For a finite tuple a , Theorem 5.4.5 says that $a \downarrow_C^t B$ if and only if $\text{tp}(a/C \cup B)$ does not divide over C . We compare the relation $\downarrow^t$ with the axioms of Definition 5.4.7, read with arbitrary models M, N of T_{RG} as bases, and with one further axiom, *anti-reflexivity* (AR): if $a \downarrow_M a$ for a finite tuple a , then $a \subseteq M$. It reappears for stable theories in Theorem 6.2.2.

Theorem 5.4.9 (Independence in the Rado graph). *The following hold in T_{RG} :*

- (a) *The relation $\perp^t$ satisfies all the axioms of Definition 5.4.7 except (UN), and it satisfies (AR).*
- (b) *The relation $\perp^t$ does not satisfy (UN).*
- (c) (Independence theorem.) *Let C be a small set and b_1, b_2, c_1, c_2 finite tuples. If $b_1 \perp_C^t b_2$, $c_1 \perp_C^t b_1$, $c_2 \perp_C^t b_2$ and $c_1 \equiv_C c_2$, then there is c with $c \equiv_{Cb_1} c_1$, $c \equiv_{Cb_2} c_2$ and $c \perp_C^t b_1 b_2$.*

Proof. (a) Invariance, monotonicity, finite character, symmetry and anti-reflexivity are immediate from the definition. *Base monotonicity:* if $A \cap B \subseteq M$ and $M \subseteq N \subseteq B$, then $A \cap B \subseteq N$. *Transitivity:* if $M \subseteq N$, $A \cap N \subseteq M$ and $A \cap B \subseteq N$, then $A \cap B \subseteq A \cap N \subseteq M$, so $A \cap (N \cup B) \subseteq M$. *Local character:* given a finite tuple a and a model N , the downward Löwenheim–Skolem theorem gives a model $M \preceq N$ with $a \cap N \subseteq M$ and $|M| \leq \kappa$, and then $a \perp_M^t N$. *Existence:* given A , M and B , let X be the graph on $M \cup B$ together with fresh vertices a'_v , one for each coordinate a_v of A outside M (equal coordinates receiving equal vertices), where the edges on $M \cup B$ are those of $\mathbb{M}$, the vertices a'_v are joined to each other and to M as the a_v are, and no a'_v is joined to a vertex of $B \setminus M$. Lemma 5.4.3 embeds X into $\mathbb{M}$ over $M \cup B$, and the resulting tuple A' satisfies $A' \equiv_M A$ by (5.1) and $A' \cap B \subseteq M$.

(b) Let M be a model, and let b be a vertex not in M . By Lemma 5.4.3 there are vertices $a, a' \notin M \cup \{b\}$, adjacent to no vertex of M , with a adjacent to b and a' not. Then $a \equiv_M a'$ by (5.1), and $a \perp_M^t b$ and $a' \perp_M^t b$, but $a \not\equiv_{Mb} a'$.

(c) Let I be the set of indices v with $(c_1)_v \notin C$; since $c_1 \equiv_C c_2$, it is also the set of indices with $(c_2)_v \notin C$, and $(c_1)_v = (c_2)_v$ for $v \notin I$. Let X be the graph on $C \cup b_1 \cup b_2$, with the edges of $\mathbb{M}$, together with fresh vertices c'_v for $v \in I$ (equal coordinates receiving equal vertices), where the new vertices are joined to each other and to $C \cup b_1$ as the coordinates of c_1 are, and to $b_2 \setminus C$ as the coordinates of c_2 are. The two prescriptions overlap only on C , because $b_1 \cap b_2 \subseteq C$, and on C (and among the new vertices) they agree, because $c_1 \equiv_C c_2$. By Lemma 5.4.3, there is an embedding g of X into $\mathbb{M}$ fixing $C \cup b_1 \cup b_2$. Let c be the tuple with $c_v = g(c'_v)$ for $v \in I$ and $c_v = (c_1)_v$ for $v \notin I$. The coordinates $(c_1)_v$ with $v \in I$ lie outside $C \cup b_1$, since $c_1 \cap b_1 \subseteq C$, and the coordinates c_v with $v \in I$ lie outside $C \cup b_1 \cup b_2$. Hence the map fixing $C \cup b_1$ and sending c_1 to c is a bijection preserving edges and non-edges, and $c \equiv_{Cb_1} c_1$ by (5.1). In the same way $c \equiv_{Cb_2} c_2$, and $c \cap (b_1 \cup b_2) \subseteq C$. $\odot$

Figure 5.7 shows the construction of (c) in the smallest interesting case.

Stability and simplicity, side by side

In a stable theory, as we show in Chapter 6, every type over a model M has exactly one independent extension to $M \cup B$: the heir, given by the definition scheme of the type. In the Rado graph, a type over M has many independent extensions to $M \cup \{b\}$, one for each way of joining the new vertices to b , and uniqueness fails (see Theorem 5.4.9(b)). In its place stands the independence theorem (see Theorem 5.4.9(c)): independent extensions over two independent sets can always be realized together. Dividing is measured by coordinates (see Theorem 5.4.5), each dividing step uses up one coordinate, and so no formula can have the tree property (see Theorem 5.4.6).

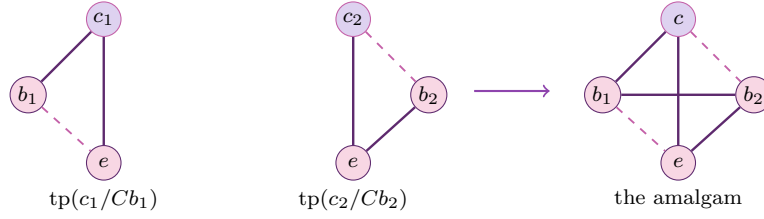


FIGURE 5.7. The independence theorem in the Rado graph, with $C = \{e\}$ (see Theorem 5.4.9(c)). Solid lines are edges and dashed lines non-edges. The vertices c_1 and c_2 have the same type over C (both are adjacent to e); the vertex c_1 is adjacent to b_1 , and c_2 is not adjacent to b_2 . The vertex c inherits its edges to $C \cup \{b_1\}$ from c_1 and to $C \cup \{b_2\}$ from c_2 . The two prescriptions can only conflict on C , where they agree. A second vertex, joined to b_1 and b_2 in the same way but differing elsewhere, would be just as good: the amalgam exists, but it is not unique.

5.5. Fraïssé limits

The goal of this section is to find the property of finite graphs on which Section 5.4 depends, and to prove that every Fraïssé limit with this property is simple. In the proofs of Theorems 5.4.5 and 5.4.9, new vertices were joined to old ones as prescribed by several types at once, and Lemma 5.4.3 embedded the resulting graph into $\mathbb{M}$. Every graph is allowed in the random graph, so the prescriptions only had to be consistent with one another. In a Fraïssé limit only the structures of a given class are allowed, and the question becomes whether the structure assembled from the prescriptions belongs to the class. In the proof of the independence theorem, three structures are given pairwise, on $C \cup c \cup b_1$, on $C \cup c \cup b_2$ and on $C \cup b_1 \cup b_2$, and they must be completed to a structure on the union. This is a 3-amalgamation problem.

Throughout this section, the language L is finite and relational. For an L -structure N , the *age* of N is the class of finite L -structures that embed into N .

Definition 5.5.1. A *Fraïssé class* is a nonempty class $\mathcal{K}$ of finite L -structures, closed under isomorphism, with the following properties:

- (a) *hereditary property*: every substructure of a structure in $\mathcal{K}$ belongs to $\mathcal{K}$;
- (b) *joint embedding property*: any two structures in $\mathcal{K}$ embed into a common structure in $\mathcal{K}$;
- (c) *amalgamation property*: if $A, B_1, B_2 \in \mathcal{K}$ and $f_i: A \rightarrow B_i$ are embeddings, then there are $B \in \mathcal{K}$ and embeddings $g_i: B_i \rightarrow B$ with $g_1 \circ f_1 = g_2 \circ f_2$.

An L -structure N is *ultrahomogeneous* if every isomorphism between finite substructures of N extends to an automorphism of N . Given a Fraïssé class $\mathcal{K}$, a structure N has the *extension property* for $\mathcal{K}$ if, whenever $A \subseteq B$ belong to $\mathcal{K}$, with $|B \setminus A| = 1$ and A possibly empty, every embedding of A into N extends to an embedding of B into N .

Fact 5.5.2 (Fraïssé [Fra54]; see [Hod93, Theorems 7.1.2 and 7.4.1]). *Let $\mathcal{K}$ be a Fraïssé class.*

- (a) *There is a countable ultrahomogeneous L -structure $M_{\mathcal{K}}$ whose age is $\mathcal{K}$, and it is unique up to isomorphism.*
- (b) *The theory $T_{\mathcal{K}}$ of $M_{\mathcal{K}}$ is $\aleph_0$ -categorical and admits elimination of quantifiers.*

- (c) An L -structure N is a model of $T_{\mathcal{K}}$ if and only if every finite substructure of N belongs to $\mathcal{K}$ and N has the extension property for $\mathcal{K}$.

The structure $M_{\mathcal{K}}$ is the *Fraïssé limit* of $\mathcal{K}$.

Example 5.5.3. The following are Fraïssé classes; the amalgamation property is checked by taking unions (orienting the new pairs, for tournaments) or, for linear orders, by interleaving.

- (1) The finite sets, in the language with equality only. The limit is a countably infinite set. Its theory is stable: by quantifier elimination, the type of an n -tuple over a set A is determined by the equalities among its coordinates and with elements of A , so there are at most $|A| + \aleph_0$ such types; the theory is λ -stable for every infinite λ , and stable by Lemma 5.1.2.
- (2) The finite linear orders. The limit is $(\mathbb{Q}, <)$.
- (3) The finite graphs. The limit is the Rado graph, and $T_{\mathcal{K}}$ is T_{RG} .
- (4) The finite tournaments (complete graphs with an orientation of each edge). The limit is the *random tournament*.
- (5) For $r \geq 2$, the finite r -uniform hypergraphs. The limit is the *random r -hypergraph*.
- (6) The finite triangle-free graphs. The limit is the *Henson graph* $\mathcal{H}$.

In the proofs of Section 5.4, the new vertices were always fresh: no new vertex was identified with an old one. In Definition 5.5.4 below, we ask the same of amalgamation in $\mathcal{K}$. For Fraïssé classes, we treat structures as sets with relations and say that a structure D *extends* a structure B if B is a substructure of D .

Definition 5.5.4. Let $\mathcal{K}$ be a Fraïssé class.

- (a) The class $\mathcal{K}$ has *disjoint 2-amalgamation* if whenever $B_1, B_2 \in \mathcal{K}$ have a common substructure A with $B_1 \cap B_2 = A$, some structure in $\mathcal{K}$ on $B_1 \cup B_2$ extends B_1 and B_2 .
- (b) The class $\mathcal{K}$ has *disjoint 3-amalgamation* if the following holds. Let B_1, B_2, B_3 be structures with a common substructure $A \in \mathcal{K}$ and with $B_l \cap B_m = A$ for $l \neq m$, and let $B_{12}, B_{13}, B_{23} \in \mathcal{K}$ be structures on $B_1 \cup B_2$, $B_1 \cup B_3$, $B_2 \cup B_3$ such that B_{lm} extends B_l and B_m . Then some structure in $\mathcal{K}$ on $B_1 \cup B_2 \cup B_3$ extends B_{12} , B_{13} and B_{23} .

In a 3-amalgamation problem every tuple contained in the union of two of the sets B_l already has its relations prescribed, by one of the structures B_{lm} ; what remains free is the relations on tuples that meet all three of the sets $B_l \setminus A$. Figure 5.8 shows a 3-amalgamation problem in general, and two problems without solution.

Example 5.5.5. All classes of Example 5.5.3 have disjoint 2-amalgamation. For graphs, triangle-free graphs and hypergraphs, take the union of the two structures, with no relations between $B_1 \setminus A$ and $B_2 \setminus A$; a triangle in the union that is contained neither in B_1 nor in B_2 would need two adjacent vertices, one in each of $B_1 \setminus A$ and $B_2 \setminus A$. For tournaments, orient the new pairs from B_1 to B_2 . For linear orders, put the elements of $B_1 \setminus A$ before those of $B_2 \setminus A$ within each interval determined by A . The classes (1), (3), (4) and (5) have disjoint 3-amalgamation. For sets, graphs and tournaments, every relation involves at most two elements, so in a 3-amalgamation

problem every relation is already prescribed and the union of B_{12} , B_{13} and B_{23} is a solution. For r -uniform hypergraphs, declare that no hyperedge meets all three of the sets $B_l \setminus A$.

The classes (2) and (6) do not have disjoint 3-amalgamation. Take $A = \emptyset$ and one-element sets $B_1 = \{x\}$, $B_2 = \{y\}$, $B_3 = \{z\}$. For linear orders, let B_{12} , B_{23} , B_{13} say that $x < y$, $y < z$ and $z < x$: no linear order extends them. For triangle-free graphs, let each B_{lm} be an edge: the only graph on $\{x, y, z\}$ extending them is a triangle.

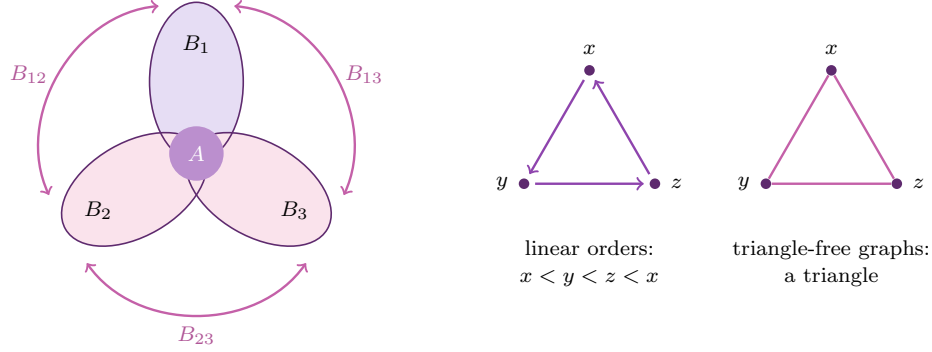


FIGURE 5.8. Left: a disjoint 3-amalgamation problem (see Definition 5.5.4(b)). The sets B_1, B_2, B_3 meet pairwise in A ; the structures B_{12}, B_{13}, B_{23} are given on the three pairwise unions, and a solution is a structure in $\mathcal{K}$ on the whole union extending all three. Right: two problems without solution, with $A = \emptyset$ and one point in each B_l (see Example 5.5.5). Each pair is consistent, and only the triple is forbidden.

For the rest of this section, we fix a Fraïssé class $\mathcal{K}$, put $T = T_{\mathcal{K}}$, and let $\mathbb{M}$ be a monster model of T . By quantifier elimination, (5.1) holds with “preserves edges and non-edges” replaced by “is an isomorphism of substructures”; we refer to this version as (5.1) as well. The two tools of Section 5.4, the embedding of graphs over a set (see Lemma 5.4.3) and the extraction of indiscernible subsequences by Ramsey’s theorem (see Lemma 5.4.4), generalize directly.

Lemma 5.5.6. *Let C be a small subset of $\mathbb{M}$, and let X be an L -structure whose universe is the disjoint union of C and a small set X_0 , whose restriction to C is the substructure of $\mathbb{M}$ on C , and whose finite substructures all belong to $\mathcal{K}$. Then there is an embedding $g: X \rightarrow \mathbb{M}$ with $g(c) = c$ for every $c \in C$.*

Proof. Argue as in the proof of Lemma 5.4.3, with the atomic formulas of L in place of edges. Given finite sets $C_0 \subseteq C$ and $\{v_1, \dots, v_m\} \subseteq X_0$, the substructures of X on $C_0 \cup \{v_1, \dots, v_l\}$, for $l \leq m$, belong to $\mathcal{K}$, and each extends the previous one by one element. Applying the extension property of Fact 5.5.2(c) m times, starting from the inclusion of C_0 into $\mathbb{M}$, gives an embedding of the substructure on $C_0 \cup \{v_1, \dots, v_m\}$ into $\mathbb{M}$ over C_0 . $\odot$

Lemma 5.5.7. *Let D be a finite subset of $\mathbb{M}$, and let $(b_i)_{i < \omega}$ be tuples of the same finite length. Then there is an infinite set $H \subseteq \omega$ such that $(b_i)_{i \in H}$ is indiscernible over D .*

Proof. Let r be the larger of 2 and the maximal arity of a symbol of L . Argue as in the proof of Lemma 5.4.4, colouring each r -element set $i_1 < \dots < i_r$ by the diagram over D of

$(b_{i_1}, \dots, b_{i_r})$, that is, by the set of atomic formulas with parameters from D , and of their negations, satisfied by it. Let H be infinite and homogeneous. Each atomic formula involves at most r of the tuples. If it involves the tuples $b_{j_1}, \dots, b_{j_s}$, with $j_1 < \dots < j_s$ in H and $s \leq r$, then its truth value is read off from the diagram of any increasing r -tuple from H that begins with $j_1, \dots, j_s$, and so depends only on the position of the tuples involved. Hence all increasing n -tuples from H have the same diagram over D , and the sequence is indiscernible over D by (5.1). $\odot$

The first half of the proof of Theorem 5.4.5 needed an indiscernible sequence of distinct conjugates of a vertex, and in the Rado graph it was written down by hand. In general we obtain it from disjoint 2-amalgamation and the extraction of indiscernible sequences (see Lemma 5.2.2).

Lemma 5.5.8. *Assume that $\mathcal{K}$ has disjoint 2-amalgamation. Let C be a small set, and let d be an element of $\mathbb{M}$ not in C . Then there is a sequence $(d_i)_{i < \omega}$ of distinct elements, indiscernible over C , with $d_0 = d$.*

Proof. First, there are infinitely many distinct realizations of $\text{tp}(d/C)$. By compactness and saturation, it suffices to find, for every finite $C_0 \subseteq C$ and every n , distinct elements with the same diagram over C_0 as d . Let Y be the substructure of $\mathbb{M}$ on $C_0 \cup \{d\}$. By $n-1$ applications of disjoint 2-amalgamation, there is a structure in $\mathcal{K}$ on $C_0 \cup \{d^1, \dots, d^n\}$, with new elements d^l , in which each $C_0 \cup \{d^l\}$ is a copy of Y over C_0 . By Lemma 5.5.6, it embeds into $\mathbb{M}$ over C_0 . Let $(d'_i)_{i < \omega}$ be distinct realizations of $\text{tp}(d/C)$.

Next, apply Lemma 5.2.2 over C to the sequence (d'_i) . Its Ehrenfeucht–Mostowski type over C contains every formula of $\text{tp}(d/C)$, in one variable, and the formula $y_1 \neq y_2$, so the lemma gives a sequence (d''_i) , indiscernible over C , of distinct realizations of $\text{tp}(d/C)$. An automorphism of $\mathbb{M}$ over C sending d''_0 to d carries it to the required sequence. $\odot$

With these lemmas, the arguments of Section 5.4 go through for Fraïssé limits: disjoint 2-amalgamation supplies the conjugates that make a type divide, and disjoint 3-amalgamation supplies the common realizations that keep it from dividing.

Theorem 5.5.9. *Let $\mathcal{K}$ be a Fraïssé class in a finite relational language with disjoint 2-amalgamation and disjoint 3-amalgamation, and let $T = T_{\mathcal{K}}$.*

- (1) *For a finite tuple a and small sets C, B , the type $\text{tp}(a/C \cup B)$ divides over C if and only if $a \cap B \not\subseteq C$.*
- (2) *If a is a tuple of length n and $D_0 \subseteq \dots \subseteq D_m$ are small sets such that $\text{tp}(a/D_{j+1})$ divides over D_j for every $j < m$, then $m \leq n$. The theory T is simple.*
- (3) *The relation $\perp^t$ of Definition 5.4.8 satisfies all the axioms of Definition 5.4.7 except possibly (UN), together with (AR) and the independence theorem stated in Theorem 5.4.9(c).*

Proof. (1) If $a \cap B \not\subseteq C$, argue as in the first part of the proof of Theorem 5.4.5, with Lemma 5.5.8 in place of the explicit sequence (d''_i) .

Suppose that $a \cap B \subseteq C$. Let $\varphi(x, b)$ belong to $\text{tp}(a/C \cup B)$, with b a finite tuple from $C \cup B$, and let $(b_i)_{i < \omega}$ be indiscernible over C with $b_0 = b$. Let $\Sigma(x)$ be the set of formulas $\psi(x, b_i)$, for $i < \omega$ and $\psi(x, y)$ an $L(C)$ -formula with $\psi(x, b) \in \text{tp}(a/C \cup b)$. A realization a'' of Σ satisfies $a''b_i \equiv_C ab$ for every i , and in particular every $\varphi(x, b_i)$, so it suffices to show, by saturation, that every finite subset Σ_0 of Σ is realized. Here, unlike in the Rado graph,

the structure to be embedded is not determined by the prescriptions, and 3-amalgamation supplies it.

The set Σ_0 involves finitely many indices $i_1 < \dots < i_h$ and finitely many formulas, which by quantifier elimination we may take to be quantifier-free with parameters from a finite set $C_0 \subseteq C$; enlarge C_0 to contain $a \cap C$ and $b \cap C$. Let F be the set of elements not in C that belong to two distinct terms of the sequence. By the claim in the proof of Theorem 5.4.5, which used only indiscernibility, each element of F is a coordinate that is constant along the sequence, so $F \subseteq b$ and $F \subseteq b_i$ for every i . Since $b_i \cap C = b \cap C$ for every i , we obtain $b_i \cap b_j \subseteq C_0 \cup F$ for $i \neq j$. Put $A = C_0 \cup F$.

Let I be the set of indices v with $a_v \notin C$; as before, $a_v \notin b$ for $v \in I$. Take fresh elements a'_v for $v \in I$, equal exactly when the a_v are, and let a' be the tuple with these coordinates and with $a'_v = a_v$ for $v \notin I$. For $l \leq h$, let π_l be the map from $A \cup b_{i_l} \cup a'$ onto $A \cup b \cup a$ that fixes A , sends b_{i_l} to b coordinatewise, and sends a' to a . It is a well defined bijection: a coordinate $b_{i_l}[p]$ lies in A if and only if $b[p]$ does, and then the two are equal (for coordinates in C_0 because $b_{i_l} \equiv_C b$, and for coordinates in F because these are constant); the map is injective on b_{i_l} because $b_{i_l} \equiv_C b$; and the coordinates a_v with $v \in I$ lie outside $A \cup b$, because $a \cap b \subseteq C$ and $F \subseteq b$. Let Z_l be the structure on $A \cup b_{i_l} \cup a'$ that makes π_l an isomorphism onto the substructure of $\mathbb{M}$ on $A \cup b \cup a$. Then Z_l belongs to $\mathcal{K}$, its restriction to $A \cup b_{i_l}$ is the substructure of $\mathbb{M}$ (because π_l there is the restriction of an automorphism of $\mathbb{M}$ over C sending b_{i_l} to b , and fixing F), and its restriction to $A \cup a'$ does not depend on l .

By induction on $l \leq h$, we build $Y_l \in \mathcal{K}$ on $A \cup a' \cup b_{i_1} \cup \dots \cup b_{i_l}$ extending $Z_1, \dots, Z_l$ and the substructure of $\mathbb{M}$ on $A \cup b_{i_1} \cup \dots \cup b_{i_l}$. Put $Y_1 = Z_1$. For $l \geq 2$, apply disjoint 3-amalgamation over A to

$$B_1 = A \cup a', \quad B_2 = A \cup b_{i_1} \cup \dots \cup b_{i_{l-1}}, \quad B_3 = A \cup b_{i_l},$$

with $B_{12} = Y_{l-1}$, $B_{13} = Z_l$, and B_{23} the substructure of $\mathbb{M}$ on $B_2 \cup B_3$. The sets B_1, B_2, B_3 meet pairwise in A : the elements a'_v with $v \in I$ are fresh, and $b_{i_l} \cap b_{i_j} \subseteq A$. The three structures agree on B_1, B_2 and B_3 , by the properties of the Z_l . A solution is the required Y_l (see Figure 5.9).

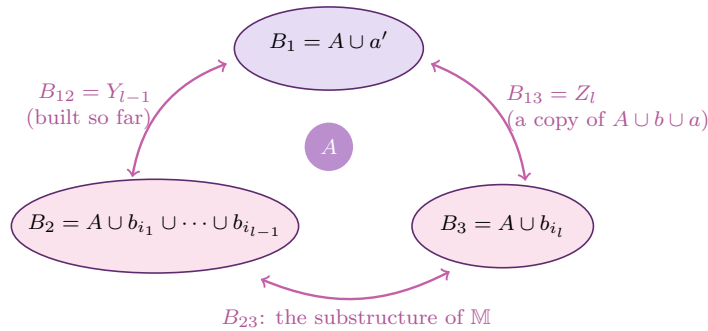


FIGURE 5.9. The inductive step in the proof of Theorem 5.5.9(1). The fresh copy a' of a has already been joined to the first $l - 1$ terms (the structure Y_{l-1}); it must now be joined to b_{i_l} as a is joined to b (the structure Z_l), while the terms of the sequence keep the structure they have in $\mathbb{M}$. Disjoint 3-amalgamation over A glues the three prescriptions into Y_l . In the Rado graph nothing has to be glued, because every relation is binary and every graph is allowed.

By Lemma 5.5.6, applied to Y_h over the finite set $A \cup b_{i_1} \cup \dots \cup b_{i_h}$, there is an embedding g of Y_h into $\mathbb{M}$ fixing this set. Put $a'' = g(a')$. For each l , the map $g \circ \pi_l^{-1}$ is an isomorphism from the substructure on $A \cup b \cup a$ onto the substructure on $A \cup b_{i_l} \cup a''$ that fixes C_0 , sends b to b_{i_l} , and sends a to a'' . Hence $a''b_{i_l}$ and ab satisfy the same quantifier-free formulas over C_0 , and a'' realizes Σ_0 .

(2) The first statement follows from (1) as in the proof of Theorem 5.4.6, and the second follows from the first as there, with Lemma 5.5.7 in place of Lemma 5.4.4.

(3) The axioms other than existence are checked as in the proof of Theorem 5.4.9(a). Existence and the independence theorem are proved as in Theorem 5.4.9(a) and (c), with compactness reducing each to finitely many formulas over a finite part C_0 of the base, and with the structure to be embedded supplied by amalgamation. For existence, given A , M and B , let A_0 and B_0 be finite parts of A and B , and let $C_0 \subseteq M$ be finite and contain $A_0 \cap M$ and $B_0 \cap M$. Apply disjoint 2-amalgamation over the substructure on C_0 to the substructure on $C_0 \cup B_0$ and to a copy over C_0 of the substructure on $C_0 \cup A_0$ whose new elements are disjoint from B_0 . For the independence theorem, apply disjoint 3-amalgamation over the substructure on C_0 , where C_0 contains $(b_1 \cup b_2 \cup c_1) \cap C$, to

$$B_1 = C_0 \cup c', \quad B_2 = C_0 \cup b_1, \quad B_3 = C_0 \cup b_2,$$

where c' is a fresh copy of the coordinates of c_1 outside C , B_{12} makes $C_0 \cup c' \cup b_1$ a copy of $C_0 \cup c_1 \cup b_1$, B_{13} makes $C_0 \cup c' \cup b_2$ a copy of $C_0 \cup c_2 \cup b_2$, and B_{23} is the substructure of $\mathbb{M}$. The sets meet pairwise in C_0 because $b_1 \cap b_2 \subseteq C$, and B_{12} and B_{13} agree on B_1 because $c_1 \equiv_C c_2$. Embed the solution over $C_0 \cup b_1 \cup b_2$ by Lemma 5.5.6, and conclude by saturation. ☺

By Example 5.5.5, Theorem 5.5.9 applies to the Rado graph, the random tournament and the random r -hypergraphs, and also to the pure infinite set, which is stable. The rational order has disjoint 2-amalgamation but not disjoint 3-amalgamation, and it is not simple by Proposition 5.3.2. The Henson graph shows that disjoint 3-amalgamation cannot be dropped even when the forbidden configuration is a triangle rather than an order.

Example 5.5.10. Let $\mathcal{K}$ be the class of finite triangle-free graphs, so that $T_{\mathcal{K}}$ is the theory of the Henson graph $\mathcal{H}$. We show that the formula $\varphi(x; y_1, y_2)$, defined as $E(x, y_1) \wedge E(x, y_2)$, has the 2-tree property.

Let G be the graph with vertices u_σ, v_σ , for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, whose edges are the pairs $\{u_{\sigma \smallfrown \langle i \rangle}, v_{\sigma \smallfrown \langle j \rangle}\}$ with $\sigma \in \omega^{<\omega}$ and $i < j$. Every edge joins a vertex u_τ to a vertex $v_{\tau'}$, so G is bipartite and in particular triangle-free; by Lemma 5.5.6 with $C = \emptyset$, we may assume that G is an induced subgraph of $\mathbb{M}$. Put $b_\sigma = (u_\sigma, v_\sigma)$.

Siblings are 2-inconsistent. Let $i < j$. A solution of $\varphi(x, b_{\sigma \smallfrown \langle i \rangle}) \wedge \varphi(x, b_{\sigma \smallfrown \langle j \rangle})$ would be adjacent to $u_{\sigma \smallfrown \langle i \rangle}$ and to $v_{\sigma \smallfrown \langle j \rangle}$, which are adjacent to each other, and would form a triangle with them; every finite substructure of a model of $T_{\mathcal{K}}$ is triangle-free.

Branches are consistent. Two distinct nodes of a branch have different lengths and so are never siblings. Hence, for $\eta \in \omega^\omega$ and $n \geq 1$, the $2n$ vertices $u_{\eta \smallfrown m}, v_{\eta \smallfrown m}$ ($1 \leq m \leq n$) are pairwise nonadjacent. The graph obtained from them by adding one vertex adjacent to all of them is triangle-free, and Lemma 5.5.6 embeds it over these vertices; the new vertex satisfies $\varphi(x, b_{\eta \smallfrown m})$ for every $m \leq n$. By Remark 5.2.8, condition (a) of Definition 5.2.7 follows.

Hence the theory $T_{\mathcal{K}}$ is not simple. It also has the independence property: for pairwise nonadjacent vertices $b_1, \dots, b_n$ and a set $S \subseteq \{1, \dots, n\}$, adding a vertex adjacent exactly

to the b_l with $l \in S$ creates no triangle, so the edge relation shatters $\{b_1, \dots, b_n\}$. Figure 5.10 shows the siblings of one node.

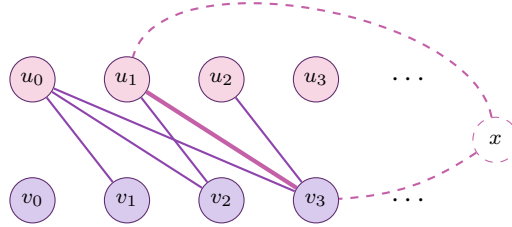


FIGURE 5.10. The siblings of one node in Example 5.5.10, written $b_i = (u_i, v_i)$. The vertex u_i is joined to v_j exactly when $i < j$: the bipartite graph is a ladder (compare Figure 1.1). A common solution x of $\varphi(x, b_1)$ and $\varphi(x, b_3)$ would close the triangle x, u_1, v_3 (pink), so siblings are 2-inconsistent. Along a branch, the pairs are pairwise nonadjacent, and a common neighbour exists.



Remark 5.5.11. The proof of Theorem 5.5.9(2) shows that a type of an n -tuple admits at most n successive dividing steps. In the language of simplicity theory, this says that the theory T is *supersimple*, and that every type of a single element has *SU-rank* at most 1. A theory is supersimple if the type of every finite tuple over every set B does not divide over some finite subset of B . The *SU-rank* of a type p over A is at least $\alpha + 1$ if p has an extension of *SU-rank* at least α that divides over A , and it is at least a limit ordinal λ if it is at least every $\alpha < \lambda$ (see [Wag00] or [Cas11]). Kruckman [Kru19] studied Fraïssé classes with disjoint n -amalgamation for every n , and showed that their limits have pseudofinite theories. The class of finite triangle-free graphs is the basic example of a *free amalgamation* class, in which the union of two structures over a common substructure, with no new relations, is always an amalgam. Shelah showed that the Henson graph has the property SOP_3 of his hierarchy of strict order properties, but not SOP_4 [She96, Claim 2.8], and Conant [Con17] developed the model theory of free amalgamation theories in general, showing in particular that they are NSOP_4 .

5.6. The real-valued case

Chapters 1 and 4 each closed by replacing formulas with bounded functions on arbitrary sets, and the same can be done for the tree property. Two things are lost in the passage. There is no compactness, so a tree must carry its own witnesses, branch by branch; and there is no definability in the sense of Chapter 1, so the proof that stable theories are simple, which counted definitions (see Proposition 5.3.1), has nothing to count. What takes the place of a definition is an approximation by polynomials in instances (see Theorem 1.2.11), and approximations can be counted just as well, provided that we count them only up to a tolerance. The tolerance is supplied by a gap between the values that the tree requires on its branches and the values that it forbids among siblings. We treat the case $k = 2$ of Definition 5.2.7, in which siblings are pairwise incompatible.

Definition 5.6.1. Let A, B be sets, let $\varphi: A \times B \rightarrow [0, 1]$ be a function, and let $r < s$ be reals. The function φ has the *2-tree property with gap* (r, s) if there are $b_\sigma \in B$, for $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, such that the following hold:

- (a) for every $\sigma \in \omega^{<\omega}$ with $\sigma \neq \emptyset$, there is $a \in A$ with $\varphi(a, b_{\sigma|_m}) \geq s$ for $1 \leq m \leq |\sigma|$;
- (b) for every $\tau \in \omega^{<\omega}$ and all $i \neq j$, there is no $a \in A$ with $\varphi(a, b_{\tau \smallfrown \langle i \rangle}) > r$ and $\varphi(a, b_{\tau \smallfrown \langle j \rangle}) > r$.

The function φ has the *2-tree property* if it has the 2-tree property with gap (r, s) for some reals $r < s$.

Condition (a) asks for witnesses to the finite parts of the branches only; on arbitrary sets there is nothing that would turn them into witnesses for whole branches. Only gaps with $0 \leq r < s \leq 1$ matter for a $[0, 1]$ -valued function: a gap with $s > 1$ violates condition (a), and a gap with $r < 0$ violates condition (b), since condition (a) provides elements of A and every value exceeds r . For a formula the two definitions agree. Fix reals $0 \leq r < s \leq 1$; for a $\{0, 1\}$ -valued function, a value is at least s , or larger than r , exactly when it is 1. A formula $\varphi(x, y)$ has the 2-tree property of Definition 5.2.7 if and only if, for some model M , the function φ on $M^{|x|} \times M^{|y|}$ has the 2-tree property with gap (r, s) . Indeed, if the tuples b_σ witness the 2-tree property in $\mathbb{M}$, a model $M \preceq \mathbb{M}$ containing all of them (by the downward Löwenheim–Skolem theorem) contains witnesses for the finite parts of the branches, by elementarity, and pairs of siblings remain inconsistent in M . Conversely, every model of T embeds elementarily in $\mathbb{M}$, and a tree as in Definition 5.6.1 in a model $M \preceq \mathbb{M}$ is a tree in $\mathbb{M}$ whose finite branches are consistent and whose siblings are 2-inconsistent; Remark 5.2.8 gives the 2-tree property.

The proof that a stable function has no such tree counts branches, and the branches are counted in binary trees of infinite height, the real-valued form of the trees of Definition 2.1.2. In Lemma 5.6.3 below, we show that every 2-tree contains one.

Definition 5.6.2. Let $\varphi: A \times B \rightarrow [0, 1]$ be a function, and let $r < s$ be reals. A *binary tree of infinite height for φ with gap* (r, s) consists of families $(b_\sigma : \sigma \in 2^{<\omega})$ in B and $(a_\tau : \tau \in 2^{<\omega})$ in A such that, for all $\sigma, \tau \in 2^{<\omega}$,

$$\sigma \smallfrown \langle 1 \rangle \sqsubseteq \tau \implies \varphi(a_\tau, b_\sigma) \geq s, \quad \sigma \smallfrown \langle 0 \rangle \sqsubseteq \tau \implies \varphi(a_\tau, b_\sigma) \leq r.$$

Lemma 5.6.3. *If a function $\varphi: A \times B \rightarrow [0, 1]$ has the 2-tree property with gap (r, s) , then it has a binary tree of infinite height with gap (r, s) .*

Proof. Let (b_σ) witness the 2-tree property. Define $f: 2^{<\omega} \rightarrow \omega^{<\omega}$ by recursion: $f(\emptyset) = \emptyset$, $f(\sigma \smallfrown \langle 1 \rangle) = f(\sigma) \smallfrown \langle 0 \rangle$ and $f(\sigma \smallfrown \langle 0 \rangle) = f(\sigma) \smallfrown \langle 1 \rangle$. Then f preserves initial segments, and $f(\sigma \smallfrown \langle 0 \rangle)$ and $f(\sigma \smallfrown \langle 1 \rangle)$ are siblings. For $\sigma \in 2^{<\omega}$ put $b'_\sigma = b_{f(\sigma \smallfrown \langle 1 \rangle)}$, and for $\tau \in 2^{<\omega}$ let $a'_\tau \in A$ satisfy $\varphi(a'_\tau, b_{f(\tau)|_m}) \geq s$ for $1 \leq m \leq |f(\tau)|$, as condition (a) allows (for $\tau = \emptyset$ any element of A will do).

Let $\sigma, \tau \in 2^{<\omega}$ be given. If $\sigma \smallfrown \langle 1 \rangle \sqsubseteq \tau$, then $f(\sigma \smallfrown \langle 1 \rangle) \sqsubseteq f(\tau)$, so b'_σ lies on the branch below $f(\tau)$ and $\varphi(a'_\tau, b'_\sigma) \geq s$. If $\sigma \smallfrown \langle 0 \rangle \sqsubseteq \tau$, then $f(\sigma \smallfrown \langle 0 \rangle) \sqsubseteq f(\tau)$, so $\varphi(a'_\tau, b_{f(\sigma \smallfrown \langle 0 \rangle)}) \geq s > r$. The nodes $f(\sigma \smallfrown \langle 0 \rangle)$ and $f(\sigma \smallfrown \langle 1 \rangle)$ are siblings, and condition (b) gives $\varphi(a'_\tau, b'_\sigma) \leq r$. Thus the families (b'_σ) and (a'_τ) form a binary tree of infinite height with gap (r, s) . $\odot$

Figure 5.11 shows the embedding f . At the node $f(\sigma)$, the binary tree keeps two children of the ω -branching tree. The child 0 carries the parameter b'_σ of the binary node σ , and a leaf that turns right at σ passes through it. A leaf that turns left passes through the child 1 instead, takes values at least s there, and so, by the incompatibility of siblings, takes values at most r at the child 0.

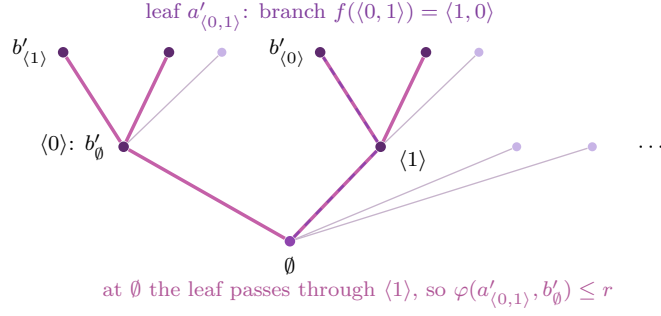


FIGURE 5.11. Lemma 5.6.3. The binary tree lives inside the ω -branching tree on the children 0 and 1 of the nodes it uses (pink); the other children (pale) are discarded. The parameter of the binary node σ is the child 0 of $f(\sigma)$. A leaf that turns right at σ passes through this child; a leaf that turns left passes through its sibling, the child 1, and the incompatibility of siblings forces the value at the child 0 down to at most r .

Theorem 5.6.4. *A stable function $\varphi: A \times B \rightarrow [0, 1]$ has no binary tree of infinite height with a gap. In particular, a stable function does not have the 2-tree property.*

Proof. The idea is to count. A binary tree of infinite height has $2^{\aleph_0}$ branches, and the types along distinct branches differ by at least $s - r$ at some node. A stable function, on the other hand, has only countably many approximate definitions over a countable set, and two branches with nearly the same definition cannot differ by $s - r$ anywhere.

Suppose that (b_σ) and (a_τ) form a binary tree of infinite height for φ with gap (r, s) , and put $\varepsilon = (s - r)/5$. Let $A_0 = \{a_\tau : \tau \in 2^{<\omega}\}$ and $B_0 = \{b_\sigma : \sigma \in 2^{<\omega}\}$, and let φ_0 be the restriction of φ to $A_0 \times B_0$. Every instance of Definition 1.2.5 for φ_0 is an instance for φ , so φ_0 is stable.

© **Step 1: types along branches.** Fix a nonprincipal ultrafilter $\mathcal{U}$ on ω , and for $\eta \in 2^\omega$ put $p_\eta = \lim_{n, \mathcal{U}} \text{ltp}_{\varphi_0}(a_{\eta|n}) \in \text{LS}(\varphi_0)$. Let $\sigma \sqsubseteq \eta$ be given. For every $n > |\sigma|$ we have $\sigma \smallfrown \langle \eta(|\sigma|) \rangle \sqsubseteq \eta|n$, so $\varphi(a_{\eta|n}, b_\sigma)$ is at least s if $\eta(|\sigma|) = 1$ and at most r if $\eta(|\sigma|) = 0$, and the same bound holds for $p_\eta(b_\sigma)$. If $\eta \neq \eta'$ and σ is the longest common initial segment of η and η' , then $\eta(|\sigma|) \neq \eta'(|\sigma|)$, and therefore

$$(5.2) \quad |p_\eta(b_\sigma) - p_{\eta'}(b_\sigma)| \geq s - r.$$

© **Step 2: approximate definitions.** Let $\hat{\varphi}_0$ be the separately continuous function of Proposition 1.2.8 for φ_0 ; as in the proof of Proposition 4.2.2, we have $\hat{\varphi}_0(p, \text{rtp}_{\varphi_0}(b)) = p(b)$ for all $p \in \text{LS}(\varphi_0)$ and $b \in B_0$. Let $\eta \in 2^\omega$ be given. By Theorem 1.2.11, in the form given by its proof, there are $a_1, \dots, a_k \in A_0$ and a real polynomial P with $|\hat{\varphi}_0(p_\eta, q) - P(q(a_1), \dots, q(a_k))| \leq \varepsilon$ for all $q \in \text{RS}(\varphi_0)$. Taking $q = \text{rtp}_{\varphi_0}(b)$ gives

$$|p_\eta(b) - P(\varphi(a_1, b), \dots, \varphi(a_k, b))| \leq \varepsilon \quad \text{for all } b \in B_0.$$

Each monomial is bounded by 1 on $[0, 1]^k$, so changing the coefficients of P by small enough amounts gives a polynomial P_η with rational coefficients with $|P - P_\eta| \leq \varepsilon$ on $[0, 1]^k$. Put $\bar{a}_\eta = (a_1, \dots, a_k)$. Then $|p_\eta(b) - P_\eta(\varphi(a_1, b), \dots, \varphi(a_k, b))| \leq 2\varepsilon$ for all $b \in B_0$.

© **Step 3: counting.** The set A_0 is countable and there are countably many polynomials with rational coefficients, so there are only countably many pairs $(\bar{a}_\eta, P_\eta)$, while 2^ω is uncountable. Hence there are $\eta \neq \eta'$ with $(\bar{a}_\eta, P_\eta) = (\bar{a}_{\eta'}, P_{\eta'})$, and then $|p_\eta(b) - p_{\eta'}(b)| \leq 4\varepsilon < s - r$ for every $b \in B_0$. This contradicts (5.2). The last statement follows from Lemma 5.6.3. ☺

For formulas, Theorem 5.6.4 gives a second proof of the case $k = 2$ of Proposition 5.3.1. If $\varphi(x, y)$ is stable, then it is stable on every model (see Remark 1.2.2), so the corresponding $\{0, 1\}$ -valued function on every model is stable in the sense of Definition 1.2.5 (see Proposition 1.2.4). By Theorem 5.6.4 this function has no 2-tree property with any gap, and by the paragraph following Definition 5.6.1 the formula φ does not have the 2-tree property. The two proofs count different things: definitions over a model of size λ in Proposition 5.3.1, approximate definitions over a countable set here. The second count is the real-valued form of the counting in the proof of (1) $\Rightarrow$ (2) in Theorem 2.5.1: a binary tree of infinite height is a Cantor scheme of types, and stability leaves room only for countably many types up to a tolerance.

The converse of the last statement of Theorem 5.6.4 fails. Let M be a model of T_{RG} . By the extension axioms (see Example 4.1.4), every sequence of distinct vertices of M is independent for the edge relation, in the sense of Definition 4.2.1, so the edge relation is not NIP and hence, by Proposition 4.2.2, not stable. On the other hand, the random graph is simple (see Theorem 5.4.6), so by the paragraph following Definition 5.6.1 the edge relation does not have the 2-tree property on any model.

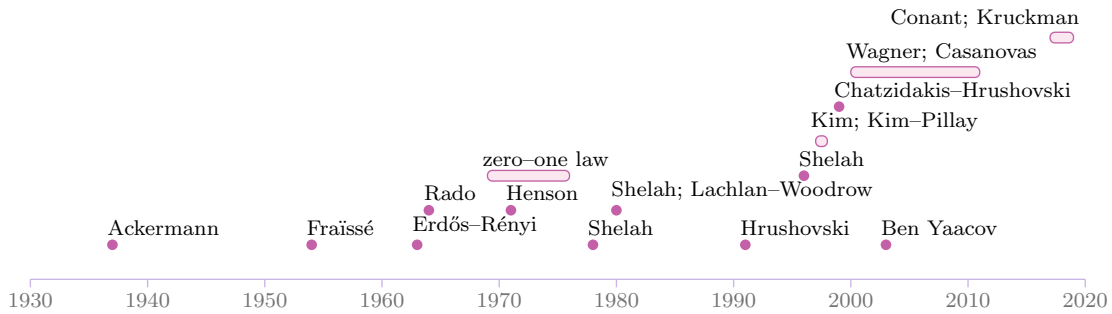
5.7. Historical remarks

- ⌘ **Section 5.1.** Working inside a single large saturated and strongly homogeneous model goes back to Shelah [She90], who wrote $\mathfrak{C}$ for this model. The name *monster model* became standard later. The notion of ω -stability comes from Morley's categoricity theorem [Mor65]. The study of λ -stability for arbitrary λ , including the fact that stability is equivalent to λ -stability for all λ with $\lambda^{[T]} = \lambda$, is due to Shelah [She71, She90].
- ⌘ **Section 5.2.** Dividing, forking and the tree property were introduced by Shelah in *Classification Theory* ([She90, Chapter III]; first edition 1978), and the simple theories, those in which no formula has the tree property, were named and studied in his paper [She80]. Indiscernible sequences go back to Ehrenfeucht and Mostowski, and were made into a standard tool of classification theory by Morley and Shelah [She90]. For the next fifteen years the simple unstable theories received little attention. They returned to the center of model theory with Hrushovski's work on pseudofinite fields, circulated as a manuscript from 1991 and published in [Hru02], whose theories are simple and unstable, and with the work of Chatzidakis and Hrushovski on difference fields [CH99].
- ⌘ **Section 5.3.** That stable theories are simple, and that the strict order property implies the tree property, go back to Shelah [She90, Chapter III], [She80]; the dichotomy of Fact 5.3.3 is in [She90, Chapter II]. Kim's thesis and his paper [Kim98] proved that forking and dividing agree in simple theories and that forking independence is symmetric; Kim and Pillay [KP97] proved the independence theorem, abstracting it from Hrushovski's work on pseudofinite fields, and characterized simple theories by their independence relation. The title of this chapter is that of their survey [KP98]. The monographs of Wagner [Wag00] and Casanovas [Cas11] develop the theory in full.
- ⌘ **Section 5.4.** The graph of Example 5.4.1 is implicit in Ackermann's coding of the hereditarily finite sets by natural numbers [Ack37]: the relation “the i -th binary digit of j is 1” encodes membership, and symmetrizing it gives the Rado graph. Erdős and Rényi [ER63]

proved that a random countable graph, with each edge chosen independently with probability $\frac{1}{2}$, is isomorphic with probability 1 to a fixed graph, and Rado [Rad64] gave the explicit construction of Example 5.4.1 and proved that the graph is universal for countable graphs. The theory T_{RG} is also the almost sure theory of finite graphs: by the zero-one law of Glebskiĭ, Kogan, Liogon'kiĭ and Talanov [GKL+69] and of Fagin [Fag76], a first-order sentence holds in almost all finite graphs if and only if it holds in the Rado graph. Cameron's survey [Cam97] collects the many faces of this graph.

✂ **Section 5.5.** Fraïssé [Fra54] introduced ages and proved Fact 5.5.2(a), generalizing Cantor's characterization of $(\mathbb{Q}, <)$ as the unique countable dense linear order without endpoints. Henson [Hen71] constructed the homogeneous K_n -free graphs, and Lachlan and Woodrow [LW80] proved that the countably infinite ultrahomogeneous graphs are, up to complementation, the disjoint unions of complete graphs of equal size, the Rado graph, and the Henson graphs. Disjoint n -amalgamation, as a tool for studying the theories of Fraïssé limits, was studied systematically by Kruckman [Kru19].

✂ **Section 5.6.** Simplicity for structures with real-valued relations was developed by Ben Yaacov [Ben03] in the framework of compact abstract theories, which includes continuous logic. The argument of Theorem 5.6.4 is elementary by comparison: it combines the polynomial approximation of Theorem 1.2.11 with the counting of branches in the proof of Theorem 2.5.1.



- 1980

•

Shelah; Lachlan–Woodrow. Shelah [She80] names and studies the simple theories, those without the tree property. Lachlan and Woodrow [LW80] classify the countable ultrahomogeneous graphs.
- 1991

•

Hrushovski. Hrushovski’s manuscript on pseudofinite fields, published in [Hru02], shows that their theories are simple and proves an independence theorem for them.
- 1996

•

Shelah. Shelah [She96] introduces the hierarchy of properties SOP_n ; the Henson graph has SOP_3 .
- 1997–1998

•

Kim; Kim–Pillay. Kim [Kim98] proves that forking equals dividing in simple theories and that forking is symmetric. Kim and Pillay [KP97] prove the independence theorem; their survey [KP98] is titled *From stability to simplicity*.
- 1999

•

Chatzidakis–Hrushovski. Chatzidakis and Hrushovski [CH99] develop the model theory of difference fields, a central example of a simple unstable theory.
- 2000–2011

•

Wagner; Casanovas. The monographs of Wagner [Wag00] and Casanovas [Cas11].
- 2003

•

Ben Yaacov. Ben Yaacov [Ben03] extends simplicity to compact abstract theories, a framework that includes continuous logic.
- 2017–2019

•

Conant; Kruckman. Conant [Con17] develops free amalgamation theories; Kruckman [Kru19] studies disjoint n -amalgamation and pseudofinite countably categorical theories.

Chapter 6

Stability and notions of independence

Ladders, trees and definitions all describe stability from inside a single formula. Independence describes stability globally: in this chapter we show that the stable theories are exactly those carrying a notion of independence, written $A \downarrow_M B$ and read “ A is independent from B over M ”, that obeys a short list of natural axioms. The most important of these are finite character, symmetry, and the existence and uniqueness of independent extensions. The idea is that, given a type p over a model M and a set B , an independent extension of p to $M \cup B$ adds no information beyond p . Stability is precisely the condition that such an extension always exists and is unique.

Both directions rest on Chapter 2. To obtain independence from stability we use definability of types: the independent extension is the *heir*, the extension of p given by the same definition scheme as p . The converse is a counting argument of the kind used there: uniqueness and local character together bound the number of types.

We keep the conventions of Section 5.1, and we measure independence against the axioms of Definition 5.4.7. In Chapter 5 the random graph carried a relation with every axiom but uniqueness; here uniqueness takes center stage.

6.1. Definitions and heirs

In a stable theory every type over a model has a definition scheme, one definition for each formula (see Lemma 6.1.1). A definition can be evaluated at parameters outside the model, so the scheme extends the type to larger sets of parameters, and the result is the heir (see Definition 6.1.2). The section closes with the fact that will make independence symmetric: definitions over a model commute (see Lemma 6.1.4). In Subsection 6.1.1 we show that this commutation is the exchange of iterated ultrafilter limits that defines stability in Section 1.2.

Lemma 6.1.1. *Assume that T is stable, and let $p \in S_x(M)$ be given, where x may be infinite. For every $L(M)$ -formula $\varphi(x', y)$, with x' a finite subtuple of x and y finite, there is an $L(M)$ -formula $d_p\varphi(y)$ such that for all $b \in M$, we have*

$$\varphi(x', b) \in p \iff M \models d_p\varphi(b).$$

Any two formulas with this property are equivalent in $\mathbb{M}$.

Proof. Write $\varphi(x', y) = \varphi_0(x', y, m)$ with $\varphi_0 \in L$ and $m \in M$. The set of formulas $\varphi_0(x', c)^u$ ($c \in M$) that belong to p is a complete φ_0 -type over M in the sense of Definition 1.1.1. Since φ_0 does not have the order property, Theorem 1.1.6 shows that this φ_0 -type has a definition $\psi(z)$ over M , which is a positive Boolean combination of instances of φ_0 . Here the tuple z corresponds to the parameter variables y followed by those replaced by m . Then the formula $d_p\varphi(y) := \psi(y, m)$ works. If $\psi_1(y)$ and $\psi_2(y)$ both work, then we have $M \models \forall y (\psi_1 \leftrightarrow \psi_2)$. This is an $L(M)$ -sentence, so this sentence also holds in $\mathbb{M}$. ☺

The family $(d_p\varphi)_\varphi$ given by Lemma 6.1.1, determined up to equivalence in $\mathbb{M}$, is called the *definition scheme* of p .

Definition 6.1.2. Assume that T is stable. Let $p \in S_x(M)$ be given, and let B be small. Define

$$p|B := \{\varphi(x', b) : \varphi(x', y) \in L(M), b \in B, \mathbb{M} \models d_p\varphi(b)\}.$$

The set $p|B$ is called the *heir* of p over MB .

Every $L(MB)$ -formula has the form $\varphi(x', b)$ with $\varphi \in L(M)$ and $b \in B$, so the heir $p|B$ is a set of $L(MB)$ -formulas. Figure 6.1 shows the construction. The next lemma shows that the heir is in fact a complete type extending p , and that heirs are compatible with enlarging the parameter set and the base model.

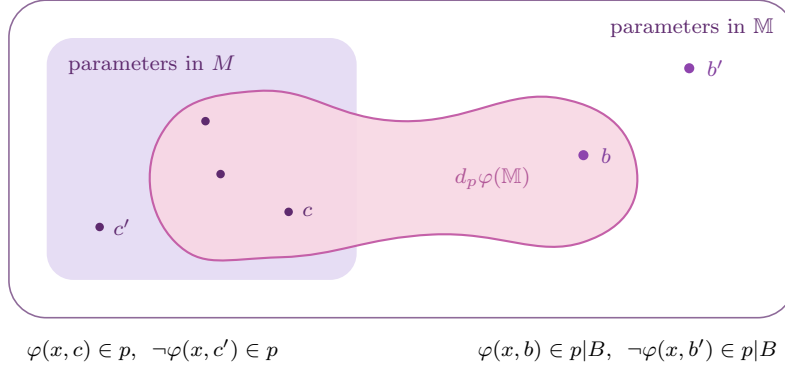


FIGURE 6.1. Heirs (see Definition 6.1.2). The definition $d_p\varphi$ is a formula with parameters in M , so it defines a set of parameters in all of $\mathbb{M}$ (pink). Inside M (shaded) it records which instances $\varphi(x, c)$ belong to p ; outside M the same set decides which instances $\varphi(x, b)$ belong to the heir $p|B$. The heir is the type p evaluated, by its own definitions, at parameters that M has never seen.

Lemma 6.1.3. Assume that T is stable, let $p \in S_x(M)$ be given, and let B and D be small sets. Then the following hold:

- (a) The set $p|B$ belongs to $S_x(MB)$ and extends p .
- (b) If $B \subseteq D$, then the type $p|B$ is the restriction of $p|D$ to MB .
- (c) If N is a model with $M \subseteq N$, then the type $p|N$ has the same definition scheme as p , and we have $(p|N)|B = p|NB$.

Proof. (a) *Completeness.* Let $\varphi \in L(M)$ be given. Since p is complete, for every $c \in M$ exactly one of $\varphi(x', c)$, $\neg\varphi(x', c)$ lies in p . Hence we have $M \models \forall y (d_p(\neg\varphi)(y) \leftrightarrow \neg d_p\varphi(y))$, and the same holds in $\mathbb{M}$. Thus for $b \in B$ exactly one of $\varphi(x', b)$, $\neg\varphi(x', b)$ belongs to $p|B$.

Consistency. Let $\varphi_i(x_i, b_i) \in p|B$ be given for $i < n$, and put $\chi(\bar{x}; y_0, \dots, y_{n-1}) = \bigwedge_{i < n} \varphi_i(x_i, y_i)$, where $\bar{x}$ lists the variables in the x_i . For $\bar{c} \in M$, the formula $\chi(\bar{x}, \bar{c})$ is in p if and only if each $\varphi_i(x_i, c_i)$ is, and in that case the formula $\chi(\bar{x}, \bar{c})$ is satisfiable in M . Therefore we have

$$M \models \forall \bar{y} \left(\bigwedge_{i < n} d_p\varphi_i(y_i) \rightarrow d_p\chi(\bar{y}) \right) \quad \text{and} \quad M \models \forall \bar{y} (d_p\chi(\bar{y}) \rightarrow \exists \bar{x} \chi(\bar{x}, \bar{y})).$$

Both sentences transfer to $\mathbb{M}$. Applied to $\bar{b} = (b_0, \dots, b_{n-1})$, they give $\mathbb{M} \models \exists \bar{x} \bigwedge_i \varphi_i(x_i, b_i)$. By compactness, the set $p|B$ is consistent.

Extension. For $c \in M$ we have $\varphi(x', c) \in p \iff M \models d_p\varphi(c)$, which holds if and only if $\varphi(x', c) \in p|B$.

(b) This is immediate from the definition.

(c) By (a), the type $p|N$ extends p . For $n \in N$ we have $\varphi(x', n) \in p|N$ if and only if $\mathbb{M} \models d_p\varphi(n)$, so the formula $d_p\varphi$ is a definition of $p|N$ for $\varphi \in L(M)$. For an $L(N)$ -formula $\varphi(x', y) = \varphi_0(x', y, n)$ with $\varphi_0 \in L(M)$, the formula $d_p\varphi_0(y, n)$ defines $p|N$. By the uniqueness clause of Lemma 6.1.1, the definition scheme of $p|N$ is thus given, up to equivalence, by that of p . Now for $b \in B$ and $n \in N$, we have

$$\varphi_0(x', b, n) \in (p|N)|B \iff \mathbb{M} \models d_p\varphi_0(b, n) \iff \varphi_0(x', b, n) \in p|NB. \quad \text{☺}$$

The heart of the matter is the following lemma, which says that definitions of types over a model commute; this lemma is the only place in this chapter where the absence of the order property is used directly. To state the lemma, let $p \in S_x(M)$ and $q \in S_y(M)$ be given, with x and y finite, and let $\varphi(x, y) \in L(M)$ be given. As before, we write $d_p\varphi(y)$ for the definition of p for $\varphi(x, y)$, and we write $d_q\varphi(x)$ for the definition of q for φ regarded as a formula in the variables y , with parameter variables x .

Lemma 6.1.4 (Harrington's lemma). *Assume that T is stable, and let $p \in S_x(M)$, $q \in S_y(M)$ and $\varphi(x, y) \in L(M)$ be given, where x and y are finite. Then we have*

$$d_p\varphi(y) \in q \iff d_q\varphi(x) \in p.$$

Proof. The idea is to realize p and q alternately, so that each pair (a_i, b_j) consults one definition below the diagonal and the other on and above the diagonal; if the two definitions disagreed, the result would be a ladder. Choose recursively $a_i, b_i \in \mathbb{M}$ ($i < \omega$) such that a_i realizes $p| \{b_j : j < i\}$ and b_i realizes $q| \{a_j : j \leq i\}$. This is possible by Lemma 6.1.3(a) and saturation. In particular, we have $a_i \models p$ and $b_i \models q$.

If $j < i$, then since $a_i \models p| \{b_j\}$ and $b_j \models q$, we have

$$\mathbb{M} \models \varphi(a_i, b_j) \iff \mathbb{M} \models d_p\varphi(b_j) \iff d_p\varphi(y) \in q.$$

If $i \leq j$, then since $b_j \models q| \{a_i\}$ and $a_i \models p$, we have

$$\mathbb{M} \models \varphi(a_i, b_j) \iff \mathbb{M} \models d_q\varphi(a_i) \iff d_q\varphi(x) \in p.$$

Suppose that the two right-hand sides differ, and write $\varphi(x, y) = \varphi_0(x, y, m)$ with $\varphi_0 \in L$ and $m \in M$. Put $b'_j = b_j m$. Then either $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i \leq j$ for all i, j , or $\mathbb{M} \models \varphi_0(a_i, b'_j) \iff i > j$ for all i, j . In the first case the formula $\varphi_0(x; y, z)$ has the order property directly, and in the second by Remark 1.1.5. Either way this contradicts stability. ☹

Figure 6.2 shows the configuration built in the proof.

6.1.1. Harrington's lemma as limit exchange

Harrington's lemma is the exchange of iterated ultrafilter limits of Section 1.2, read in the language of types. We now make this precise, and in doing so obtain a second proof of the lemma that does not build a ladder by hand.

The first step is to represent types over M by ultrafilters on tuples from M . Let x and y be finite tuples of variables. Every type $p \in S_x(M)$ is finitely satisfiable in M : if $\psi(x) \in p$, then we have $\mathbb{M} \models \exists x \psi(x)$, and since ψ is an $L(M)$ -formula and $M \preceq \mathbb{M}$, some $a \in M^{|x|}$ satisfies ψ . Hence the sets $\psi(M) = \{a \in M^{|x|} : \mathbb{M} \models \psi(a)\}$, for $\psi \in p$, have the finite intersection property, and the type p is the *trace* of an ultrafilter $\mathcal{U}$ on $M^{|x|}$, in the sense that

$$\psi(x) \in p \iff \psi(M) \in \mathcal{U} \quad \text{for every } L(M)\text{-formula } \psi(x).$$

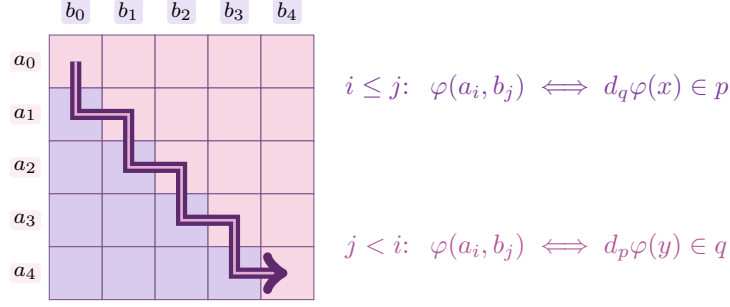


FIGURE 6.2. The proof of Lemma 6.1.4. The elements are realized in the order $a_0, b_0, a_1, b_1, \dots$ (the pink path): each a_i realizes the heir of p over the b_j already chosen, and each b_j the heir of q over the a_i already chosen. Below the diagonal (lavender) the truth value of $\varphi(a_i, b_j)$ is read off from the definition of p , and on and above it (pink) from the definition of q . Each region is constant, so if the two answers differed, the array would be the staircase of a ladder (see Figure 1.1).

Similarly, let $q \in S_y(M)$ be the trace of an ultrafilter $\mathcal{V}$ on $M^{|y|}$, and let $\varphi(x, y) \in L(M)$ be given. As in Section 1.2, we identify φ with the $\{0, 1\}$ -valued function $(a', b') \mapsto \varphi(a', b')$ on $M^{|x|} \times M^{|y|}$. For $b' \in M^{|y|}$ we have $\mathbb{M} \models d_p \varphi(b')$ if and only if $\varphi(x, b') \in p$, that is, if and only if $\varphi(M, b') \in \mathcal{U}$, that is, if and only if $\lim_{a', \mathcal{U}} \varphi(a', b') = 1$. Therefore we have

$$d_p \varphi(y) \in q \iff \{b' \in M^{|y|} : \mathbb{M} \models d_p \varphi(b')\} \in \mathcal{V} \iff \lim_{b', \mathcal{V}} \lim_{a', \mathcal{U}} \varphi(a', b') = 1,$$

and in the same way we have

$$d_q \varphi(x) \in p \iff \lim_{a', \mathcal{U}} \lim_{b', \mathcal{V}} \varphi(a', b') = 1.$$

Thus the two sides of Harrington's lemma are the statements that the two iterated limits of φ along $\mathcal{U}$ and $\mathcal{V}$ equal 1, and the lemma says exactly that these iterated limits agree. Note that no heirs, saturation or realizations are involved: the computation takes place entirely in M .

When T is stable, the limits do agree. Write $\varphi(x, y) = \varphi_0(x, y, m)$ with $\varphi_0 \in L$ and $m \in M$. The formula $\varphi_0(x, y, z)$ does not have the order property, so by Remark 1.2.2 it is stable on M , and therefore so is its restriction φ to the parameters of the form (b', m) , since a ladder for φ or $\neg \varphi$ would be one for φ_0 or $\neg \varphi_0$. By Proposition 1.2.4, the function φ is then stable in the sense of Definition 1.2.5, and by Corollary 1.2.6, applied to the families $(a')_{a' \in M^{|x|}}$ and $(b')_{b' \in M^{|y|}}$, the two iterated limits are equal. This proves Lemma 6.1.4 again.

The two proofs are the same argument at different levels of abstraction. The alternating realizations (a_i, b_j) in the first proof consult one definition below the diagonal and the other on and above it, exactly as the sequences of Proposition 1.2.3 display one iterated limit above the diagonal and the other below it; and in both cases a disagreement is converted into a ladder.

The computation also explains what the two orders of realization mean. For a small set B , the *coheir* of $\mathcal{U}$ over MB is the type

$$q_{\mathcal{U}}|B = \{\varphi(x, b) : \varphi(x, y) \in L(M), b \text{ a finite tuple from } B, \{a \in M^{|x|} : \mathbb{M} \models \varphi(a, b)\} \in \mathcal{U}\}.$$

We first collect its basic properties, which are used again in Chapter 7, and then return to the two orders of realization.

Lemma 6.1.5 (Coheirs). *Let M be a model, let $\mathcal{U}$ be an ultrafilter on $M^{|x|}$, and let B and B' be small sets.*

- (a) *The set $q_{\mathcal{U}}|B$ is a complete type over MB , and it is finitely satisfiable in M .*
- (b) *Every type over MB that is finitely satisfiable in M equals $q_{\mathcal{U}}|B$ for some ultrafilter $\mathcal{U}$ on $M^{|x|}$.*
- (c) *If $B \subseteq B'$, then we have $q_{\mathcal{U}}|B \subseteq q_{\mathcal{U}}|B'$.*
- (d) *The coheir is invariant over M : if $\sigma \in \text{Aut}(\mathbb{M}/M)$, then we have $\sigma(q_{\mathcal{U}}|B) = q_{\mathcal{U}}|\sigma(B)$.*

Proof. For an $L(MB)$ -formula $\varphi(x, b)$, write $\varphi(M, b) = \{a \in M^{|x|} : \mathbb{M} \models \varphi(a, b)\}$. These sets satisfy $(\neg\varphi)(M, b) = M^{|x|} \setminus \varphi(M, b)$ and $(\varphi \wedge \psi)(M, b) = \varphi(M, b) \cap \psi(M, b)$.

(a) Since $\mathcal{U}$ is an ultrafilter, exactly one of $\varphi(M, b)$ and its complement belongs to $\mathcal{U}$, so exactly one of $\varphi(x, b)$ and $\neg\varphi(x, b)$ belongs to $q_{\mathcal{U}}|B$. If $\varphi_1(x, b_1), \dots, \varphi_n(x, b_n)$ belong to $q_{\mathcal{U}}|B$, the intersection of the sets $\varphi_i(M, b_i)$ belongs to $\mathcal{U}$ and so is non-empty, and any of its elements is a tuple from M satisfying all n formulas. Hence the set $q_{\mathcal{U}}|B$ is finitely satisfiable in M , in particular consistent, and complete.

(b) Let p be a type over MB finitely satisfiable in M . The sets $\varphi(M, b)$, for $\varphi(x, b) \in p$, have the finite intersection property, so they are contained in an ultrafilter $\mathcal{U}'$ on $M^{|x|}$. Then we have $p \subseteq q_{\mathcal{U}'}|B$, and both are complete, so they are equal.

(c) This is immediate from the definition.

(d) Let a be a tuple from M . Since σ fixes a and the parameters of φ from M , we have $\mathbb{M} \models \varphi(a, b)$ if and only if $\mathbb{M} \models \varphi(a, \sigma(b))$. Hence we have $\varphi(M, b) = \varphi(M, \sigma(b))$, and it follows that $\varphi(x, b) \in q_{\mathcal{U}}|B$ if and only if $\varphi(x, \sigma(b)) \in q_{\mathcal{U}}|\sigma(B)$. $\odot$

We return to the two orders of realization. If b realizes $q_{\mathcal{V}}|\emptyset$ and a realizes $q_{\mathcal{U}}|\{b\}$, then unwinding the definitions gives

$$\mathbb{M} \models \varphi(a, b) \iff \lim_{a', \mathcal{U}} \lim_{b', \mathcal{V}} \varphi(a', b') = 1,$$

and realizing the two coheirs in the other order replaces this iterated limit by the one in the other order. Thus the two orders of realizing coheirs give the same type exactly when the iterated limits commute. In a stable theory, moreover, the coheir $q_{\mathcal{U}}|B$ is the heir of $q_{\mathcal{U}}|\emptyset$, because both are nonforking extensions and nonforking extensions of types over models are unique (see Remark 6.3.2 and [Pil96, TZ12]). Read in this way, the symmetry of independence proved in Section 6.2 is the commutation of ultrafilter limits: the logical counterpart of exchanging the order of two limits in analysis.

6.2. Independence relations

The axioms of Definition 5.4.7 record how the heir relation behaves. When T is stable the heir relation satisfies all of them, and conversely a relation satisfying just two of them, uniqueness and local character, forces T to be stable (see Theorem 6.2.2). In Chapter 5 we met relations satisfying all the axioms except uniqueness (see Theorems 5.4.9 and 5.5.9); the theorem below explains why uniqueness is exactly what they lack.

In terms of types, the axioms (EX) and (UN) say that a type $p \in S_x(M)$ has exactly one *independent extension* to MB , namely the type over MB of any $A \models p$ with $A \downarrow_M B$. All the other axioms are statements about which extensions of p count as independent. In a stable theory the natural candidate for this extension is the heir of Section 6.1, and we take it as the definition.

Definition 6.2.1. Assume that T is stable. For small A, B and a model M we define

$$A \downarrow_M^h B \quad : \Longleftrightarrow \quad \text{tp}(A/MB) = \text{tp}(A/M)|B.$$

In words, the type $\text{tp}(A/MB)$ is the *heir* of $\text{tp}(A/M)$, the extension defined by the same definition scheme. We call $\downarrow^h$ *heir independence*.

For finite tuples the definition becomes a test, formula by formula, which is the form used in the proofs below. If a, b are finite tuples and $p = \text{tp}(a/M)$, then every $L(Mb)$ -formula in x has the form $\varphi(x, b)$ with $\varphi(x, y) \in L(M)$. Hence we have

$$(6.1) \quad a \downarrow_M^h b \Longleftrightarrow \text{for all } \varphi(x, y) \in L(M) : \quad \mathbb{M} \models \varphi(a, b) \leftrightarrow d_p \varphi(b).$$

Theorem 6.2.2. *For a complete theory T , the following are equivalent:*

(1) *The theory T is stable.*

(2) *There is an independence relation satisfying all the axioms of Definition 5.4.7.*

(3) *There is a relation $A \downarrow_M B$ satisfying (UN) and (LOC).*

Moreover, if T is stable, the relation of Definition 6.2.1 satisfies all the axioms. This relation also satisfies

(AR) *Anti-reflexivity. If $a \downarrow_M^h a$ for a finite tuple a , then we have $a \subseteq M$.*

6.2.1. Proof of (1) $\Rightarrow$ (2)

Assume that T is stable, and let $\downarrow^h$ be the relation of Definition 6.2.1. Throughout, we write $p = \text{tp}(A/M)$. Most of the axioms follow by unwinding the definitions with the help of Lemma 6.1.3; symmetry is the one place where Harrington's lemma is needed.

(INV). Let $\sigma \in \text{Aut}(\mathbb{M})$ be given. Applying σ to the parameters of the definition scheme of p gives a definition scheme for $\sigma(p) = \text{tp}(\sigma A/\sigma M)$. Hence we have $\sigma(p)|\sigma(B) = \sigma(p|B)$. If $\text{tp}(A/MB) = p|B$, then we have $\text{tp}(\sigma A/\sigma M \sigma B) = \sigma(p|B) = \sigma(p)|\sigma B$.

(MON). The definition scheme of $\text{tp}(A'/M)$ is the part of that of p that involves only variables for A' . Hence the heir $\text{tp}(A'/M)|B'$ is the restriction of $p|B$ to these variables and to MB' (see Lemma 6.1.3(b)), which is $\text{tp}(A'/MB')$.

(FC). The condition $\text{tp}(A/MB) = p|B$ says that for every $L(M)$ -formula $\varphi(x', y)$ and every $b \in B$ we have $\mathbb{M} \models \varphi(a', b) \leftrightarrow d_p \varphi(b)$, where a' is the finite subtuple of A corresponding to x' . Each instance involves only finite $a' \subseteq A$ and $b \subseteq B$, and is exactly an instance of $a' \downarrow_M^h b$.

(BMON). Let N be a model with $M \subseteq N \subseteq B$, and suppose that $\text{tp}(A/MB) = p|B$. Then we have $\text{tp}(A/N) = p|N$ by Lemma 6.1.3(b). By Lemma 6.1.3(c), we have $\text{tp}(A/N)|B = (p|N)|B = p|NB = p|B = \text{tp}(A/NB)$, which means that $A \downarrow_N^h B$.

(TR). Let N be a model with $M \subseteq N$, and suppose that $\text{tp}(A/N) = p|N$ and $\text{tp}(A/NB) = \text{tp}(A/N)|B$. By Lemma 6.1.3(c), we have $\text{tp}(A/NB) = (p|N)|B = p|NB$, which means that $A \downarrow_M^h NB$.

(SYM). By (FC) we need only treat finite tuples a, b . Put $p = \text{tp}(a/M)$ and $q = \text{tp}(b/M)$, and assume that $a \downarrow_M^h b$. For $\varphi(x, y) \in L(M)$, using (6.1) and Lemma 6.1.4, we have

$$\mathbb{M} \models \varphi(a, b) \Longleftrightarrow \mathbb{M} \models d_p \varphi(b) \Longleftrightarrow d_p \varphi(y) \in q \Longleftrightarrow d_q \varphi(x) \in p \Longleftrightarrow \mathbb{M} \models d_q \varphi(a).$$

By (6.1) with the roles of a and b exchanged, this says $b \downarrow_M^h a$.

(EX). By Lemma 6.1.3(a) and saturation, the type $p|B$ has a realization A' . Then we have $A' \equiv_M A$ and $A' \downarrow_M^h B$.

(UN). If $A \equiv_M A'$, then both have type p over M . If both are independent from B over M , then we have $\text{tp}(A/MB) = p|B = \text{tp}(A'/MB)$.

(LOC). Let a be finite, let N be a model, and put $p = \text{tp}(a/N)$. For each L -formula $\varphi(x, y)$ fix a definition $d_p\varphi(y) \in L(N)$, and let $D \subseteq N$ be the set of all parameters used, so that we have $|D| \leq \kappa$. By downward Löwenheim–Skolem there is a model $M \preceq N$ with $D \subseteq M$ and $|M| \leq \kappa$. Put $q = \text{tp}(a/M)$. Each $d_p\varphi$ is an $L(M)$ -formula, and on M it agrees with the definition of p , so the formula $d_p\varphi$ defines q for φ . These formulas therefore form a definition scheme for q (for $L(M)$ -formulas $\varphi_0(x, y, m)$ use $d_p\varphi_0(y, m)$), and so the heir $q|N$ is defined by the same formulas as p . Since $q|N$ and p are both complete types over N , we have $q|N = p = \text{tp}(a/N)$, which means that $a \downarrow_M^h N$.

(AR). Suppose that $a \downarrow_M^h a$ with $p = \text{tp}(a/M)$, and suppose that some coordinate a_0 of a is not in M . Let $\varphi(x, y)$ be the formula $x_0 = y_0$, where x_0 and y_0 are the corresponding coordinates of x and y . For $c \in M$ we have $\varphi(x, c) \notin p$, because $a_0 \neq c_0$. Hence we have $M \models \neg \exists y d_p\varphi(y)$, and the same holds in $\mathbb{M}$. By (6.1), we have $\mathbb{M} \models \varphi(a, a) \leftrightarrow d_p\varphi(a)$. However, the formula $\varphi(a, a)$ holds and the formula $d_p\varphi(a)$ fails, a contradiction.

6.2.2. Proof of (2) $\Rightarrow$ (3) $\Rightarrow$ (1)

The implication (2) $\Rightarrow$ (3) is immediate, because a relation satisfying all the axioms satisfies (UN) and (LOC) in particular. For (3) $\Rightarrow$ (1), the idea is that the axiom (LOC) lets us describe each type over a model N by a type over a small submodel, and the axiom (UN) guarantees that this description loses no information; counting the descriptions then bounds the number of types. Let $\downarrow$ satisfy (UN) and (LOC), and put $\lambda = 2^\kappa$, so that $\lambda^\kappa = \lambda$. By Lemma 5.1.2, it suffices to show that the theory T is λ -stable.

Let A be a set with $|A| \leq \lambda$, and let x be a finite tuple of variables. By downward Löwenheim–Skolem there is a model $N \supseteq A$ with $|N| \leq \lambda$, and restriction maps $S_x(N)$ onto $S_x(A)$. Thus we need only show that $|S_x(N)| \leq \lambda$.

For each $p \in S_x(N)$ choose a realization a_p . By (LOC) choose a model $M_p \subseteq N$ with $|M_p| \leq \kappa$ and $a_p \downarrow_{M_p} N$. Consider the map

$$p \mapsto (M_p, \text{tp}(a_p/M_p)).$$

This map is injective. Indeed, suppose that $M_p = M_{p'} =: M$ and $a_p \equiv_M a_{p'}$. Since $a_p \downarrow_M N$ and $a_{p'} \downarrow_M N$, the axiom (UN) gives $a_p \equiv_N a_{p'}$, which means that $p = p'$.

The number of subsets of N of size $\leq \kappa$ is at most $\lambda^\kappa = \lambda$. For each such M we have $|S_x(M)| \leq 2^\kappa = \lambda$. Hence we have $|S_x(N)| \leq \lambda \cdot \lambda = \lambda$. ☺

6.3. Local character, uniqueness, and forking

 **Remark 6.3.1.** The bound $|M| \leq \kappa$ in (LOC) plays no special role in the proof of (3) $\Rightarrow$ (1): it may be weakened to $|M| < \kappa_0$ for any fixed cardinal κ_0 , by taking $\lambda = 2^{\kappa_0 + \kappa}$. Thus stability is equivalent to the existence of a relation for which types over models have unique independent extensions and every type is independent over a submodel of bounded size; all the other axioms then hold automatically, by (1) $\Rightarrow$ (2).

 **Remark 6.3.2.** The relation $\downarrow^h$ has a counterpart over arbitrary base sets, Shelah's non-forking independence, which Chapter 7 develops. A type over $B \supseteq C$ *forks* over C if it implies a finite disjunction of formulas each of which divides over C (see Definition 5.2.3); a type

that divides over C forks over C , and Theorem 7.3.3 shows that over models the two notions agree in simple theories. In a stable theory the relation $\downarrow^h$ of Definition 6.2.1 is *nonforking independence* restricted to models as bases. For $M \subseteq B$, the following are equivalent: the type $\text{tp}(a/B)$ does not fork over M ; this type is the heir of $\text{tp}(a/M)$; this type is definable over M ; see Pillay [Pil96] or Tent–Ziegler [TZ12]. Nonforking makes sense over arbitrary base sets C . Over such sets, nonforking satisfies all the axioms above except uniqueness, which holds only over algebraically closed sets in T^{eq} (Harnik–Harrington [HH84]). Example 3.2.6 shows the failure over $\emptyset$. In that theory there is only one 1-type over $\emptyset$, and it has at least two nonforking extensions to N : the type of a new element of C_1 and the type of a new element of C_2 . Their φ -parts are defined by the conjugate formulas $E(y, c)$ with $c \in C_1$ and with $c \in C_2$, so both types are definable almost over $\emptyset$, and neither is definable over $\emptyset$. Kim and Pillay [KP97] proved that in a stable (indeed in a simple; see Chapters 5 and 7) theory, nonforking is the *only* relation on arbitrary small sets satisfying invariance, monotonicity, transitivity, finite character, symmetry, extension, local character and the independence theorem. Developing forking over arbitrary sets requires considerably more machinery; for this reason we have taken models as base sets in this chapter.

6.4. Historical remarks

✂ **Abstract independence.** The axiomatic study of independence is older than model theory. Steinitz, in his foundational paper on the algebraic theory of fields [Ste10], introduced transcendence bases and showed that any two transcendence bases have the same cardinality. The argument is an exchange argument of the kind now called the Steinitz exchange lemma, the same one that gives the dimension of a vector space. Van der Waerden, in *Moderne Algebra* [vdW30], observed that linear dependence and algebraic dependence obey the same few rules, and derived the existence of bases and the invariance of their cardinality from an abstract notion of dependence whose central axiom is exchange. Whitney [Whi35] went a step further and introduced *matroids*, structures that isolate the abstract properties of linear dependence, and gave several equivalent axiomatizations, in terms of independent sets, bases, circuits and rank. Nakasawa developed the same theory independently in the mid-1930s, and Birkhoff [Bir35] and Mac Lane [Mac36] related matroid theory to lattices and to projective geometry.

After a period in which matroids were studied mainly by Rado and Tutte, Rota made them central to combinatorics. In his monograph with Crapo [CR70] they become *combinatorial geometries*, and a closure operator with the exchange property is called a *pregeometry*, the term still used in model theory. Rota’s view was that dependence is a geometric notion with invariants of its own, such as the Möbius function and the characteristic polynomial of the associated lattice.

Model theory met these ideas in the study of uncountably categorical theories. Baldwin and Lachlan [BL71] showed that in a strongly minimal set algebraic closure has the exchange property, so algebraic closure is a pregeometry and every model has a well-defined dimension. They used this to give a new proof of Morley’s categoricity theorem [Mor65]. Shelah’s forking [She90] carries independence far beyond pregeometries: nonforking independence is not given by a closure operator, yet in a stable theory nonforking independence has the symmetry, existence and uniqueness properties on which arguments about dimension depend. The later work of Zilber and Hrushovski on the geometry of forking turned the study of forking into a subject of its own, geometric stability theory [Pil96]. The point of view of this chapter, which characterizes independence axiomatically, as van der Waerden and Whitney

did for linear dependence, is that of Harnik and Harrington and of Kim and Pillay, discussed below.

- ⌘ **Section 6.1.** Heirs and coheirs (*héritiers* and *cohéritiers*) were introduced by Lascar and Poizat [LP79]. They gave an alternative presentation of Shelah’s forking in stable theories, based only on definability of types and finite satisfiability. Lemma 6.1.4, which says that the definitions of two types over a model commute, is due to Harrington. The lemma appears in Harnik–Harrington [HH84], and is the key step in proving symmetry of independence in stable theories.
- ⌘ **Section 6.2.** Nonforking independence was introduced by Shelah [She90], and its reformulation through heirs is the one of Lascar and Poizat recalled above. The characterization of stability through abstract independence notions is due to Harnik and Harrington [HH84]. Kim and Pillay [KP97] later extended this characterization from stable to simple theories: they proved that the simple theories are exactly those carrying a relation satisfying the axioms listed above, with the independence theorem in place of uniqueness, and that such a relation is then nonforking. Theorem 6.2.2 is the stable case over models, with uniqueness (stationarity over models) as the key axiom. The anchor symbol $\perp$ for independence was introduced by Makkai in a lecture in Jerusalem, during the Model Theory Year at the Institute for Advanced Studies (1980–81), and was adopted at once. The symbol appears in print in Makkai’s survey [Mak84].

Chapter 7

Simplicity and notions of independence

This chapter is in preparation: Sections 7.1–7.4 and the historical remarks are written, and the plan of the rest follows.

Chapter 6 characterized the stable theories as those carrying a relation of independence over models with all the properties of Definition 5.4.7, uniqueness included. In Chapter 5 the random graph, and every Fraïssé limit with disjoint 3-amalgamation, carried a relation with all these properties except uniqueness, and with the independence theorem in its place. The purpose of this chapter is to show that this is no accident of the examples. By a theorem of Kim and Pillay [KP97], a theory is simple exactly when it carries such a relation, and the relation is then unique: it is nondividing independence. As Chapter 6 turned on Harrington’s lemma, this chapter turns on three results of Kim [Kim98] and of Kim and Pillay: dividing is witnessed by any Morley sequence, independence is symmetric, and independent extensions over independent sets amalgamate.

Plan of the chapter

1. *Dividing over models.* Nondividing independence $\downarrow^d$; the characterization of nondividing by indiscernible sequences, extracted by Ramsey’s theorem and compactness (see Lemma 5.2.2); coheirs do not divide (see Lemma 7.1.3), which gives existence in every theory; coheir sequences as Morley sequences; transitivity on the left.
2. *Kim’s lemma.* In a simple theory, a formula divides over a model exactly when its instances along a Morley sequence are inconsistent. The proof reverses finite pieces of the sequence and reuses the tree construction of Lemma 5.3.5(1).
3. *Forking and symmetry.* Forking equals dividing over models, so nondividing types have nondividing extensions; independence is symmetric and transitive.
4. *The independence theorem over models*, after Casanovas [Cas11].
5. *The Kim–Pillay theorem.* A theory is simple if and only if it carries a relation over models with the properties of Definition 5.4.7 other than uniqueness, together with the independence theorem; such a relation is nondividing independence.
6. *Stability through forking.* A theory is stable if and only if nondividing independence over models satisfies uniqueness and local character; this follows from the Kim–Pillay theorem and Theorem 6.2.2.
7. *Examples and pictures.* The Rado graph and the Fraïssé limits of Chapter 5 revisited, with figures for Morley sequences, the reversal argument, and the amalgamation in the independence theorem.

8. Historical remarks.

7.1. Dividing over models

A type over $M \cup B$ that does not divide over M says nothing about B that M could not have foreseen: whatever sequence of copies of B the indiscernibility of M allows, the type can be realized simultaneously against all of them. This section turns that sentence into a lemma, and draws three consequences that hold in every theory: types finitely satisfiable in M do not divide over M , so nondividing extensions always exist; such types generate indiscernible sequences of a special kind, the Morley sequences; and nondividing is transitive on the left. Simplicity enters only in the next section.

Definition 7.1.1. For small sets A, B, C , write $A \downarrow_C^d B$ if for every finite tuple a from A the type $\text{tp}(a/CB)$ does not divide over C . We call $\downarrow_C^d$ *nondividing independence*.

A formula of $\text{tp}(a/CB)$ mentions only finitely many elements of B , and a formula that divides over C belongs to the type of a over C together with those elements. Hence $A \downarrow_C^d B$ holds if and only if $a \downarrow_C^d b$ for all finite tuples a from A and b from B ; this is finite character, built into the definition. For a finite tuple b , the notation $\text{tp}(a/Cb)$ is short for $\text{tp}(a/C \cup b)$.

Parameters from C cause no difficulty. Let $\psi(x, y) \in L(C)$ be given, and write $\psi(x, y) = \psi_0(x, y, c)$ with $\psi_0 \in L$ and c a tuple from C ; the formula $\psi(x, b)$ divides over C when $\psi_0(x, b, c)$ does, in the sense of Definition 5.2.3. A sequence indiscernible over C that begins with (b, c) has the form $((b_i, c))_{i < \omega}$, because each of its terms has the type of (b, c) over C , and conversely $((b_i, c))_{i < \omega}$ is indiscernible over C whenever $(b_i)_{i < \omega}$ is. Hence $\psi(x, b)$ divides over C if and only if $\{\psi(x, b_i) : i < \omega\}$ is inconsistent for some sequence $(b_i)_{i < \omega}$, indiscernible over C , with $b_0 = b$.

7.1.1. Nondividing and indiscernible sequences

The definition of dividing quantifies over indiscernible sequences, one formula at a time. The first lemma quantifies over sequences once and for all: a type does not divide exactly when it can be moved, without changing it, to a position from which a given indiscernible sequence still looks indiscernible (see Figure 7.1).

Lemma 7.1.2. *Let C be a small set, let a be a small tuple, and let b be a finite tuple. The following are equivalent:*

- (1) *the type $\text{tp}(a/Cb)$ does not divide over C ;*
- (2) *for every sequence $(b_i)_{i < \omega}$, indiscernible over C , with $b_0 = b$, there is a' with $a' \equiv_{Cb} a$ such that $(b_i)_{i < \omega}$ is indiscernible over $C \cup a'$.*

Proof. Write $p(x, b) = \text{tp}(a/Cb)$, where $p(x, y)$ is the set of $L(C)$ -formulas $\psi(x, y)$ with $\mathbb{M} \models \psi(a, b)$, and write $p(x, b_i)$ for the set of formulas $\psi(x, b_i)$ with $\psi \in p$.

(2) $\Rightarrow$ (1). Let $\psi(x, b)$ be a formula of $\text{tp}(a/Cb)$, with $\psi(x, y) \in L(C)$, and let $(b_i)_{i < \omega}$ be indiscernible over C with $b_0 = b$. By (2) there is $a' \equiv_{Cb} a$ such that the sequence is indiscernible over $C \cup a'$. Since $\mathbb{M} \models \psi(a', b_0)$, indiscernibility over $C \cup a'$ gives $\mathbb{M} \models \psi(a', b_i)$ for every i . Hence $\{\psi(x, b_i) : i < \omega\}$ is consistent. By the observation following Definition 7.1.1, the formula $\psi(x, b)$ does not divide over C .

(1) $\Rightarrow$ (2). Let $(b_i)_{i < \omega}$ be indiscernible over C with $b_0 = b$. The proof has two steps: first realize all the copies $p(x, b_i)$ at once, then make the sequence indiscernible over the realization by extracting a new sequence and moving it back.

© **Step 1.** The set $\Sigma(x) = \bigcup_{i < \omega} p(x, b_i)$ is consistent. Otherwise, by compactness, some finite subset is inconsistent. It involves finitely many formulas $\psi_1, \dots, \psi_r$ of p and finitely many indices $i_1, \dots, i_n$. Let ψ be the conjunction of $\psi_1, \dots, \psi_r$; then $\psi(x, b) \in p(x, b)$, and $\{\psi(x, b_{i_l}) : 1 \leq l \leq n\}$ is inconsistent. Hence the set $\{\psi(x, b_i) : i < \omega\}$ is inconsistent, so $\psi(x, b)$ divides over C , contradicting (1). Let a'' realize Σ .

© **Step 2.** Apply Lemma 5.2.2 over $C \cup a''$ to the sequence $(b_i)_{i < \omega}$. Its Ehrenfeucht–Mostowski type over $C \cup a''$ contains, for every n , the type of $(b_0, \dots, b_{n-1})$ over C , because the sequence is indiscernible over C ; it also contains every formula $\psi(a'', y_1)$ with $\psi \in p$, because a'' realizes every $p(x, b_i)$. The lemma therefore gives a sequence $(b'_i)_{i < \omega}$, indiscernible over $C \cup a''$, such that $(b'_0, \dots, b'_{n-1}) \equiv_C (b_0, \dots, b_{n-1})$ for every n and $\mathbb{M} \models \psi(a'', b'_0)$ for every $\psi \in p$. The first property gives $(b'_i)_{i < \omega} \equiv_C (b_i)_{i < \omega}$, and the second gives $a''b'_0 \equiv_C ab$. By strong homogeneity there is $\sigma \in \text{Aut}(\mathbb{M}/C)$ with $\sigma(b'_i) = b_i$ for every i . Put $a' = \sigma(a'')$. Then $a'b = \sigma(a''b'_0) \equiv_C ab$, that is, $a' \equiv_{Cb} a$, and the sequence $(b_i) = (\sigma(b'_i))$ is indiscernible over $\sigma(C \cup a'') = C \cup a'$. ☺

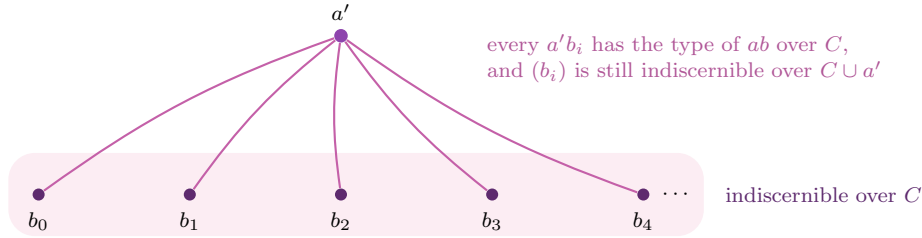


FIGURE 7.1. Lemma 7.1.2. A type $\text{tp}(a/Cb)$ that does not divide over C can be moved, over $C \cup b$, to a tuple a' that stands in the same relation to every term of a given indiscernible sequence starting with b , so that the sequence does not notice a' .

7.1.2. Coheirs do not divide

Dividing is a statement about many copies of a parameter, and a type that is finitely satisfiable in M is witnessed by single elements of M , which every copy over M must treat alike. That is the whole proof of the next lemma.

Lemma 7.1.3. *Let M be a model, let B be a small set, and let a be a small tuple such that $\text{tp}(a/MB)$ is finitely satisfiable in M . Then $a \downarrow_M^d B$.*

Proof. Let $\psi(x, b)$ be a formula of $\text{tp}(a/MB)$, with $\psi(x, y) \in L(M)$ and b a finite tuple from B , and let $(b_i)_{i < \omega}$ be indiscernible over M with $b_0 = b$. By finite satisfiability there is a tuple m from M with $\mathbb{M} \models \psi(m, b_0)$. The tuple m and the parameters of ψ lie in M , so indiscernibility over M gives $\mathbb{M} \models \psi(m, b_i)$ for every i . Hence $\{\psi(x, b_i) : i < \omega\}$ is consistent, and $\psi(x, b)$ does not divide over M . ☺

Every type over a model is finitely satisfiable in it (see Subsection 6.1.1), and such a type extends to a type over any larger set MB that is still finitely satisfiable in M . The following proposition, together with Lemma 7.1.3, gives the existence axiom (EX) for $\downarrow^d$ in every theory.

Proposition 7.1.4. *Let M be a model, and let A and B be small sets. Then there is $A' \equiv_M A$ such that $\text{tp}(A'/MB)$ is finitely satisfiable in M . In particular, we have $A' \downarrow_M^d B$.*

Proof. Let $\pi = \text{tp}(A/M)$, in variables X , and let F be the set of $L(MB)$ -formulas $\chi(X)$ that are satisfied by no tuple from M . We claim that $\pi \cup \{\neg\chi : \chi \in F\}$ is consistent. A finite subset is contained in $\{\theta, \neg\chi_1, \dots, \neg\chi_n\}$ for some $\theta \in \pi$ and $\chi_1, \dots, \chi_n \in F$. Since θ is an $L(M)$ -formula realized in $\mathbb{M}$ and $M \preceq \mathbb{M}$, some tuple m from M satisfies θ , and m satisfies each $\neg\chi_l$ because $\chi_l \in F$. By saturation, the set is realized by some A' . Then $A' \equiv_M A$, and every formula χ of $\text{tp}(A'/MB)$ is satisfied by a tuple from M , since otherwise $\chi \in F$ and $\neg\chi \in \text{tp}(A'/MB)$. Hence $\text{tp}(A'/MB)$ is finitely satisfiable in M , and Lemma 7.1.3 gives $A' \downarrow_M^d B$. $\odot$

For a finite tuple a , the types produced in this way are exactly the coheirs $q_{\mathcal{U}}|B$ of Lemma 6.1.5: a type over MB that is finitely satisfiable in M has the form $q_{\mathcal{U}}|B$ for some ultrafilter $\mathcal{U}$ on $M^{|a|}$ with $q_{\mathcal{U}}|\emptyset = \text{tp}(a/M)$.

7.1.3. Morley sequences

Realizing a coheir over and over, each time over everything realized so far, produces an indiscernible sequence in which every term is independent from the earlier ones. Such sequences are the test sequences of simplicity theory.

Definition 7.1.5. Let M be a model. A *Morley sequence over M* is a sequence $(b_i)_{i < \omega}$, indiscernible over M , such that $b_i \downarrow_M^d \{b_j : j < i\}$ for every i .

Lemma 7.1.6. Let M be a model, let b be a finite tuple, and let $\mathcal{U}$ be an ultrafilter on $M^{|b|}$ with $q_{\mathcal{U}}|\emptyset = \text{tp}(b/M)$. Put $b_0 = b$, and for $i \geq 1$ let b_i realize $q_{\mathcal{U}}|\{b_j : j < i\}$. Then $(b_i)_{i < \omega}$ is a Morley sequence over M with $b_0 = b$.

The ultrafilter exists by Lemma 6.1.5(b), since $\text{tp}(b/M)$ is finitely satisfiable in M , and the tuples b_i exist by saturation. We write $b_{<i}$ for $(b_j)_{j < i}$.

Proof. For $i \geq 1$ and $i_1 < \dots < i_n < i$, Lemma 6.1.5(c) gives $q_{\mathcal{U}}|\{b_{i_1}, \dots, b_{i_n}\} \subseteq q_{\mathcal{U}}|b_{<i} = \text{tp}(b_i/Mb_{<i})$; since the smaller set is a complete type over $M \cup \{b_{i_1}, \dots, b_{i_n}\}$ by Lemma 6.1.5(a), it equals the type of b_i over that set. In particular, every b_i realizes $q_{\mathcal{U}}|\emptyset = \text{tp}(b/M)$, for $i = 0$ by hypothesis.

We show by induction on n that all increasing n -tuples of the sequence have the same type over M . For $n = 1$ this was just observed. Let $i_1 < \dots < i_{n+1}$ and $j_1 < \dots < j_{n+1}$ be given. By the induction hypothesis and strong homogeneity there is $\sigma \in \text{Aut}(\mathbb{M}/M)$ with $\sigma(b_{i_l}) = b_{j_l}$ for $l \leq n$. By Lemma 6.1.5(d), the automorphism σ carries $q_{\mathcal{U}}|\{b_{i_1}, \dots, b_{i_n}\}$, the type of $b_{i_{n+1}}$ over $M \cup \{b_{i_1}, \dots, b_{i_n}\}$, to $q_{\mathcal{U}}|\{b_{j_1}, \dots, b_{j_n}\}$, the type of $b_{j_{n+1}}$ over $M \cup \{b_{j_1}, \dots, b_{j_n}\}$. Hence $\sigma(b_{i_{n+1}}) \equiv_{Mb_{j_1} \dots b_{j_n}} b_{j_{n+1}}$, and therefore $(b_{i_1}, \dots, b_{i_{n+1}}) \equiv_M (b_{j_1}, \dots, b_{j_{n+1}})$. Thus the sequence is indiscernible over M .

For every i , the type $\text{tp}(b_i/Mb_{<i})$ is a coheir. By Lemma 6.1.5(a), this type is finitely satisfiable in M , and by Lemma 7.1.3 we have $b_i \downarrow_M^d b_{<i}$. $\odot$

7.1.4. Transitivity on the left

Nondividing independence can be built up one coordinate block at a time on the left: if A is independent from B over C , and D is independent from B over $C \cup A$, then $A \cup D$ is independent from B over C . The proof uses the implication (1) $\Rightarrow$ (2) of Lemma 7.1.2 twice, first for A and then for D , concludes with the implication (2) $\Rightarrow$ (1), and uses no hypothesis on T .

Lemma 7.1.7. Let A, B, C, D be small sets. If $A \downarrow_C^d B$ and $D \downarrow_{CA}^d B$, then $AD \downarrow_C^d B$.

Proof. Regard A as a tuple, and let finite tuples e from D and b from B be given. By finite character it suffices to show that $\text{tp}(Ae/Cb)$ does not divide over C ; the types of the

finite tuples from $A \cup e$ are restrictions of this type. Every formula of $\text{tp}(A/Cb)$ mentions a finite tuple from A , so the hypothesis $A \downarrow_C^d B$ says that $\text{tp}(A/Cb)$ does not divide over C ; likewise $D \downarrow_{CA}^d B$ says that $\text{tp}(e/CAb)$ does not divide over $C \cup A$. We verify condition (2) of Lemma 7.1.2 for the tuple Ae .

Let $(b_i)_{i < \omega}$ be indiscernible over C with $b_0 = b$. By Lemma 7.1.2, applied to A , there is $A' \equiv_{Cb} A$ such that (b_i) is indiscernible over $C \cup A'$. Choose $\sigma \in \text{Aut}(\mathbb{M}/Cb)$ with $\sigma(A') = A$. Then the sequence $(\sigma(b_i))_{i < \omega}$ is indiscernible over $C \cup A$ and begins with $\sigma(b_0) = b$. By Lemma 7.1.2, applied to e over the base $C \cup A$, there is $e' \equiv_{CAb} e$ such that $(\sigma(b_i))$ is indiscernible over $C \cup Ae'$. Applying σ^{-1} , which fixes $C \cup b$, we see that (b_i) is indiscernible over $C \cup A'\sigma^{-1}(e')$, and

$$A'\sigma^{-1}(e') = \sigma^{-1}(Ae') \equiv_{Cb} Ae' \equiv_{Cb} Ae.$$

By Lemma 7.1.2, the type $\text{tp}(Ae/Cb)$ does not divide over C . ☺

7.1.5. The axioms that hold in every theory

We collect what this section has shown, measured against Definition 5.4.7.

Proposition 7.1.8. *In every complete theory, the relation $\downarrow^d$, with models as bases, satisfies (INV), (MON), (BMON), (FC) and (EX). If T is simple, it also satisfies (LOC).*

Proof. Invariance: an automorphism σ of $\mathbb{M}$ carries sequences indiscernible over C to sequences indiscernible over $\sigma(C)$ and preserves consistency, so $\varphi(x, b)$ divides over C if and only if $\varphi(x, \sigma(b))$ divides over $\sigma(C)$. Monotonicity is immediate from the definition, and finite character was observed after Definition 7.1.1. *Base monotonicity:* let $M \subseteq N \subseteq B$, and let a be a finite tuple from A . Since $NB = B = MB$, a formula of $\text{tp}(a/NB)$ that divided over N would divide over M by Remark 5.2.5. *Existence* is Proposition 7.1.4. *Local character:* let a be a finite tuple and N a model. If T is simple, Theorem 5.3.6 gives $A_0 \subseteq N$ with $|A_0| \leq \kappa$ such that $\text{tp}(a/N)$ does not divide over A_0 . By the downward Löwenheim–Skolem theorem there is a model $M \preceq N$ with $A_0 \subseteq M$ and $|M| \leq \kappa$, and $\text{tp}(a/N)$ does not divide over M , by Remark 5.2.5 once more. Thus $a \downarrow_M^d N$. ☺

Symmetry, transitivity and the independence theorem remain. They can fail outside simple theories; indeed Kim [Kim01] showed that symmetry of forking independence characterizes the simple theories. We shall prove all three from simplicity, through Kim's lemma, which is the subject of the next section.

7.2. Kim's lemma

A formula divides when some indiscernible sequence of copies of its parameter makes it inconsistent. The definition asks for one such sequence, and gives no hint where to look for it. In a simple theory there is no need to look: any Morley sequence will do, and so will any sequence of realizations of the type of the parameter in which each term is independent from the earlier ones. This is Kim's lemma [Kim98], and the rest of the chapter rests on it. With the preparations of Section 7.1 the proof is short. A Morley sequence, read backwards, becomes a chain of formulas each of which divides over the earlier ones, and Lemma 5.3.5(1) turns such a chain into a tree. A simple theory has no such trees, and so the instances of the formula along the sequence cannot be consistent. Throughout the section, M is a model.

7.2.1. Enlarging the base

The proof of Kim's lemma needs a formula that divides over M to divide also over M together with later terms of the sequence. Enlarging the base can only make dividing harder (see Remark 5.2.5); the next lemma shows that it does not make it harder when the new parameters are independent from the parameter of the formula. Since dividing will be counted with a fixed k , we note first that the parameter convention following Definition 7.1.1 applies to k -dividing as well, with the same proof: for $\psi(x, y) \in L(C)$, the formula $\psi(x, b)$ k -divides over C if and only if $\{\psi(x, b_i) : i < \omega\}$ is k -inconsistent for some sequence $(b_i)_{i < \omega}$, indiscernible over C , with $b_0 = b$.

Lemma 7.2.1. *Let C be a small set, let b be a finite tuple, and let e be a small tuple such that $\text{tp}(e/Cb)$ does not divide over C . Let $\varphi(x, y) \in L(C)$, and let $k \geq 2$ be given. If $\varphi(x, b)$ k -divides over C , then $\varphi(x, b)$ k -divides over $C \cup e$.*

Proof. Let $(b_i)_{i < \omega}$ be indiscernible over C , with $b_0 = b$ and $\{\varphi(x, b_i) : i < \omega\}$ k -inconsistent. By Lemma 7.1.2 there is $e' \equiv_{Cb} e$ such that (b_i) is indiscernible over $C \cup e'$. Choose $\sigma \in \text{Aut}(\mathbb{M}/Cb)$ with $\sigma(e') = e$. The sequence $(\sigma(b_i))_{i < \omega}$ is indiscernible over $\sigma(C \cup e') = C \cup e$, and it begins with $\sigma(b) = b$. Since σ fixes the parameters of φ , which lie in C , the set $\{\varphi(x, \sigma(b_i)) : i < \omega\}$ is the image under σ of a k -inconsistent set, and so it is k -inconsistent. Hence $\varphi(x, b)$ k -divides over $C \cup e$. $\odot$

7.2.2. The lemma

We write $b_{<i}$ for $(b_j)_{j < i}$, as in Section 7.1.

Theorem 7.2.2 (Kim's lemma). *Assume that T is simple. Let M be a model, let b be a finite tuple, and let $\varphi(x, y) \in L(M)$.*

- (a) *Suppose that $\varphi(x, b)$ divides over M , and let $(b_i)_{i < \omega}$ be a sequence such that $b_i \equiv_M b$ and $b_i \downarrow_M^d b_{<i}$ for every i . Then $\{\varphi(x, b_i) : i < \omega\}$ is inconsistent.*
- (b) *The following are equivalent:*
 - (1) *the formula $\varphi(x, b)$ divides over M ;*
 - (2) *for every Morley sequence $(b_i)_{i < \omega}$ over M with $b_0 = b$, the set $\{\varphi(x, b_i) : i < \omega\}$ is inconsistent;*
 - (3) *for some Morley sequence $(b_i)_{i < \omega}$ over M with $b_0 = b$, the set $\{\varphi(x, b_i) : i < \omega\}$ is inconsistent.*

Part (a) does not require the sequence to be indiscernible: it asks only that every term have the type of b over M and be independent from the earlier terms. This generality will be used in the proof of symmetry in the next section. The proof of (a) reads the sequence backwards, because the Morley condition points the wrong way: it makes each term independent from the *earlier* terms, while the tree construction of Lemma 5.3.5 needs each formula to divide over the parameters of the earlier formulas of the chain. Reversing a finite piece of the sequence makes the earlier terms of the chain the later terms of the sequence; transitivity on the left (see Lemma 7.1.7) shows that these later terms, taken together, are independent from the term at hand, and Lemma 7.2.1 then moves the dividing over to them. An infinite sequence cannot be reversed into an ω -sequence, which is why the finite form of Lemma 5.3.5 is needed (see Figure 7.2).

Proof. (a) Suppose, for a contradiction, that $\{\varphi(x, b_i) : i < \omega\}$ is consistent. By Lemma 5.2.4, applied as in the parameter convention, there is $k \geq 2$ such that $\varphi(x, b)$ k -divides over M . Write $\varphi(x, y) = \varphi_0(x, y, m)$ with $\varphi_0 \in L$ and m a finite tuple from M , and regard φ_0 as a formula $\varphi_0(x; y, z)$ with parameter variables y, z . We show that φ_0 has the k -tree property, which contradicts the simplicity of T . By Lemma 5.3.5(2), it suffices to find, for every $n \geq 1$, tuples $a_0, \dots, a_{n-1}$ as in Lemma 5.3.5(1) for the formula φ_0 . Let $n \geq 1$ be given.

© **Step 1.** For $i < l < n$, we have $b_{i+1} \dots b_l \downarrow_M^d b_i$. We argue by induction on l . For $l = i + 1$, the hypothesis $b_{i+1} \downarrow_M^d b_{<i+1}$ gives $b_{i+1} \downarrow_M^d b_i$ by monotonicity. Suppose that the assertion holds for some l with $i < l < n - 1$, and put $A = \{b_{i+1}, \dots, b_l\}$. The hypothesis on b_{l+1} says that $\text{tp}(b_{l+1}/Mb_{<l+1})$ does not divide over M . Its restriction to $M \cup A \cup b_i$ does not divide over M either, and so, by Remark 5.2.5 applied to $M \subseteq M \cup A \subseteq M \cup A \cup b_i$, it does not divide over $M \cup A$. Thus $b_{l+1} \downarrow_{MA}^d b_i$. Together with $A \downarrow_M^d b_i$, Lemma 7.1.7 gives $Ab_{l+1} \downarrow_M^d b_i$, which is the assertion for $l + 1$.

© **Step 2.** For $i < n$, the formula $\varphi(x, b_i)$ k -divides over $M \cup \{b_{i+1}, \dots, b_{n-1}\}$. Since $b_i \equiv_M b$, there is $\sigma \in \text{Aut}(\mathbb{M}/M)$ with $\sigma(b) = b_i$; it fixes the parameters of φ , and it carries a sequence witnessing that $\varphi(x, b)$ k -divides over M to one witnessing that $\varphi(x, b_i)$ k -divides over M . For $i = n - 1$ there is nothing more to show. For $i < n - 1$, Step 1 with $l = n - 1$ says that $\text{tp}(b_{i+1} \dots b_{n-1}/Mb_i)$ does not divide over M , and Lemma 7.2.1, with $e = b_{i+1} \dots b_{n-1}$, gives the assertion.

© **Step 3.** We reverse the order. For $j < n$, put $a_j = (b_{n-1-j}, m)$. Then $\{\varphi_0(x, a_j) : j < n\} = \{\varphi(x, b_i) : i < n\}$, which is consistent. Let $j < n$ be given, and let $A_j = \{a_l : l < j\}$, the set of coordinates of m and of $b_{n-j}, \dots, b_{n-1}$ (for $j = 0$, the empty set). By Step 2, with $i = n - 1 - j$, the formula $\varphi(x, b_{n-1-j})$ k -divides over $C_j = M \cup \{b_{n-j}, \dots, b_{n-1}\}$; that is, $\varphi_0(x, a_j)$ k -divides over C_j . Since $A_j \subseteq C_j$, Remark 5.2.5 shows that $\varphi_0(x, a_j)$ k -divides over A_j . Thus $a_0, \dots, a_{n-1}$ satisfy the hypotheses of Lemma 5.3.5(1) for the formula φ_0 , as required.

(b) (1) $\Rightarrow$ (2). A Morley sequence over M with $b_0 = b$ is indiscernible over M , so $b_i \equiv_M b$ for every i , and it satisfies $b_i \downarrow_M^d b_{<i}$ by definition. Part (a) applies.

(2) $\Rightarrow$ (3). There is a Morley sequence over M beginning with b , by Lemma 7.1.6.

(3) $\Rightarrow$ (1). A Morley sequence over M is indiscernible over M , so this is the definition of dividing, with the parameter convention. ☺

7.2.3. Without simplicity

In a theory that is not simple, dividing may be visible along one Morley sequence and invisible along another. The theory of $(\mathbb{Q}, <)$ shows how. It is not simple, by Proposition 5.3.2, since the formula $x < y$ has the strict order property. We use quantifier elimination as in Example 5.2.6, now with parameters: applied to a tuple together with finite tuples from a set D , it shows that the type of the tuple over D is determined by the order between its coordinates and the elements of D . For an element c , write $M < c$ if c is greater than every element of M .

Example 7.2.3. Let T be the theory of $(\mathbb{Q}, <)$, let M be a model, and let $\varphi(x; y_1, y_2)$ be the formula $y_1 < x < y_2$. Let $b = (d_1, d_2)$ with $M < d_1 < d_2$.

The formula $\varphi(x, b)$ divides over M . Choose pairs $b^i = (b_1^i, b_2^i)$ with $b^0 = b$, with $M < b_1^i < b_2^i$, and with $b_2^i < b_1^{i+1}$ for every i . Every increasing tuple of the sequence (b^i)

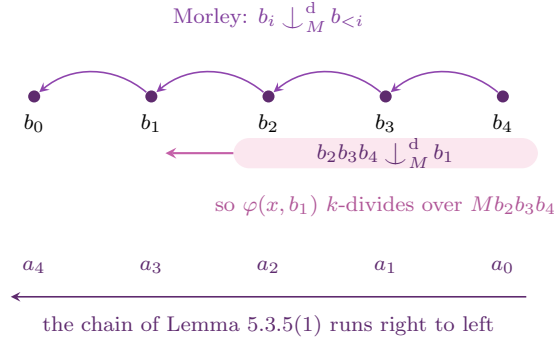


FIGURE 7.2. The proof of Kim's lemma, for $n = 5$. The Morley condition makes each term independent from the terms on its left (arrows above). Transitivity on the left makes the terms on the right of b_1 , taken together, independent from b_1 (shaded), and Lemma 7.2.1 then makes $\varphi(x, b_1)$ divide over M together with them. Read from right to left, the sequence is a chain in which every formula divides over the earlier ones, and Lemma 5.3.5(1) builds a tree of height 5 from it.

is ordered in the same way, and all its coordinates are greater than every element of M . By quantifier elimination, the sequence is indiscernible over M . The intervals (b_1^i, b_2^i) are pairwise disjoint, so the formulas $\varphi(x, b^i)$ are 2-inconsistent.

A Morley sequence along which the formula is consistent. Choose pairs $b_i = (b_{i,1}, b_{i,2})$ with $b_0 = b$, and with

$$M < b_{i+1,1} < b_{i,1} \quad \text{and} \quad b_{i,2} < b_{i+1,2}$$

for every i , so that the intervals $(b_{i,1}, b_{i,2})$ are nested and grow outward (see Figure 7.3). As before, the sequence (b_i) is indiscernible over M ; and every element of the interval (d_1, d_2) satisfies all the formulas $\varphi(x, b_i)$, so these are consistent. It remains to show that $b_i \downarrow_M^d b_{<i}$ for $i \geq 1$ (for $i = 0$ there is nothing to prove), and we verify condition (2) of Lemma 7.1.2 for the finite tuple $c = b_{<i}$.

Let $(c_j)_{j < \omega}$ be indiscernible over M with $c_0 = c$. Every coordinate of c is greater than every element of M , and each c_j has the type of c over M , so the same holds for every coordinate of every c_j . By saturation there are a'_1, a'_2 with $M < a'_1$, with a'_1 smaller than every coordinate of every c_j , and with a'_2 greater than all these coordinates. Put $a' = (a'_1, a'_2)$. For $j_1 < \dots < j_r$, the order between the coordinates of a' , the coordinates of $c_{j_1}, \dots, c_{j_r}$, and the elements of M is determined by the order between the coordinates of $c_{j_1}, \dots, c_{j_r}$ and the elements of M , which does not depend on $j_1 < \dots < j_r$. By quantifier elimination, (c_j) is indiscernible over $M \cup a'$. Finally, $b_{i,1}$ lies above M and below every coordinate of c , and $b_{i,2}$ lies above every coordinate of c , exactly as a'_1 and a'_2 do; so $a' \equiv_{M_c} b_i$. By Lemma 7.1.2, we have $b_i \downarrow_M^d b_{<i}$, and (b_i) is a Morley sequence over M .

The coheir sequence sees the dividing. Let $\mathcal{U}$ be an ultrafilter on M^2 with $q_{\mathcal{U}}|_{\emptyset} = \text{tp}(b/M)$, and let $(b'_i)_{i < \omega}$ be the Morley sequence of Lemma 7.1.6, with $b'_0 = b$ and $b'_i = (b'_{i,1}, b'_{i,2})$. Let $j < i$ be given. Since $b'_j \equiv_M b$, we have $M < b'_{j,1}$, so every pair (m_1, m_2) from M satisfies $m_2 < b'_{j,1}$, and the formula $y_2 < b'_{j,1}$ belongs to $q_{\mathcal{U}}|_{b'_{<i}} = \text{tp}(b'_i/Mb'_{<i})$.

Hence $b'_{i,2} < b'_{j,1}$: each interval lies entirely below the earlier ones, and the formulas $\varphi(x, b'_i)$ are 2-inconsistent, as Kim's lemma would require.

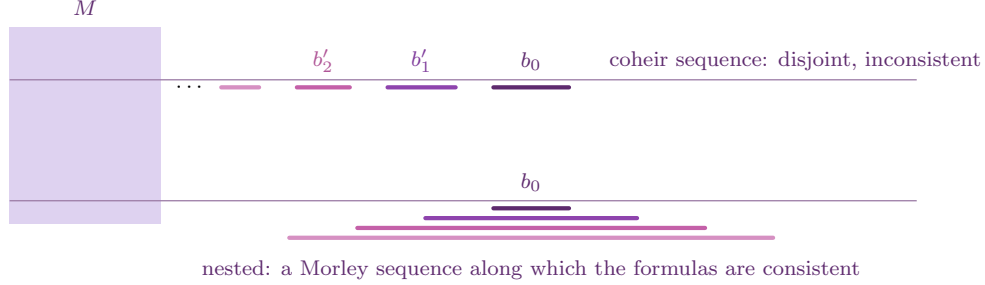


FIGURE 7.3. Example 7.2.3, in the theory of $(\mathbb{Q}, <)$. The model M is shaded, and every interval lies above it. Both sequences of intervals are Morley sequences over M beginning with the same interval $b_0 = b$. Along the coheir sequence (top) the intervals move down toward M and are pairwise disjoint; along the nested sequence (bottom) they grow outward and all contain b_0 . Kim's lemma says that in a simple theory this cannot happen.

Kim's lemma is the form in which simplicity enters the rest of the chapter. In the next section we use it to show that a formula forks over a model exactly when it divides, so that nondividing types have nondividing extensions to any larger set, and that nondividing independence is symmetric and transitive.

7.3. Forking, symmetry and transitivity

Dividing is easy to test and hard to extend; forking is easy to extend and hard to test. A formula divides when one sequence of its conjugates is inconsistent, which can be checked; but a type that does not divide over C need not have an extension to a larger set that does not divide over C , because a disjunction of formulas that divide need not divide. Forking closes dividing under disjunctions and implications, and so makes extension automatic, at the price of an extra quantifier. In a simple theory, over a model, the two notions coincide, and one has the advantages of both. From this and Kim's lemma, symmetry follows in a few lines, and transitivity follows from symmetry and Lemma 7.1.7. Throughout the section, M and N are models.

7.3.1. Forking

Definition 7.3.1. Let C be a small set, and let $\varphi(x, y) \in L(C)$.

- (a) The formula $\varphi(x, b)$ *forks over* C if there are $r \geq 1$ and formulas $\psi_1(x, c_1), \dots, \psi_r(x, c_r)$, with $\psi_l(x, z_l) \in L(C)$, each of which divides over C (in the sense of the parameter convention following Definition 7.1.1), such that

$$\mathbb{M} \models \forall x \left(\varphi(x, b) \rightarrow \bigvee_{l=1}^r \psi_l(x, c_l) \right).$$

- (b) Let $B \supseteq C$ be small. A type p over B *forks over* C if $p \vdash \bigvee_{l=1}^r \psi_l(x, c_l)$ for some formulas $\psi_l(x, c_l)$ as in (a).

This agrees with Remark 6.3.2. A formula that divides over C forks over C (take $r = 1$ and $\psi_1 = \varphi$), and so does every formula that implies a formula forking over C , and every finite disjunction of formulas forking over C . By compactness, a type that is closed under finite conjunctions, in particular a complete type, forks over C if and only if one of its formulas forks over C . Hence a complete type over B that forks over C divides over C if forking and dividing agree for formulas.



Remark 7.3.2. Forking and dividing differ in general. The standard example is the theory of the dense circular order, with a ternary relation $R(x, y, z)$, read “going around the circle in the positive direction from x , one meets y before z ” (see [TZ12, Chapter 7]). Choose three distinct elements c_1, c_2, c_3 in this order around the circle. Every element is equal to some c_l or lies on one of the three open arcs between them, so the formula $x = x$ implies the disjunction of the six formulas $x = c_l$ and $R(c_l, x, c_{l+1})$ (indices modulo 3). Each of the six divides over $\emptyset$. Indeed, by quantifier elimination for this theory, all pairs of distinct points have the same type over $\emptyset$, and a sequence of points $e_0, e_1, \dots$ in positive order around the circle, or of pairs (e_i, f_i) with $e_0, f_0, e_1, f_1, \dots$ in positive order, is indiscernible over $\emptyset$. An automorphism moves e_0 to c_l , or (e_0, f_0) to (c_l, c_{l+1}) , and the formulas $x = e_i$, like the formulas $R(e_i, x, f_i)$ of the pairwise disjoint arcs, are 2-inconsistent. Hence $x = x$ forks over $\emptyset$. It does not divide over $\emptyset$, since it has no parameters and is consistent. Over models of a simple theory this cannot happen, as we now show.

7.3.2. Forking equals dividing

Theorem 7.3.3. *Assume that T is simple. Let b be a finite tuple, and let $\varphi(x, y) \in L(M)$. Then $\varphi(x, b)$ forks over M if and only if $\varphi(x, b)$ divides over M . Consequently, for a small tuple a and a small set B , the type $\text{tp}(a/MB)$ forks over M if and only if it divides over M .*

Proof. A formula that divides over M forks over M . For the converse, suppose that $\varphi(x, b)$ does not divide over M , and let formulas $\psi_1(x, c_1), \dots, \psi_r(x, c_r)$ be given, with $\psi_l(x, c_l) \in L(M)$, such that $\varphi(x, b)$ implies their disjunction. We show that some $\psi_l(x, c_l)$ does not divide over M .

Let $c = (b, c_1, \dots, c_r)$, and let $(c^i)_{i < \omega}$ be a Morley sequence over M with $c^0 = c$, which exists by Lemma 7.1.6. Write $c^i = (b^i, c_1^i, \dots, c_r^i)$, with the blocks of the lengths of $b, c_1, \dots, c_r$. The sequence $(b^i)_{i < \omega}$ is indiscernible over M , and $b^i \downarrow_M^d b^{<i}$ for every i , by monotonicity on both sides; so it is a Morley sequence over M with $b^0 = b$. The formula $\varphi(x, b)$ does not divide over M , so the set $\{\varphi(x, b^i) : i < \omega\}$ is consistent; let a realize it.

The $L(M)$ -formula $\forall x(\varphi(x, y) \rightarrow \bigvee_l \psi_l(x, z_l))$ is satisfied by c , and $c^i \equiv_M c$; so $\varphi(x, b^i)$ implies $\bigvee_l \psi_l(x, c_l^i)$ for every i . Hence for every i there is l with $\mathbb{M} \models \psi_l(a, c_l^i)$, and by the pigeonhole principle there are a single l and an infinite set $I \subseteq \omega$ such that $\mathbb{M} \models \psi_l(a, c_l^i)$ for every $i \in I$. Let $i_0 < i_1 < \dots$ enumerate I , and put $d_k = c_{l^{i_k}}^{i_k}$. As before, $(d_k)_{k < \omega}$ is a Morley sequence over M , and a realizes $\{\psi_l(x, d_k) : k < \omega\}$. By Kim’s lemma (see Theorem 7.2.2(b)), the formula $\psi_l(x, d_0)$ does not divide over M . Since $d_0 \equiv_M c_l$, an automorphism of $\mathbb{M}$ over M carries d_0 to c_l , and so $\psi_l(x, c_l)$ does not divide over M either.

The statement about types follows from the observation after Definition 7.3.1. ☺

The first use of forking is the one it was invented for.

Corollary 7.3.4 (Extension). *Assume that T is simple. Let a be a small tuple, and let $B \subseteq B'$ be small sets such that $a \downarrow_M^d B$. Then there is $a' \equiv_{MB} a$ with $a' \downarrow_M^d B'$.*

Proof. Let $\Sigma(x)$ be the union of $\text{tp}(a/MB)$ and the set of formulas $\neg\psi(x, c)$ such that $\psi(x, z) \in L(M)$, the tuple c is from B' , and $\psi(x, c)$ divides over M . If Σ were inconsistent,

then by compactness, and since $\text{tp}(a/MB)$ is closed under conjunctions, some formula $\varphi(x, b)$ of $\text{tp}(a/MB)$ would imply a finite disjunction of formulas dividing over M ; so $\varphi(x, b)$ would fork over M , and by Theorem 7.3.3 it would divide over M , contradicting $a \downarrow_M^d B$. By saturation, the set Σ is realized by some a' . Then $a' \equiv_{MB} a$, and no formula of $\text{tp}(a'/MB')$ divides over M , since the negation of each formula of this kind over MB' belongs to Σ . Hence $a' \downarrow_M^d B'$. ☺

7.3.3. Symmetry

Symmetry says that if a tells us nothing new about b , then b tells us nothing new about a . The proof turns a failure of symmetry into a sequence of copies of a , each independent from the earlier ones and all in the same relation to b , and Kim's lemma forbids such a sequence (see Figure 7.4).

Theorem 7.3.5 (Symmetry). *Assume that T is simple, and let A and B be small sets. If $A \downarrow_M^d B$, then $B \downarrow_M^d A$.*

Proof. By finite character, we may assume that A and B are finite tuples a and b . Suppose, for a contradiction, that $a \downarrow_M^d b$ and that $\text{tp}(b/Ma)$ divides over M . Let $\psi(y, a)$ be a formula of $\text{tp}(b/Ma)$, with $\psi(y, z) \in L(M)$, that divides over M .

We choose a_i for $i < \omega$ such that $a_i \equiv_{Mb} a$ and $a_i \downarrow_M^d ba_{<i}$. Put $a_0 = a$, which satisfies both conditions because $a \downarrow_M^d b$. Given $a_{<i}$ with $i \geq 1$, apply Corollary 7.3.4 to a , with $\{b\}$ and $\{b\} \cup a_{<i}$ in place of B and B' ; it gives $a_i \equiv_{Mb} a$ with $a_i \downarrow_M^d ba_{<i}$.

Every a_i has the type of a over M , and $a_i \downarrow_M^d a_{<i}$ by monotonicity. Since $\psi(y, a)$ divides over M , Kim's lemma (see Theorem 7.2.2(a)) says that $\{\psi(y, a_i) : i < \omega\}$ is inconsistent. On the other hand, the tuple b satisfies $\psi(y, a)$, and $a_i \equiv_{Mb} a$, so b satisfies $\psi(y, a_i)$ for every i and realizes the set. This is a contradiction. ☹

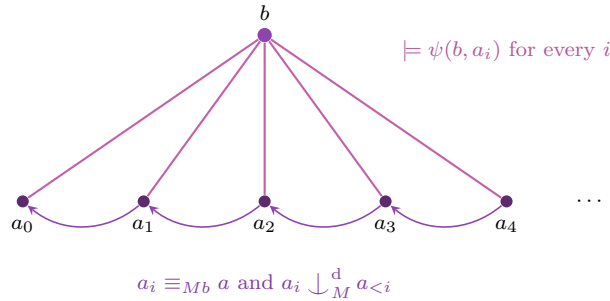


FIGURE 7.4. The proof of symmetry (see Theorem 7.3.5). If $a \downarrow_M^d b$ but a formula $\psi(y, a)$ of $\text{tp}(b/Ma)$ divides over M , then extension produces copies a_i of a over $M \cup b$, each independent from the earlier ones. The tuple b realizes every $\psi(y, a_i)$, while Kim's lemma says that no tuple realizes all of them.

7.3.4. Transitivity, and the axioms that hold in simple theories

With symmetry in hand, transitivity reduces to transitivity on the left (see Lemma 7.1.7): one turns the relations around, applies that lemma, and turns the result around again. The theorem below collects the axioms that nondividing independence satisfies in a simple theory.

Theorem 7.3.6. *Assume that T is simple. Then the relation $\downarrow^d$, with models as bases, satisfies (INV), (MON), (BMON), (TR), (FC), (SYM), (EX) and (LOC).*

Proof. All but transitivity and symmetry are in Proposition 7.1.8, and symmetry is Theorem 7.3.5. *Transitivity:* let $M \subseteq N$, and suppose that $A \downarrow_M^d N$ and $A \downarrow_N^d B$. By symmetry, we have $N \downarrow_M^d A$ and $B \downarrow_N^d A$; since $M \cup N = N$, the second relation says that $B \downarrow_{MN}^d A$. Lemma 7.1.7, with M , N , A and B in the roles of C , A , B and D , gives $NB \downarrow_M^d A$, and symmetry gives $A \downarrow_M^d NB$. 

By Theorem 7.3.3, over models, nondividing independence is nonforking independence; we keep the notation $\downarrow^d$. Of the properties of Definition 5.4.7, only uniqueness is missing, and it must fail in general: uniqueness together with local character implies stability (see Theorem 6.2.2), while the random graph is simple and unstable (see Theorem 5.4.6). In its place, simple theories have the independence theorem, which the next section proves.

7.4. The independence theorem

Uniqueness fails in an unstable theory: a type over a model may have many independent extensions to a larger set, and nothing singles one of them out. What survives is a weaker and, as it turns out, more characteristic property. Two independent extensions of the same type, one to $M \cup a$ and one to $M \cup b$, can always be realized together, independently, provided only that a and b are themselves independent over M . This is the independence theorem of Kim and Pillay [KP97]; in the random graph it is the amalgamation of Theorem 5.4.9(c). In a stable theory it is a consequence of uniqueness; in a simple theory it has to be earned.

The proof moves a parameter step by step. Suppose that a formula $\varphi(x, a) \wedge \psi(x, b')$ does not divide over M , where b' has the type of b over M ; we want the same of $\varphi(x, a) \wedge \psi(x, b)$. If b' and b begin an indiscernible sequence, the parameter can be moved from b' to b (see Proposition 7.4.5); in general a third tuple, a coheir over everything in sight, serves as a stepping stone between them. The step itself rests on a fact about conjunctions along indiscernible sequences (see Proposition 7.4.4), and here a second idea enters: an arbitrary indiscernible sequence over M can be replaced by a Morley sequence over a larger model, obtained as a coheir of a reversed copy of the sequence, and to Morley sequences Kim's lemma applies. Throughout the section, T is simple and M is a model.

7.4.1. Two tools

The first tool extends the corollary on extension (see Corollary 7.3.4) from complete types to partial ones. A set of formulas is *closed under conjunctions* if it contains the conjunction of any two of its members.

Lemma 7.4.1. *Let $B \subseteq B'$ be small sets, and let $\pi(x)$ be a set of $L(MB)$ -formulas such that no finite conjunction of members of π divides over M . Then some a realizes π with $a \downarrow_M^d B'$.*

Proof. Arguing as in the proof of Corollary 7.3.4, let $\Sigma(x)$ be the union of π and the set of formulas $\neg\chi(x, c)$ with $\chi(x, z) \in L(M)$, the tuple c from B' , and $\chi(x, c)$ dividing over M . If Σ were inconsistent, some finite conjunction of members of π would imply a finite disjunction of formulas dividing over M (or be inconsistent, and then divide over M); so it would fork over M , and by Theorem 7.3.3 it would divide over M . Hence Σ is consistent, and any realization a of Σ satisfies $a \downarrow_M^d B'$. 

We shall also use a consequence of Theorem 7.3.3 that does not depend on the lemma: a formula implied by a formula that does not divide over M does not divide over M either, since forking is closed under implication (see the remarks after Definition 7.3.1) and forking equals dividing over models (see Theorem 7.3.3). The same holds over any model in place of M .

The second tool turns indiscernible sequences around and attaches companions to their terms. We say that tuples d_0, d_1 *begin an indiscernible sequence over M* if there are d_i for $2 \leq i < \omega$ such that $(d_i)_{i < \omega}$ is indiscernible over M .

Lemma 7.4.2. *If d_0, d_1 begin an indiscernible sequence over M , then so do d_1, d_0 .*

Proof. Let $(d_i)_{i < \omega}$ be indiscernible over M . By compactness and saturation there is a sequence $(d'_i)_{i \in \mathbb{Z}}$ such that $(d'_{i_1}, \dots, d'_{i_n}) \equiv_M (d_0, \dots, d_{n-1})$ for every n and all $i_1 < \dots < i_n$ in $\mathbb{Z}$: a finite part of the required type mentions finitely many indices, and an increasing map from them into ω realizes it. Put $e_i = d'_{1-i}$ for $i < \omega$. An increasing tuple of (e_i) is a decreasing tuple of (d'_i) , and every decreasing n -tuple of (d'_i) has the type of $(d_{n-1}, \dots, d_0)$ over M ; hence $(e_i)_{i < \omega}$ is indiscernible over M , and $(e_0, e_1) \equiv_M (d_1, d_0)$. An automorphism of $\mathbb{M}$ over M carrying (e_0, e_1) to (d_1, d_0) carries (e_i) to the required sequence. $\odot$

Lemma 7.4.3. *Suppose that finite tuples b_0, b_1 begin an indiscernible sequence over M , and let a be a tuple with $a \perp_{Mb_0}^d b_1$. Then there is a^* such that the pairs (b_0, a) and (b_1, a^*) begin an indiscernible sequence over M .*

Proof. Let $(b_i)_{i < \omega}$ be indiscernible over M . The sequence $(b_i)_{1 \leq i < \omega}$ is indiscernible over $C = M \cup b_0$ and begins with b_1 , and $\text{tp}(a/Cb_1)$ does not divide over C . By Lemma 7.1.2 there is $a' \equiv_{Cb_1} a$ such that $(b_i)_{i \geq 1}$ is indiscernible over $C \cup a'$. Applying an automorphism of $\mathbb{M}$ over $C \cup b_1$ that carries a' to a , and replacing b_i by its image for $i \geq 2$, we may assume that $(b_i)_{i \geq 1}$ is indiscernible over $M \cup b_0 \cup a$; the whole sequence remains indiscernible over M .

For $n < \omega$ the sequences $(b_i)_{i < \omega}$ and $(b_{n+i})_{i < \omega}$ have the same type over M , so there is $\sigma_n \in \text{Aut}(\mathbb{M}/M)$ with $\sigma_n(b_i) = b_{n+i}$ for every i . Put $a_n = \sigma_n(a)$. For $n < m$ we have

$$(b_n, a_n, b_m) = \sigma_n(b_0, a, b_{m-n}) \equiv_M (b_0, a, b_{m-n}) \equiv_M (b_0, a, b_1),$$

the last step because $(b_i)_{i \geq 1}$ is indiscernible over $M \cup b_0 \cup a$. Apply Lemma 5.2.2 over M to the sequence of pairs $((b_n, a_n))_{n < \omega}$. Its Ehrenfeucht–Mostowski type over M contains $\text{tp}(b_0, a, b_1/M)$, read as a type of the first coordinate of one pair, the second coordinate of that pair, and the first coordinate of a later pair. The lemma gives an indiscernible sequence $((b_n^*, a_n^*))_{n < \omega}$ over M with $(b_0^*, a_0^*, b_1^*) \equiv_M (b_0, a, b_1)$. An automorphism of $\mathbb{M}$ over M that carries (b_0^*, a_0^*, b_1^*) to (b_0, a, b_1) carries a_1^* to the required a^* . $\odot$

7.4.2. Conjunctions along indiscernible sequences

If $\psi(x, b)$ does not divide over M and (b_i) is an indiscernible sequence over M beginning with b , the formulas $\psi(x, b_i)$ are consistent, by the definition of dividing. The next proposition says more: finitely many of them together still do not divide. For a Morley sequence this is an application of Kim’s lemma to blocks of the sequence (see Step 3 of the proof below). For an arbitrary indiscernible sequence we first replace it by a Morley sequence over a larger model, with the same type over M .

Proposition 7.4.4. *Let $\psi(x, y) \in L(M)$, let b be a finite tuple such that $\psi(x, b)$ does not divide over M , and let $(b_i)_{i < \omega}$ be indiscernible over M with $b_0 = b$. Then for every $n \geq 1$ the formula $\bigwedge_{i < n} \psi(x, b_i)$ does not divide over M .*

Proof. ☉ **Step 1: a reversed copy.** By compactness and saturation, as in the proof of Lemma 7.4.2, there is a sequence $(h_j)_{j < \omega}$ such that $(h_{j_0}, \dots, h_{j_{n-1}}) \equiv_M (b_0, \dots, b_{n-1})$ for every n and all $j_0 > j_1 > \dots > j_{n-1}$: the decreasing tuples of (h_j) have the types of the increasing tuples of (b_i) . Let N be a small model containing $M \cup \{h_j : j < \omega\}$.

☉ **Step 2: a Morley sequence over N .** Let $\mathcal{D}$ be a nonprincipal ultrafilter on ω , and let $\mathcal{V}$ be the ultrafilter on $N^{|y|}$ consisting of the sets X with $\{j : h_j \in X\} \in \mathcal{D}$. Let t_0 realize $q_{\mathcal{V}}|\emptyset$, the type over N determined by $\mathcal{V}$, and for $k \geq 1$ let t_k realize $q_{\mathcal{V}}|\{t_l : l < k\}$. By Lemma 7.1.6, applied over the model N , the sequence $(t_k)_{k < \omega}$ is a Morley sequence over N .

We claim that $(t_0, \dots, t_{n-1}) \equiv_M (b_0, \dots, b_{n-1})$ for every n . By induction on n , the definition of the coheir gives, for every $\theta(y_0, \dots, y_{n-1}) \in L(N)$,

$$\mathbb{M} \models \theta(t_0, \dots, t_{n-1}) \iff \left\{ j_{n-1} : \left\{ j_{n-2} : \dots \left\{ j_0 : \mathbb{M} \models \theta(h_{j_0}, \dots, h_{j_{n-1}}) \right\} \in \mathcal{D} \dots \right\} \in \mathcal{D} \right\} \in \mathcal{D}.$$

Indeed, t_{n-1} satisfies $\theta(t_0, \dots, t_{n-2}, y)$ exactly when the set of j with $\mathbb{M} \models \theta(t_0, \dots, t_{n-2}, h_j)$ belongs to $\mathcal{D}$, and for each j the formula $\theta(y_0, \dots, y_{n-2}, h_j)$ has parameters in N . Let $\theta \in L(M)$ be given, and suppose that $\mathbb{M} \models \theta(b_0, \dots, b_{n-1})$; then θ holds at every tuple $(h_{j_0}, \dots, h_{j_{n-1}})$ with $j_0 > j_1 > \dots > j_{n-1}$. The set $\{j : j > j'\}$ belongs to $\mathcal{D}$ for every j' , because $\mathcal{D}$ is nonprincipal. Hence, by induction from the innermost set outward, for all $j_{r+1} > \dots > j_{n-1}$ the set of j_r at the level of nesting r contains $\{j_r : j_r > j_{r+1}\}$, and so belongs to $\mathcal{D}$; at the outermost level we obtain $\mathbb{M} \models \theta(t_0, \dots, t_{n-1})$. Applying this to $\neg\theta$ gives the converse. This proves the claim. In particular $\psi(x, t_0)$ does not divide over M .

☉ **Step 3: over N .** By Lemma 7.4.1 there is a_0 with $\mathbb{M} \models \psi(a_0, t_0)$ and $a_0 \downarrow_M^d N \cup \{t_0\}$. Let $p(x, y)$ be the set of $L(N)$ -formulas $\theta(x, y)$ with $\mathbb{M} \models \theta(a_0, t_0)$, so that $p(x, t_0) = \text{tp}(a_0/Nt_0)$, and write $p(x, t_k)$ for the set of formulas $\theta(x, t_k)$ with $\theta \in p$. By Remark 5.2.5, the type $p(x, t_0)$ does not divide over N . Let $\theta(x, t_0) \in p$ be given. Since $\theta(x, t_0)$ does not divide over N and (t_k) is indiscernible over N with first term t_0 , the set $\{\theta(x, t_k) : k < \omega\}$ is consistent. The blocks $e_m = (t_{mn}, \dots, t_{mn+n-1})$, for $m < \omega$, form an indiscernible sequence over N , and $e_m \downarrow_N^d e_{<m}$: indeed, for $r < n - 1$, if $(t_{mn}, \dots, t_{mn+r}) \downarrow_N^d t_{<mn}$, then $t_{mn+r+1} \downarrow_{N \cup \{t_{mn}, \dots, t_{mn+r}\}}^d t_{<mn}$ by monotonicity and Remark 5.2.5, and Lemma 7.1.7 gives $(t_{mn}, \dots, t_{mn+r+1}) \downarrow_N^d t_{<mn}$; so (e_m) is a Morley sequence over N . The formulas $\bigwedge_{k < n} \theta(x, t_{mn+k})$, for $m < \omega$, are consistent together. Hence Kim's lemma, in the form of Theorem 7.2.2(b) applied over N , shows that $\bigwedge_{k < n} \theta(x, t_k)$ does not divide over N .

Therefore no finite conjunction of members of $\bigcup_{k < n} p(x, t_k)$ divides over N : each such conjunction is implied by a formula $\bigwedge_{k < n} \theta(x, t_k)$ with $\theta \in p$, because p is closed under conjunctions. By Lemma 7.4.1, applied over N , some a realizes this union with $a \downarrow_N^d \{t_k : k < n\}$.

☉ **Step 4: back to M .** The type $p(x, t_0)$ contains $\text{tp}(a_0/N)$, so $a \equiv_N a_0$, and hence $a \downarrow_M^d N$. Transitivity (see Theorem 7.3.6) gives $a \downarrow_M^d N \cup \{t_k : k < n\}$. The tuple a satisfies $\psi(x, t_k)$ for every $k < n$, because $\psi(x, t_0) \in p$; so $\bigwedge_{k < n} \psi(x, t_k)$ belongs to a type that does not divide over M , and does not divide over M . By the claim of Step 2, the same holds for $\bigwedge_{k < n} \psi(x, b_k)$. ☺

7.4.3. Moving a parameter

The tools are now in place for the step announced at the start of the section: a parameter is moved from the first term of an indiscernible sequence to the second, by giving the second

term a companion (see Lemma 7.4.3) and applying Proposition 7.4.4 to the resulting sequence of pairs.

Proposition 7.4.5. *Let $\varphi(x, z), \psi(x, y) \in L(M)$, and let a, b_0, b_1 be finite tuples such that b_0, b_1 begin an indiscernible sequence over M and $a \perp_{Mb_0}^d b_1$. If $\varphi(x, a) \wedge \psi(x, b_0)$ does not divide over M , then $\varphi(x, a) \wedge \psi(x, b_1)$ does not divide over M .*

Proof. By Lemma 7.4.3 there is a^* such that (b_0, a) and (b_1, a^*) begin an indiscernible sequence over M . Apply Proposition 7.4.4 to the formula $\chi(x; y, z) = \psi(x, y) \wedge \varphi(x, z)$ and this sequence of pairs: the formula $\chi(x; b_0, a) \wedge \chi(x; b_1, a^*)$ does not divide over M . It implies $\varphi(x, a) \wedge \psi(x, b_1)$, which therefore does not divide over M . ☺

Two tuples with the same type over a model need not begin an indiscernible sequence; a coheir over both of them joins them (see Figure 7.5).

Proposition 7.4.6. *Let $\varphi(x, z), \psi(x, y) \in L(M)$, and let a, b, b' be finite tuples with $b \equiv_M b'$ and $a \perp_M^d bb'$. If $\varphi(x, a) \wedge \psi(x, b')$ does not divide over M , then $\varphi(x, a) \wedge \psi(x, b)$ does not divide over M .*

Proof. Let $\mathcal{U}$ be an ultrafilter on $M^{|b|}$ with $q_{\mathcal{U}}|\emptyset = \text{tp}(b/M) = \text{tp}(b'/M)$ (see Lemma 6.1.5(b)), and let e realize $q_{\mathcal{U}}| \{a, b, b'\}$. Then e realizes $q_{\mathcal{U}}| \{b'\}$ and $q_{\mathcal{U}}| \{b\}$, so by Lemma 7.1.6 each of the pairs b', e and b, e begins an indiscernible sequence over M , and by Lemma 7.4.2 so does the pair e, b . The type $\text{tp}(e/Mabb')$ is finitely satisfiable in M , so $e \perp_M^d abb'$ by Lemma 7.1.3. By symmetry we have $bb' \perp_M^d a$, and $e \perp_{Mbb'}^d a$ by monotonicity and Remark 5.2.5; Lemma 7.1.7 and symmetry give $a \perp_M^d bb'e$. Consequently $a \perp_{Mb'}^d e$ and $a \perp_{Me}^d b$.

Proposition 7.4.5, applied to the pair b', e , shows that $\varphi(x, a) \wedge \psi(x, e)$ does not divide over M ; applied to the pair e, b , it shows that $\varphi(x, a) \wedge \psi(x, b)$ does not divide over M . ☺

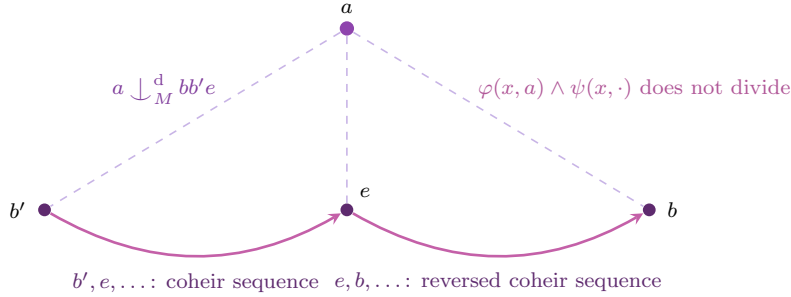


FIGURE 7.5. Proposition 7.4.6. A coheir e over $M \cup \{a, b, b'\}$ joins b' to b : the pair b', e begins an indiscernible sequence, and so, after reversal, does the pair e, b . Along each of the two steps Proposition 7.4.5 carries non-dividing of $\varphi(x, a) \wedge \psi(x, \cdot)$ from one end to the other.

7.4.4. The theorem

Proposition 7.4.6 carries the weight of the theorem. What remains is to find, for given formulas over A and over B , a copy of the parameter from B for which the conjunction is known not to divide, and such that the parameter from A is independent over M from the original and the copy together, as the proposition requires.

Theorem 7.4.7 (Independence theorem). *Assume that T is simple. Let M be a model, let A and B be small sets with $A \downarrow_M^d B$, and let c_1, c_2 be tuples with $c_1 \equiv_M c_2$, $c_1 \downarrow_M^d A$ and $c_2 \downarrow_M^d B$. Then there is c with $c \equiv_{MA} c_1$, $c \equiv_{MB} c_2$ and $c \downarrow_M^d AB$.*

Proof. Let $\Sigma(x)$ be the closure under conjunctions of the union of the types $\text{tp}(c_1/MA)$ and $\text{tp}(c_2/MB)$. By Lemma 7.4.1 it suffices to show that no formula of Σ divides over M . Such a formula is implied by a formula $\varphi(x, a) \wedge \psi(x, b)$ with $\varphi(x, a) \in \text{tp}(c_1/Ma)$, $\psi(x, b) \in \text{tp}(c_2/Mb)$, $\varphi, \psi \in L(M)$, and a and b finite tuples from A and B . Fix such formulas and tuples; by monotonicity, $a \downarrow_M^d b$, $c_1 \downarrow_M^d a$ and $c_2 \downarrow_M^d b$.

Let $\sigma \in \text{Aut}(M/M)$ carry c_2 to c_1 , and put $b_0 = \sigma(b)$. Then $c_1 b_0 \equiv_M c_2 b$, and $b_0 \downarrow_M^d c_1$ because $b \downarrow_M^d c_2$ by symmetry. By Corollary 7.3.4 there is $b' \equiv_{Mc_1} b_0$ with $b' \downarrow_M^d c_1 a b$. Thus $c_1 b' \equiv_M c_2 b$, and in particular $b' \equiv_M b$.

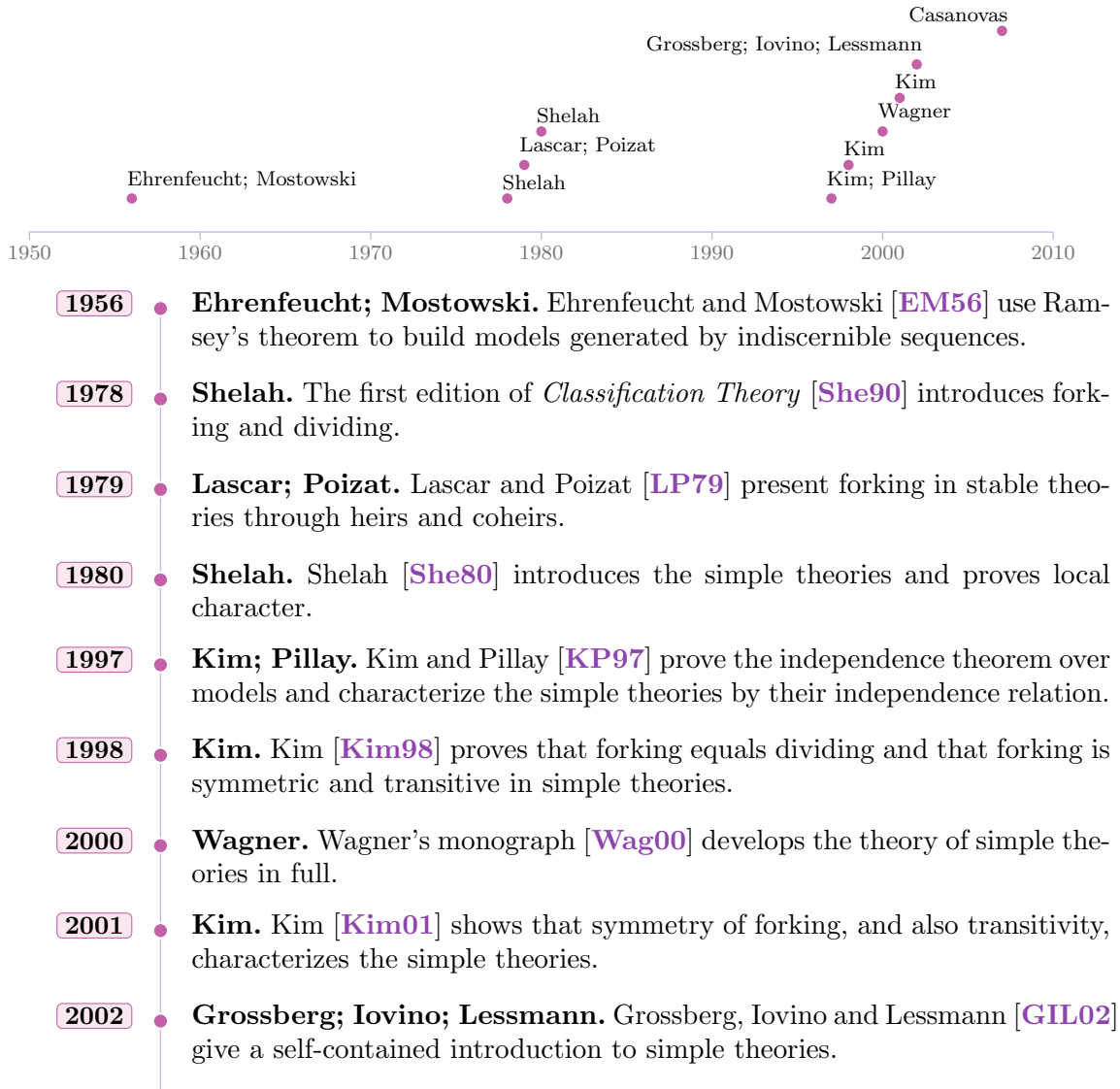
We have $a \downarrow_M^d c_1$ by symmetry, and $b' \downarrow_{Ma}^d c_1$ by monotonicity and Remark 5.2.5, so Lemma 7.1.7 and symmetry give $c_1 \downarrow_M^d a b'$. Hence the formula $\varphi(x, a) \wedge \psi(x, b')$, which belongs to $\text{tp}(c_1/Mab')$ because $c_1 b' \equiv_M c_2 b$, does not divide over M . Similarly, $b \downarrow_M^d a$ and $b' \downarrow_{Mb}^d a$ give $b b' \downarrow_M^d a$, that is, $a \downarrow_M^d b b'$. By Proposition 7.4.6, the formula $\varphi(x, a) \wedge \psi(x, b)$ does not divide over M . $\odot$

In the Rado graph, Theorem 5.4.9(c) is the analogue for trivial independence, over arbitrary small sets, and Theorem 5.5.9 proves the corresponding statement for Fraïssé limits with disjoint 3-amalgamation. With Theorem 7.4.7, nondividing independence over models has every property of Definition 5.4.7 except uniqueness, together with the independence theorem. That these properties characterize the simple theories, and determine the relation, is the Kim–Pillay theorem, the subject of a section still in preparation (see the plan at the start of the chapter).

7.5. Historical remarks

- ✂ **Section 7.1.** Dividing and forking were introduced by Shelah in the first edition of *Classification Theory* [She90], where they serve stable theories; dividing is the combinatorial notion, and forking is its closure under finite disjunctions, designed so that nonforking extensions always exist. Indiscernible sequences go back to Ehrenfeucht and Mostowski [EM56], who extracted them with Ramsey’s theorem, as in Lemma 5.2.2, and Morley [Mor65] made them a central tool of the subject; sequences whose terms are independent from the earlier ones are named after him. That finitely satisfiable types do not divide, and the use of coheirs to produce such sequences, come from the work of Lascar and Poizat [LP79] on forking in stable theories. The characterization of nondividing by indiscernible sequences (see Lemma 7.1.2) is standard; we follow the presentation of Grossberg, Iovino and Lessmann [GIL02].
- ✂ **Section 7.2.** Simple theories were introduced by Shelah [She80], who showed that local character holds in them, and for some fifteen years they received little attention. Kim [Kim98] revived the subject. He proved that in a simple theory a formula divides exactly when it divides along some, equivalently every, Morley sequence, and deduced that forking and dividing agree and that independence is symmetric and transitive. In a later paper [Kim01] he showed that each of symmetry and transitivity of forking characterizes the simple theories.

- ⌘ **Section 7.3.** Forking and dividing differ in general, and the dense circular order is the usual example (see [TZ12, Chapter 7]). The derivation of symmetry from Kim’s lemma, by a sequence of copies of a that a single tuple b cannot follow, goes back to Kim [Kim98].
- ⌘ **Section 7.4.** The independence theorem is due to Kim and Pillay [KP97], who abstracted it from Hrushovski’s work on pseudofinite fields [Hru02]. They proved it for Lascar strong types over arbitrary sets, which over a model are simply types, and that is the form in which it appears in the monographs of Wagner [Wag00] and Casanovas [Cas11], in Casanovas’s notes [Cas07], and in the primer of Grossberg, Iovino and Lessmann [GIL02]. Our proof follows the route of Casanovas: a parameter is moved along indiscernible sequences (see Propositions 7.4.5 and 7.4.6), and the moving rests on the fact that conjunctions along an indiscernible sequence do not divide (see Proposition 7.4.4). Casanovas proves that fact over a set that is not a model, using symmetry and transitivity of forking over arbitrary sets. To stay with models, we have replaced that step by an argument specific to these notes: the indiscernible sequence is traded for a Morley sequence over a larger model, obtained as a coheir of a reversed copy of the sequence, to which Kim’s lemma applies directly.



- 2007** • **Casanovas.** Casanovas [**Cas07**] streamlines the basic theory, including the independence theorem for Lascar strong types.

Appendix A

Grothendieck's double limit criterion

Grothendieck's double limit criterion is used at several points in these notes: in Subsection 1.2.4 and in Remarks 1.2.14 and 4.2.6. This appendix states the criterion in the form we need and explains how the criterion relates to the stability of functions (see Definition 1.2.5). Throughout, we fix a compact Hausdorff space K , and $C(K)$ denotes the Banach space of continuous real-valued functions on K with the supremum norm. The *weak topology* on $C(K)$ is the coarsest topology making every bounded linear functional continuous, and the *pointwise topology* is the one induced by the product topology of $\mathbb{R}^K$. Point evaluations $f \mapsto f(x)$ are bounded linear functionals, so the weak topology is finer than the pointwise topology.

A.1. The criterion

Definition A.1.1. A bounded set $H \subseteq C(K)$ satisfies the *double limit condition* if for all sequences (x_m) in K and (f_n) in H and all ultrafilters $\mathcal{U}, \mathcal{V}$ on $\mathbb{N}$, we have

$$\lim_{m, \mathcal{U}} \lim_{n, \mathcal{V}} f_n(x_m) = \lim_{n, \mathcal{V}} \lim_{m, \mathcal{U}} f_n(x_m).$$

In the language of Definition 1.2.5, the double limit condition says that the evaluation function $K \times H \rightarrow \mathbb{R}$, $(x, f) \mapsto f(x)$, is stable. Grothendieck stated the condition with ordinary limits; the next lemma shows that the two formulations are equivalent.

Lemma A.1.2. *A bounded set $H \subseteq C(K)$ satisfies the double limit condition if and only if*

$$\lim_m \lim_n f_n(x_m) = \lim_n \lim_m f_n(x_m)$$

for all sequences (x_m) in K and (f_n) in H for which both iterated limits exist.

Proof. If both ordinary iterated limits exist, these limits are equal to the iterated limits along any nonprincipal ultrafilters, so the double limit condition implies the second condition.

Conversely, suppose the double limit condition fails for (x_m) , (f_n) , $\mathcal{U}$ and $\mathcal{V}$, with iterated limits $\alpha \neq \beta$. After an affine change of scale we may assume that all values lie in $[0, 1]$. Proposition 1.2.3, applied to the function $(x, f) \mapsto f(x)$, gives subsequences, which we still denote (x_m) and (f_n) , with $|f_n(x_m) - \alpha| \leq 2^{1-m}$ for $m < n$ and $|f_n(x_m) - \beta| \leq 2^{1-n}$ for $n < m$. By a diagonal argument we may pass to further subsequences along which $\lim_n f_n(x_m)$ exists for every m and $\lim_m f_n(x_m)$ exists for every n ; the inequalities are preserved, with the indices renumbered. Then $\lim_n f_n(x_m)$ is within 2^{1-m} of α , so $\lim_m \lim_n f_n(x_m) = \alpha$, and symmetrically $\lim_n \lim_m f_n(x_m) = \beta$. Hence the second condition fails. ☹

Theorem A.1.3 (Grothendieck). *Let $H \subseteq C(K)$ be bounded. The following are equivalent:*

- (1) *The set H is relatively weakly compact in $C(K)$.*
- (2) *Every function in the pointwise closure of H in $\mathbb{R}^K$ is continuous.*
- (3) *The set H satisfies the double limit condition.*

The equivalence of (2) and (3) is an exchange of ultrafilter limits, together with Proposition 1.2.3; the measure-theoretic input and the angelicity of pointwise compact sets enter only in (2) $\Rightarrow$ (1).

Proof. (2) $\Rightarrow$ (3). Let (x_m) , (f_n) , $\mathcal{U}$, $\mathcal{V}$ be given, and put $x = \lim_{m,\mathcal{U}} x_m \in K$ and $f = \lim_{n,\mathcal{V}} f_n \in \mathbb{R}^K$. By (2), the function f is continuous. Using the continuity of f and of each f_n , we obtain

$$\lim_{m,\mathcal{U}} \lim_{n,\mathcal{V}} f_n(x_m) = \lim_{m,\mathcal{U}} f(x_m) = f(x) = \lim_{n,\mathcal{V}} f_n(x) = \lim_{n,\mathcal{V}} \lim_{m,\mathcal{U}} f_n(x_m).$$

(3) $\Rightarrow$ (2). Suppose that f is in the pointwise closure of H but is not continuous at some $x \in K$. Then $f = \lim_{i,\mathcal{U}} f_i$ for a family $(f_i)_{i \in I}$ in H and an ultrafilter $\mathcal{U}$ on I , and there are a family $(x_j)_{j \in J}$ in K and an ultrafilter $\mathcal{V}$ on J with $\lim_{j,\mathcal{V}} x_j = x$ and $\lim_{j,\mathcal{V}} f(x_j) = \beta \neq f(x)$. (Fix $\varepsilon > 0$ such that every neighborhood O of x contains a point x_O with $|f(x_O) - f(x)| \geq \varepsilon$, and let $\mathcal{V}$ be an ultrafilter on the set of neighborhoods of x containing, for each O , the set of neighborhoods contained in O .) By continuity of each f_i , we have

$$\lim_{i,\mathcal{U}} \lim_{j,\mathcal{V}} f_i(x_j) = \lim_{i,\mathcal{U}} f_i(x) = f(x), \quad \lim_{j,\mathcal{V}} \lim_{i,\mathcal{U}} f_i(x_j) = \lim_{j,\mathcal{V}} f(x_j) = \beta.$$

After an affine change of scale, Proposition 1.2.3 and the proof of Corollary 1.2.6 turn these families into sequences whose iterated limits along a nonprincipal ultrafilter are β and $f(x)$. This contradicts (3).

(1) $\Rightarrow$ (2). Let W be the weak closure of H , which is weakly compact by (1). The identity map from $C(K)$ with the weak topology to $C(K)$ with the pointwise topology is continuous, so W is pointwise compact, hence closed in $\mathbb{R}^K$. Hence the pointwise closure of H is contained in $W \subseteq C(K)$.

(2) $\Rightarrow$ (1). Let $\bar{H}$ be the pointwise closure of H . This set is compact in $\mathbb{R}^K$, because H is bounded, and is contained in $C(K)$ by (2). We need only show that the weak and pointwise topologies agree on $\bar{H}$, since then $\bar{H}$ is weakly compact. By the Riesz representation theorem, the weak topology is generated by the maps $h \mapsto \int h d\mu$ with μ a Radon measure on K . On the uniformly bounded set $\bar{H}$ each such map is sequentially continuous for the pointwise topology, by Lebesgue's dominated convergence theorem. Grothendieck proved that a pointwise compact subset of $C(K)$ is *angelic*: every point in the closure of a subset is the limit of a sequence from that subset [Gro52]; see also Pryce [Pry71]. On an angelic space sequentially continuous functions are continuous, which completes the proof. ☺

The simplest family of continuous functions with a discontinuous pointwise limit already shows the two iterated limits parting company.

Example A.1.4. Let $K = [0, 1]$ and $H = \{f_n : n \geq 1\}$ with $f_n(x) = x^n$. The pointwise limit of (f_n) is 0 on $[0, 1)$ and 1 at 1, which is not continuous, so (2) of Theorem A.1.3 fails. Condition (3) fails along $x_m = 1 - 1/m$: for each m we have $\lim_n f_n(x_m) = 0$, while for each n we have $\lim_m f_n(x_m) = 1$, so the two iterated limits are 0 and 1 (see Figure A.1). Hence H is not relatively weakly compact in $C[0, 1]$.

Weak compactness becomes a statement about sequences through a classical theorem of Eberlein and Šmulian, and the criterion then yields convergent subsequences.

Fact A.1.5 (Eberlein–Šmulian). *In a Banach space, every sequence in a relatively weakly compact set has a weakly convergent subsequence [Šmu40, Ebe47].*

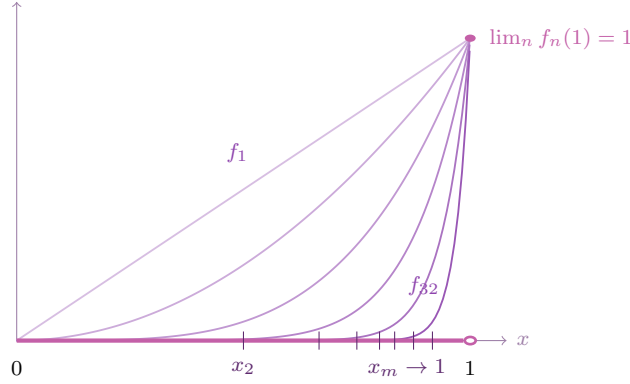


FIGURE A.1. Example A.1.4: the functions $f_n(x) = x^n$ (purple, darker for larger n) converge pointwise to a function (pink) that is 0 on $[0, 1)$ and jumps to 1 at $x = 1$. Along the points $x_m = 1 - 1/m$, letting $n \rightarrow \infty$ first gives 0, and letting $m \rightarrow \infty$ first gives 1.

Corollary A.1.6. *If a bounded set $H \subseteq C(K)$ satisfies the double limit condition, then every sequence in H has a subsequence that converges pointwise on K , to a continuous function.*

Proof. By Theorem A.1.3, the set H is relatively weakly compact. By Fact A.1.5, every sequence in H has a subsequence converging weakly to some $f \in C(K)$. Point evaluations are bounded linear functionals, so the subsequence converges pointwise to f . ☺

The criterion meets stability through the spaces of types of Section 1.2. For $\varphi: A \times B \rightarrow [0, 1]$, recall the evaluation maps $e_b: \text{LS}(\varphi) \rightarrow [0, 1]$, given by $e_b(p) = p(b)$, and $e_a: \text{RS}(\varphi) \rightarrow [0, 1]$, given by $e_a(q) = q(a)$, of Subsection 1.2.4.

Proposition A.1.7. *Let $\varphi: A \times B \rightarrow [0, 1]$ be stable. Then $\{e_b : b \in B\}$ is relatively weakly compact in $C(\text{LS}(\varphi))$, and $\{e_a : a \in A\}$ is relatively weakly compact in $C(\text{RS}(\varphi))$. More precisely, the pointwise closure of $\{e_b : b \in B\}$ consists of the functions $\hat{\varphi}(\cdot, q)$ with $q \in \text{RS}(\varphi)$, where $\hat{\varphi}$ is the function of Proposition 1.2.8.*

Proof. By (1.2) with a constant family, we have $e_b = \hat{\varphi}(\cdot, \text{rtp}_\varphi(b))$. If $q = \lim_{j, \gamma} \text{rtp}_\varphi(b_j)$, then by continuity of $\hat{\varphi}(p, \cdot)$ we get $\lim_{j, \gamma} e_{b_j}(p) = \hat{\varphi}(p, q)$ for every p . Thus every pointwise limit of functions e_b is of the form $\hat{\varphi}(\cdot, q)$, and conversely every such function is a pointwise limit of this kind. These functions are continuous on $\text{LS}(\varphi)$ by separate continuity. Hence condition (2) of Theorem A.1.3 holds, and $\{e_b : b \in B\}$ is relatively weakly compact. The statement for $\{e_a : a \in A\}$ is proved symmetrically. ☺

Proposition A.1.7 places the type spaces of a stable function among the most rigid compact spaces of topology. Recall that a compact space is an *Eberlein compactum* if the space is homeomorphic to a weakly compact subset of a Banach space with the weak topology.

Corollary A.1.8. *Let $\varphi: A \times B \rightarrow [0, 1]$ be stable. Then $\text{RS}(\varphi)$ and $\text{LS}(\varphi)$ are Eberlein compacta. In particular, every right φ -type is the limit of a sequence of realized right types, and likewise for left types.*

Proof. Let H be the pointwise closure of $\{e_b : b \in B\}$ in $\mathbb{R}^{\text{LS}(\varphi)}$. By Proposition A.1.7, the weak closure W of $\{e_b : b \in B\}$ in $C(\text{LS}(\varphi))$ is weakly compact. The identity map from W with the weak topology to W with the coarser pointwise topology is a continuous bijection from a compact space to a Hausdorff space, so the two topologies agree on W . In

particular, the set W is pointwise compact, hence pointwise closed, so $H \subseteq W$; and $W \subseteq H$ because the weak topology is finer than the pointwise topology. Thus $H = W$, and on H the weak and pointwise topologies agree. The map $q \mapsto \hat{\varphi}(\cdot, q)$ from $\text{RS}(\varphi)$ onto H is continuous for the pointwise topology, by the separate continuity of $\hat{\varphi}$, and injective, because $\hat{\varphi}(\text{ltp}_\varphi(a), q) = q(a)$ for every $a \in A$. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism, so $\text{RS}(\varphi)$ is homeomorphic to the weakly compact set H . The same argument, with the roles of A and B exchanged, applies to $\text{LS}(\varphi)$. Finally, the set H is a pointwise compact subset of $C(\text{LS}(\varphi))$, hence angelic (see the discussion before Fact A.1.5), so each point of H is the limit of a sequence from $\{e_b : b \in B\}$; through the homeomorphism, each $q \in \text{RS}(\varphi)$ is the limit of a sequence of realized types $\text{rtp}_\varphi(b_n)$. ☺

Sequences are a genuine restriction. In the unstable structure $(\omega_1, <)$ with $\varphi(x, y)$ the formula $x < y$, the left type p given by $p(\beta) = 0$ for every $\beta < \omega_1$ (an element above every countable ordinal) is a limit of realized types along an ultrafilter, but not along any sequence: if $\alpha_n < \omega_1$ for every n , then $\alpha = \sup_n \alpha_n < \omega_1$, and the realized types $\text{ltp}_\varphi(\alpha_n)$ all take the value 1 at $\alpha + 1$, so no subsequence of them converges to p .

A.2. Historical remarks

✂ **Section A.1.** Grothendieck proved the criterion of Theorem A.1.3 in a paper of 1952 [Gro52], announced two years earlier [Gro50]; the same paper proves that pointwise compact subsets of $C(K)$ are angelic. The Eberlein–Šmulian theorem is due to Šmulian [Šmu40] and Eberlein [Ebe47]. Later proofs and extensions of the criterion, by Namioka, Pták, Young, Simons, Pryce and König, are listed in the timeline in the historical remarks on Subsection 1.2.1. The reading of the criterion as a statement about stable functions, which is the point of view of Definition A.1.1 and Proposition A.1.7, is due to Ben Yaacov [Ben14] and Pillay [Pil17] for formulas, following Iovino [Iov99] and Krivine–Maurey [KM81].

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